TODOs list

p. 16 "regularity condition" for quicksylver needs more precision.
p. 20 "regularity condition" for derivations needs more precision.
p. 35 Add some performance numbers/charts from Quicksylver/derivation calculations from the algorithm implementation in Julia

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1. Introduction

A scalar records one measurement. A vector records a magnitude and direction, so it has one axis, where an axis means an independent mode of access. A matrix has two axes, and many scientific and engineering settings require tensors with three or more axes. Tensors encapsulate all of these abstractions in one unifying concept. As a result, they are used in various disciplines to record information for some fixed reference frame. Some complementary perspectives studying tensors include [Bro97] [KB09] [Lan12] [RS18] [DLDMV00] [Tuc66].

Below we give two examples.

The first is the stress tensor in mechanics. For an object such as a steel beam supporting traffic, we can track how the internal force depends on the orientation of a tiny surface element inside the material. At each point this gives a 3×3 Cauchy stress tensor: diagonal entries capture normal compression/tension, while off-diagonal entries capture shear [GFA10]. To model deformation laws, one often maps one stress state to another via a four axes tensor, since each output component can depend on each input component across two input and two output axes.

The second comes from recommender systems, of which the Netflix tensor is the most famous. The starting point is to store a user $\times$ movie matrix of ratings. The factorization of this matrix was central to recommenders competing for the Netflix Prize [KBV09]. But recommendation systems may need more axes, for example, by considering preferences over time, and considering multiple devices. This necessitates higher valence tensors, for instance with axes user $\times$ movie $\times$ time. Tensor factorization methods explicitly model and work with these higher valence tensors [Kar+10].

1.1. Main Results.

We briefly describe the two main problems tackled by my dissertation.

The first is about efficient computing of algebraic invariants of tensors. They have been key to discovering basis-independent cluster patterns in tensors [BKW24], decomposing p-groups [Wil09], computing endomorphisms of modules [BL08], and advancing isomorphism testing [IQ19], [BMW17] [BW12] [BMW22].

Throughout, $\mathbb{F}$ will denote a field. Suppose we are given a 3-tensor $T \in \mathbb{F}^{m\times n\times r}$ which represents trilinear measurements in a fixed basis. Let the (i,j,k)th component be denoted $T[i,j,k]$, and let $T_1,\dots,T_r \in \mathbb{M}_{m\times n}(\mathbb{F})$ be slices using the Matlab/Julia inspired slice notation, where $T_i := T[:,:,i]$. For $Z = (z_{ji}) \in \mathbb{M}_r(\mathbb{F})$, define T^Z as the tensor with slices

$$(T^Z)_i := \sum_{j=1}^{r} z_{ji} T_j.$$

Equations for three algebraic invariants (adjoint, centroid, and derivation) we study are:

$$\begin{aligned}
\mathrm{Adj}(T) &:= \{(X,Y) \in \mathbb{M}_m(\mathbb{F}) \times \mathbb{M}_n(\mathbb{F}) : (\forall i),\ XT_i = T_iY\ \},\\
\mathrm{Cen}(T) &:= \{(X,Y,Z) \in \mathbb{M}_m(\mathbb{F}) \times \mathbb{M}_n(\mathbb{F}) \times \mathbb{M}_r(\mathbb{F}) : (\forall i),\ XT_i = T_iY = (T^Z)_i\},\\
\mathrm{Der}(T) &:= \{(X,Y,Z) \in \mathbb{M}_m(\mathbb{F}) \times \mathbb{M}_n(\mathbb{F}) \times \mathbb{M}_r(\mathbb{F}) : (\forall i),\ XT_i + T_iY = (T^Z)_i\}.
\end{aligned} \tag{1.1}$$

Each corresponds to a linear system. Our first main result, Theorem 3.33, proves that, given $T \in \mathbb{F}^{a\times b\times c}$, there is an algorithm for computing the derivation algebra of a tensor asymptotically faster than solving the full linear system in certain scenarios.

The second result is about understanding structure when we combine two tensors into one via multiplication. Multiplication is a natural way to inductively build up data from smaller pieces. For

instance, if A and B are $\mathbb{F}$-algebras, then $A \otimes_{\mathbb{F}} B$ is again an algebra. An example of this is the extensions of scalars construction $A \otimes_{\mathbb{F}} L$ for a field extension $L/\mathbb{F}$. We define a product of tensors that serves the same role for tensors by combining interactions while preserving multilinearity. We are given tensors as bilinear maps

$$s : \mathbb{F}^m \times \mathbb{F}^n \rightarrowtail \mathbb{F}^l, \qquad t : \mathbb{F}^p \times \mathbb{F}^q \rightarrowtail \mathbb{F}^r.$$

A convention in this dissertation is to give bilinear maps with lower case letters, and multiway arrays as upper case letters. Define the product tensor

$$s \otimes t : (\mathbb{F}^m \otimes \mathbb{F}^p) \times (\mathbb{F}^n \otimes \mathbb{F}^q) \rightarrowtail \mathbb{F}^l \otimes \mathbb{F}^r$$

on coordinate vectors by

$$(s \otimes t)(e_i \otimes f_a,\ e_j \otimes f_b) \coloneqq s(e_i, e_j) \otimes t(f_a, f_b),$$

and extend bilinearly.

Example 1.1. Consider the multiplication tensors of $\mathbb{F}[x]/(x^2+1)$ and the multiplication tensor of $\mathbb{F}[y]/(y^2+5y+10)$. Each algebra is given a $2 \times 2 \times 2$ multiplication table, and their combined multiplication is as shown below.

	1	x
1	1	x
x	x	-1

$\otimes$

	1	y
1	1	y
y	y	$-5y-10$

$=$

	1	x	y	xy
1	1	x	y	xy
x	x	-1	xy	$-y$
y	y	xy	$-5y-10$	$-5xy-10x$
xy	xy	$-y$	$-5xy-10x$	$5y+10$

Our second result, Theorem 5.1, is a theorem that states given 3-tensors s and t, we can understand $\mathrm{Der}(s \otimes t)$ by individually understanding the adjoint, centroid, and derivation algebras of s and t.

2. Preliminaries and Prior Work

Given $\mathbb{F}$-vector spaces U, V, and W, a function $f : U \times V \to W$ is additive if $f(u+u', v) = f(u,v) + f(u',v)$, and $f(u, v+v') = f(u,v) + f(u,v')$ for all u, u' in U and v, v' in V. A function f is $\mathbb{F}$-bilinear if f is additive and satisfies $f(ku, v) = f(u, kv) = kf(u,v)$ for all k in $\mathbb{F}$, u in U, and v in V. We write $f : U \times V \rightarrowtail W$ ($\rightarrowtail$ for bilinear). When context is clear, we elide the prefix $\mathbb{F}$.

Example 2.1. Examples of bilinear maps include:

(1) If A is an $\mathbb{F}$-algebra, then multiplication map $\mu : A \times A \rightarrowtail A$ is bilinear.
(2) For positive integers m, n, p, the rectangular matrix multiplication map

$$\mathbb{M}_{m\times n}(\mathbb{F}) \times \mathbb{M}_{n\times p}(\mathbb{F}) \rightarrowtail \mathbb{M}_{m\times p}(\mathbb{F}), \qquad (X, Y) \mapsto XY$$

is bilinear.
(3) For $U = \mathbb{F}^2$ and $V = \mathbb{F}^3$, any bilinear form $t : U \times V \rightarrowtail \mathbb{F}$ can be represented by a matrix $M \in \mathbb{M}_{2\times 3}(\mathbb{F})$ via $t(u,v) = u^T M v$.

The above extends to explain $\mathbb{F}$-trilinear and $\mathbb{F}$-multilinear in general. Throughout this dissertation, $\mathbb{F}$-bilinear, trilinear, and multilinear maps are also called *tensors*.

For vector spaces U and V, the *tensor product* of U and V is the vector space $U \otimes V$ with the canonical map $\varphi : U \times V \rightarrowtail U \otimes V$, such that for every bilinear map f with domain $U \times V$, there is a unique induced linear map $\hat{f}$ satisfying $f = \hat{f} \circ \varphi$. For vectors $s \in U$ and $t \in V$, the notation $s \otimes t$ denotes the image of (s,t) under φ. The elements $s \otimes t$ are called pure tensors. Next we describe subspaces of a tensor product space.

Lemma 2.2. Let U, V be vector spaces, and $A \leq U$ and $X \leq V$ be subspaces, with inclusion maps $\iota_A : A \hookrightarrow U$ and $\iota_X : X \hookrightarrow V$. Then inside $U \otimes V$,

$$A \otimes X = (U \otimes X) \cap (A \otimes V).$$

Proof. The intersection of $U \otimes X$ and $A \otimes V$ in $U \otimes V$ is the pullback of the two inclusion maps. Thus it suffices to show the following square is also a pullback:

$$\begin{tikzcd} A \otimes X \arrow[r, "\iota_A \otimes \mathrm{id}_X"] \arrow[d, "\mathrm{id}_A \otimes \iota_X"'] & U \otimes X \arrow[d, "\mathrm{id}_U \otimes \iota_X"] \\ A \otimes V \arrow[r, "\iota_A \otimes \mathrm{id}_V"'] & U \otimes V \end{tikzcd}$$

Let T be a vector space with maps $q_1 : T \to A \otimes V$ and $q_2 : T \to U \otimes X$ such that $(\iota_A \otimes \mathrm{id}_V) \circ q_1 = (\mathrm{id}_U \otimes \iota_X) \circ q_2$. Then we must construct the unique u making both triangles commute in the below diagram

$$\begin{tikzcd} T \arrow[dr, dashed, "u"] \arrow[drr, bend left, "q_2"] \arrow[ddr, bend right, "q_1"'] & & \\ & A \otimes X \arrow[r, "\iota_A \otimes \mathrm{id}_X"] \arrow[d, "\mathrm{id}_A \otimes \iota_X"'] & U \otimes X \arrow[d, "\mathrm{id}_U \otimes \iota_X"] \\ & A \otimes V \arrow[r, "\iota_A \otimes \mathrm{id}_V"'] & U \otimes V \end{tikzcd}$$

Let $q : V \to V/X$ be the quotient map. Since $q \circ \iota_X = 0$, we know $(\mathrm{id}_U \otimes q) \circ (\mathrm{id}_U \otimes \iota_X) = 0$.

Also, as elements in $\mathrm{Hom}(A \otimes V, U \otimes (V/X))$,

$$(\mathrm{id}_U \otimes q) \circ (\iota_A \otimes \mathrm{id}_V) = (\iota_A \otimes \mathrm{id}_{V/X}) \circ (\mathrm{id}_A \otimes q), \tag{2.1}$$

because both sides send the pure tensor $a \otimes v$ to $\iota_A(a) \otimes q(v)$. Therefore

$$\begin{aligned} 0 = (\mathrm{id}_U \otimes q) \circ (\mathrm{id}_U \otimes \iota_X) \circ q_2 &= (\mathrm{id}_U \otimes q) \circ (\iota_A \otimes \mathrm{id}_V) \circ q_1 \\ &= (\iota_A \otimes \mathrm{id}_{V/X}) \circ (\mathrm{id}_A \otimes q) \circ q_1. \end{aligned}$$

Where the last equality is by (2.1). Since $\iota_A \otimes \mathrm{id}_{V/X}$ is injective, $(\mathrm{id}_A \otimes q) \circ q_1 = 0$. This means $\mathrm{Im}(q_1) \leq \mathrm{Ker}(\mathrm{id}_A \otimes q)$.

The following is always a short exact sequence of vector spaces:

$$0 \to X \xrightarrow{\iota_X} V \xrightarrow{q} V/X \to 0.$$

Since tensoring over a field is exact, tensoring with A gives

$$0 \to A \otimes X \xrightarrow{\mathrm{id}_A \otimes \iota_X} A \otimes V \xrightarrow{\mathrm{id}_A \otimes q} A \otimes (V/X) \to 0.$$

Hence $\mathrm{Ker}(\mathrm{id}_A \otimes q) = \mathrm{Im}(\mathrm{id}_A \otimes \iota_X)$.

As $\mathrm{Im}(q_1) \leq \mathrm{Ker}(\mathrm{id}_A \otimes q) = \mathrm{Im}(\mathrm{id}_A \otimes \iota_X)$, there is a unique $u : T \to A \otimes X$ with $q_1 = (\mathrm{id}_A \otimes \iota_X) \circ u$. Finally, we need to show $q_2 = (\iota_A \otimes \mathrm{id}_X) \circ u$. We have

$$\begin{aligned} (\mathrm{id}_U \otimes \iota_X) \circ (\iota_A \otimes \mathrm{id}_X) \circ u &= (\iota_A \otimes \mathrm{id}_V) \circ (\mathrm{id}_A \otimes \iota_X) \circ u \\ &= (\iota_A \otimes \mathrm{id}_V) \circ q_1 \\ &= (\mathrm{id}_U \otimes \iota_X) \circ q_2, \end{aligned}$$

Therefore $(\mathrm{id}_U \otimes \iota_X) \circ \big((\iota_A \otimes \mathrm{id}_X) \circ u - q_2\big) = 0$. As $\mathrm{id}_U \otimes \iota_X$ is injective, we get $q_2 = (\iota_A \otimes \mathrm{id}_X) \circ u$ as needed. □

2.1. **Algebras and Actions on 3-tensors.** Existing work such as [Jac10], [Mya90], [BW14], and [Wil16] highlights the role of the three algebras we saw in Equation (1.1). In this subsection, we expand on that and describe some operations and properties that we need for the future sections.

As before U, V, and W are finite-dimensional $\mathbb{F}$-vector spaces. Let $\mathrm{End}(U) = \mathrm{Hom}(U, U)$ be the $\mathbb{F}$-linear endomorphisms of a vector space U, and let $b : U \times V \rightarrowtail W$ be a bilinear map. For $\alpha \in \mathrm{End}(U)$, $\beta \in \mathrm{End}(V)$, and $\gamma \in \mathrm{End}(W)$, define the *axis actions*

$$b \circ_2 \alpha : U \times V \rightarrowtail W, \qquad (u, v) \mapsto b(\alpha u, v),$$

$$b \circ_1 \beta : U \times V \rightarrowtail W, \qquad (u,v) \mapsto b(u, \beta v),$$
$$b \circ_0 \gamma : U \times V \rightarrowtail W, \qquad (u,v) \mapsto \gamma(b(u,v)).$$

Definition 2.3. The left, middle, and right nuclei of b are respectively

$$\mathcal{L}(b) := \{(\alpha,\gamma) \in \mathrm{End}(U) \times \mathrm{End}(W) : b \circ_2 \alpha = b \circ_0 \gamma\},$$
$$\mathcal{M}(b) := \{(\alpha,\beta) \in \mathrm{End}(U)^{\mathrm{op}} \times \mathrm{End}(V) : b \circ_2 \alpha = b \circ_1 \beta\},$$
$$\mathcal{R}(b) := \{(\beta,\gamma) \in \mathrm{End}(V) \times \mathrm{End}(W) : b \circ_1 \beta = b \circ_0 \gamma\}.$$

The need to op for $\mathcal{M}(b)$ is since if $(\alpha,\beta),(\alpha',\beta') \in \mathcal{M}(b)$, then $b((\alpha'\alpha)u, v) = b(\alpha u, \beta' v) = b(u, (\beta\beta')v)$, so the product is $(\alpha,\beta)(\alpha',\beta') = (\alpha'\alpha, \beta\beta')$. Notice $\mathcal{M}(b)$ is the coordinate free description of the adjoint algebra in Equation (1.1).

The centroid of b is

$$\mathcal{C}(b) := \{(\alpha,\beta,\gamma) \in \mathrm{End}(U) \times \mathrm{End}(V) \times \mathrm{End}(W) : b \circ_2 \alpha = b \circ_1 \beta = b \circ_0 \gamma\},$$

and the derivation algebra of b is

$$\mathrm{Der}(b) := \{(\alpha,\beta,\gamma) \in \mathrm{End}(U) \times \mathrm{End}(V) \times \mathrm{End}(W) : b \circ_2 \alpha + b \circ_1 \beta = b \circ_0 \gamma\}.$$

Remark 2.4. Each of $\mathcal{L}(b)$, $\mathcal{M}(b)$, $\mathcal{R}(b)$, $\mathcal{C}(b)$, $\mathrm{Der}(b)$ is an $\mathbb{F}$-vector subspace of the corresponding product of endomorphism spaces. Furthermore, $\mathcal{L}(b)$, $\mathcal{M}(b)$, $\mathcal{R}(b)$, and $\mathcal{C}(b)$ are associative algebras, with $\mathcal{C}(b)$ also commutative when b is fully nondegenerate, while $\mathrm{Der}(b)$ is a Lie algebra. [BMW20, Section 1.2].

2.2. **Radicals and non-degeneracy of tensors.** Let $b : U \times V \rightarrowtail W$ be a bimap. For $u \in U$ and $v \in V$, let

$$b(u,V) := \{b(u,v') : v' \in V\},$$
$$b(U,v) := \{b(u',v) : u' \in U\},$$
$$b(U,V) := \mathrm{span}\{b(u',v') : u' \in U,\ v' \in V\}.$$

The *left radical*, *right radical*, and *coradical* of a bimap b are respectively

$$\mathrm{Rad}_L(b) := \{u \in U : b(u,V) = 0\},$$
$$\mathrm{Rad}_R(b) := \{v \in V : b(U,v) = 0\},$$
$$\mathrm{Corad}(b) := W/b(U,V).$$

We call b *fully nondegenerate* if $\mathrm{Rad}_L(b) = \mathrm{Rad}_R(b) = 0$ and $\mathrm{Corad}(b) = 0$.
We now define the *axis actions*

$$\Lambda_2^b : \mathrm{End}(U) \to \mathrm{Bil}(U,V;W), \quad \alpha \mapsto b \circ_2 \alpha,$$
$$\Lambda_1^b : \mathrm{End}(V) \to \mathrm{Bil}(U,V;W), \quad \beta \mapsto b \circ_1 \beta,$$
$$\Lambda_0^b : \mathrm{End}(W)^{\mathrm{op}} \to \mathrm{Bil}(U,V;W), \quad \gamma \mapsto b \circ_0 \gamma.$$

The superscript is suppressed when context is clear.

Lemma 2.5. If $b : U \times V \rightarrowtail W$ is fully nondegenerate, then Λ_2, Λ_1, and Λ_0 are injective.

Proof. If $\Lambda_2(\alpha) = 0$, then $b(\alpha u, v) = 0$ for all $u \in U$ and $v \in V$, so $\alpha u \in \mathrm{Rad}_L(b)$ for every u. Since $\mathrm{Rad}_L(b) = 0$, we get $\alpha = 0$. The proof for Λ_1 is the same. For Λ_0, if $\Lambda_0(\gamma) = 0$, then $\gamma(b(u,v)) = 0$ for all u, v, so γ vanishes on $\mathrm{span}(b(U,V)) = W$. Hence $\gamma = 0$. □

Fact 2.6. For any bimap $b : U \times V \rightarrowtail W$, there is a canonical induced fully nondegenerate bimap

$$b/\sqrt{b} : \big(U/\mathrm{Rad}_L(b)\big) \times \big(V/\mathrm{Rad}_R(b)\big) \rightarrowtail b(U,V),$$

defined by

$$\big(b/\sqrt{b}\big)\big(u + \mathrm{Rad}_L(b), v + \mathrm{Rad}_R(b)\big) := b(u,v).$$

The invariants of $b/\sqrt{b}$ gives insight into b. In fact, every $(\alpha,\gamma) \in \mathcal{L}(b)$ satisfies

$$\alpha(\mathrm{Rad}_L(b)) \le \mathrm{Rad}_L(b), \qquad \gamma\big(b(U,V)\big) \le b(U,V).$$

The proof is as follows. Let $u \in \mathrm{Rad}_L(b)$. Then $b(u,v) = 0$ for all v, so $\gamma(b(u,v)) = \gamma(0) = 0$. But $\gamma(b(u,v)) = b(\alpha u, v)$ as well, so $\alpha u \in \mathrm{Rad}_L(b)$. The proof for $\gamma(b(U,V)) \leq b(U,V)$ is analogous.

Define the map

$$\rho_L : \mathcal{L}(b) \to \mathcal{L}(b/\sqrt{b}), \qquad (\alpha, \gamma) \mapsto (\bar{\alpha}, \gamma|_{b(U,V)}),$$

where $\bar{\alpha}$ is induced by α on $U/\mathrm{Rad}_L(b)$. We check the image, $(\bar{\alpha}, \gamma|_{b(U,V)})$, is in $\mathcal{L}(b/\sqrt{b})$: for all $u \in U$, $v \in V$,

$$\begin{aligned}\big(b/\sqrt{b}\big)\big(\bar{\alpha}(u + \mathrm{Rad}_L(b)), v + \mathrm{Rad}_R(b)\big) &= b(\alpha u, v) = \gamma(b(u,v)) \\ &= \gamma|_{b(U,V)}\Big(\big(b/\sqrt{b}\big)\big(u + \mathrm{Rad}_L(b), v + \mathrm{Rad}_R(b)\big)\Big),\end{aligned}$$

so $(\bar{\alpha}, \gamma|_{b(U,V)}) \in \mathcal{L}(b/\sqrt{b})$. We now show ρ_L is surjective. Given $(\bar{\alpha}, \gamma_0) \in \mathcal{L}(b/\sqrt{b})$, choose the lift $\alpha \in \mathrm{End}(U)$ of $\bar{\alpha}$ with $\alpha|_{\mathrm{Rad}_L(b)} = 0$, and choose $\gamma \in \mathrm{End}(W)$ with $\gamma|_{b(U,V)} = \gamma_0$ and $\gamma = 0$ on a chosen complement of $b(U,V)$. Then for all $u \in U$ and $v \in V$,

$$\begin{aligned}b(\alpha u, v) &= \big(b/\sqrt{b}\big)\big(\bar{\alpha}(u + \mathrm{Rad}_L(b)), v + \mathrm{Rad}_R(b)\big) \\ &= \gamma_0\Big(\big(b/\sqrt{b}\big)\big(u + \mathrm{Rad}_L(b), v + \mathrm{Rad}_R(b)\big)\Big) = \gamma(b(u,v)).\end{aligned}$$

Hence $(\alpha, \gamma) \in \mathcal{L}(b)$ as needed. The kernel of ρ_L are all pairs of endomorphisms where the first component acts like zero modulo $\mathrm{Rad}_L(b)$ (hence $\alpha(U) \subseteq \mathrm{Rad}_L(b)$) and the second component is zero restricted to $b(U,V)$. Thus,

$$\mathrm{Ker}(\rho_L) = \mathrm{Hom}(U, \mathrm{Rad}_L(b)) \times \{\gamma \in \mathrm{End}(W) : \gamma|_{b(U,V)} = 0\} \cong \mathrm{Hom}(U, \mathrm{Rad}_L(b)) \times \mathrm{Hom}(\mathrm{Corad}(b), W).$$

Hence we have the below exact sequence

$$0 \to \mathrm{Hom}(U, \mathrm{Rad}_L(b)) \times \mathrm{Hom}(\mathrm{Corad}(b), W) \to \mathcal{L}(b) \xrightarrow{\rho_L} \mathcal{L}(b/\sqrt{b}) \to 0.$$

2.3. **Product of tuples of endomorphisms.** For $k \in \{0,1,2\}$, let U_k and V_k be finite-dimensional $\mathbb{F}$-vector spaces. We use the Kronecker isomorphisms

$$\kappa_k : \mathrm{End}(U_k) \otimes \mathrm{End}(V_k) \xrightarrow{\sim} \mathrm{End}(U_k \otimes V_k),$$

defined by $\kappa_k(\alpha \otimes f)(u \otimes v) := \alpha u \otimes f v$ for all $u \in U_k$ and $v \in V_k$. The following is notation for combining tuples of endomorphisms.

Definition 2.7. Let K be finite. For each $k \in K$, let U_k, V_k be finite-dimensional $\mathbb{F}$-vector spaces. Define

$$\boxtimes : \Big(\prod_{k \in K} \mathrm{End}(U_k)\Big) \times \Big(\prod_{k \in K} \mathrm{End}(V_k)\Big) \to \prod_{k \in K} \mathrm{End}(U_k \otimes V_k)$$

by

$$((\alpha_k)_{k \in K}, (f_k)_{k \in K}) \longmapsto (\kappa_k(\alpha_k \otimes f_k))_{k \in K} =: (\alpha_k)_{k \in K} \boxtimes (f_k)_{k \in K}.$$

For subspaces $A \leq \prod_{k \in K} \mathrm{End}(U_k)$ and $B \leq \prod_{k \in K} \mathrm{End}(V_k)$, define

$$A \boxtimes B := \mathrm{span}\{a \boxtimes b : a \in A,\ b \in B\} \leq \prod_{k \in K} \mathrm{End}(U_k \otimes V_k).$$

2.4. **Product of tensors.** As described in Section 1, there is a need to combine two tensors into one using a product like construction.

Definition 2.8. Fix two bimaps

$$s : U_2 \times U_1 \rightarrowtail U_0, \qquad t : V_2 \times V_1 \rightarrowtail V_0.$$

The *tensor product bimap*

$$s \otimes t : (U_2 \otimes V_2) \times (U_1 \otimes V_1) \rightarrowtail U_0 \otimes V_0$$

is the unique bilinear map satisfying $(s \otimes t)((u_2 \otimes v_2), (u_1 \otimes v_1)) = s(u_2, u_1) \otimes t(v_2, v_1)$.

Lemma 2.9. If s and t are fully nondegenerate bimaps, then $s \otimes t$ is fully nondegenerate.

Proof. For nondegeneracy of $U_2 \otimes V_2$, write $x = \sum_j u_{2,j} \otimes v_{2,j} \in U_2 \otimes V_2$ and assume $(s \otimes t)(x, (u_1 \otimes v_1)) = 0$ for all $u_1 \in U_1$ and $v_1 \in V_1$. We need to show $x = 0$. Define $\lambda_s : U_2 \to \mathrm{Hom}(U_1, U_0)$ by $\lambda_s(u_2)(u_1) := s(u_2, u_1)$ and $\lambda_t : V_2 \to \mathrm{Hom}(V_1, V_0)$ by $\lambda_t(v_2)(v_1) := t(v_2, v_1)$. Identify $\mathrm{Hom}(U_1, U_0) \otimes \mathrm{Hom}(V_1, V_0)$ with $\mathrm{Hom}(U_1 \otimes V_1,\ U_0 \otimes V_0)$ via $(f \otimes g)(u_1 \otimes v_1) = f(u_1) \otimes g(v_1)$. Under this identification, the assumption gives $(\lambda_s \otimes \lambda_t)(x) = 0$. Since $\mathrm{Ker}(\lambda_s) = \mathrm{Rad}_L(s) = 0$ and $\mathrm{Ker}(\lambda_t) = \mathrm{Rad}_L(t) = 0$, both maps are injective, so $\lambda_s \otimes \lambda_t$ is injective and $x = 0$. The argument for $U_1 \otimes V_1$ is analogous.

For the coradical, write $u_0 = \sum_a s(u_{2,a}, u_{1,a})$ and $v_0 = \sum_b t(v_{2,b}, v_{1,b})$. Then

$$\begin{aligned} u_0 \otimes v_0 &= \left(\sum_a s(u_{2,a}, u_{1,a})\right) \otimes \left(\sum_b t(v_{2,b}, v_{1,b})\right) \\ &= \sum_{a,b} (s \otimes t)((u_{2,a} \otimes v_{2,b}), (u_{1,a} \otimes v_{1,b})), \end{aligned}$$

so $\mathrm{Corad}(s \otimes t) = 0$. □

Remark 2.10. If s and t are fully nondegenerate, then the maps

$$\begin{aligned} \Lambda_2^{s\otimes t} &: \mathrm{End}(U_2 \otimes V_2) \to \mathrm{Bil}(U_2 \otimes V_2, U_1 \otimes V_1; U_0 \otimes V_0), \\ \Lambda_1^{s\otimes t} &: \mathrm{End}(U_1 \otimes V_1) \to \mathrm{Bil}(U_2 \otimes V_2, U_1 \otimes V_1; U_0 \otimes V_0), \\ \Lambda_0^{s\otimes t} &: \mathrm{End}(U_0 \otimes V_0)^{\mathrm{op}} \to \mathrm{Bil}(U_2 \otimes V_2, U_1 \otimes V_1; U_0 \otimes V_0) \end{aligned}$$

are injective by Lemmas 2.5 and 2.9.

Next we come to a key correspondence that allows us to analyze axis actions on tensor product bimaps by *linear algebra* techniques. Recall for X, Y vector spaces, and $(X \otimes Y, \varphi : X \times Y \rightarrowtail X \otimes Y)$ the tensor product space of X and Y, the element $\varphi(x, y) \in X \otimes Y$ is the pure tensor often written as $x \otimes y$. As the notation $x \otimes y$ can be ambiguous when x and y are themselves bimaps, when wanting to refer to the pure tensor element in the tensor product vector space, we write $\varphi(x, y)$.

Let $\mathcal{U} := \mathrm{Bil}(U_2, U_1; U_0)$, $\mathcal{V} := \mathrm{Bil}(V_2, V_1; V_0)$, and define the bilinear map

$$\Theta : \mathcal{U} \times \mathcal{V} \to \mathrm{Bil}(U_2 \otimes V_2, U_1 \otimes V_1; U_0 \otimes V_0), \quad \Theta(s, t) := s \otimes t \text{ as the tensor product bimap.}$$

The universal property of the tensor product space $\mathcal{U} \otimes \mathcal{V}$ gives a unique induced linear map

$$\widetilde{\Theta} : \mathcal{U} \otimes \mathcal{V} \to \mathrm{Bil}(U_2 \otimes V_2, U_1 \otimes V_1; U_0 \otimes V_0).$$

Write $u_k := \dim(U_k)$ and $v_k := \dim(V_k)$ for all $k \in \{0, 1, 2\}$. Then

$$\begin{aligned} \dim(\mathcal{U} \otimes \mathcal{V}) &= \dim(\mathcal{U}) \dim(\mathcal{V}) = (u_2 u_1 u_0)(v_2 v_1 v_0), \\ \dim(\mathrm{Bil}(U_2 \otimes V_2, U_1 \otimes V_1; U_0 \otimes V_0)) &= (u_2 v_2)(u_1 v_1)(u_0 v_0). \end{aligned}$$

These dimensions agree, so $\widetilde{\Theta}$ is an isomorphism. We shall need the notation $\Xi := \widetilde{\Theta}^{-1}$. Also recall the Kronecker isomorphism $\kappa_k : \mathrm{End}(U_k) \otimes \mathrm{End}(V_k) \to \mathrm{End}(U_k \otimes V_k)$ given by $\kappa_k(\alpha \otimes f)(u \otimes v) = \alpha u \otimes f v$.

Lemma 2.11. Following the notation above, let $s \in \mathcal{U}$, $t \in \mathcal{V}$ be bimaps, and $(\mathcal{U} \otimes \mathcal{V}, \varphi : \mathcal{U} \times \mathcal{V} \rightarrowtail \mathcal{U} \otimes \mathcal{V})$ the tensor product space of $\mathcal{U}$ with $\mathcal{V}$. For all $k \in \{0, 1, 2\}$, $\alpha \in \mathrm{End}(U_k)$, and $f \in \mathrm{End}(V_k)$,

$$\Xi\big((s \otimes t) \circ_k \kappa_k(\alpha \otimes f)\big) = \varphi(s \circ_k \alpha, t \circ_k f).$$

Proof. By definition of $\widetilde{\Theta}$,

$$\widetilde{\Theta}\big(\varphi(s \circ_k \alpha, t \circ_k f)\big) = \Theta(s \circ_k \alpha, t \circ_k f).$$

Evaluating both this bimap and $(s \otimes t) \circ_k \kappa_k(\alpha \otimes f)$ on pure tensors shows they agree, so applying Ξ gives the claim. □

As a consequence, for $k \in \{0,1,2\}$ and $x \in \mathrm{End}(U_k \otimes V_k)$ with decomposition $x = \sum_j \kappa_k(\alpha_j \otimes f_j)$, linearity gives

$$\begin{aligned} \Xi\big((s \otimes t) \circ_k x\big) &= \sum_j \Xi\big((s \otimes t) \circ_k \kappa_k(\alpha_j \otimes f_j)\big) \\ &= \sum_j \varphi(s \circ_k \alpha_j, t \circ_k f_j). \end{aligned} \tag{2.2}$$

2.5. **Prior work.** Existing research places importance on the algebras coming form Equation (1.1). Myasnikov, Wilson, and others in [Mya90], [Wil16], [Wil12], [MM10], [Wil09] proves for the bimap b the algebras $\mathrm{Adj}(b)$ and $\mathrm{Cen}(b)$ control direct sum decompositions and automorphisms of t, using them to prove properties for the originating algebraic structures. The most famous special case for the adjoint equation is the theory of Kronecker modules, which in our terminology is when the number of depth slices is fixed at 2 [Kro90]. Recent work [BKW24] also shows that derivations also controls the original tensor's null patterns, limiting the subspaces on which the tensor is non-zero.

There has also been prior work on understanding derivations of tensors coming from the multiplication tensors of algebras. Leger and Luks study generalized derivations, quasiderivations, and quasicentroids as extensions of algebra derivations [LL00], and Novotný and Hřivnák study (α, β, γ)-derivations together with associated invariant functions [NH08]. Closer to our results, Brešar studies the derivations of tensor products of nonassociative algebras [Bre17], Benkart and Osborn studies the derivations and automorphisms of nonassociative matrix algebras [BO81] and Azam studies the derivations of tensor products of algebras [Aza08].

In the next three sections, we describe in detail the problems we investigated and contributions we made as part of this dissertation.

3. Quick Derivation Solutions

The descriptions of the adjoint, centroid, and derivation algebras in Equation (1.1) are linear in the unknown matrix entries. So each admit a naive solution as a large system of linear equations. In collaboration with Joshua Maglione and James B. Wilson, we obtain an asymptotically faster algorithm for computing the adjoint algebra, inspired by analogous results for matrices known as the Bartels-Stewart algorithm [BS72]. In this chapter, I state and prove results from this joint work using a novel approach. I also state and prove my own original result, which concerns the derivation algebra, following a similar approach.

Theorem 3.21. For cubic $(n \times n \times n)$ tensors, under the regularity conditions (established in Definition 3.32), SimultaneousSylvesterSystem is solved in the same cost as the full deterministic verification. This is $O(n^4)$ cost, as compared to the naive $O(n^7)$ cost by solving as a linear system of equations.

Theorem 3.33. For cubic $(n \times n \times n)$ tensors, under the regularity conditions (established in Definition 3.38), DerivationSystem is solved in cost $O(n^{4.5})$. This is an additional $n^{1/2}$ overhead compared to the deterministic verification cost of $O(n^4)$, as compared to the naive $O(n^7)$ cost by solving as a linear system of equations.

Our strategy in this section is to reduce each large problem to a set of smaller linear algebra problems. Their solutions are affine subspaces, which we define below.

3.1. **Preliminaries.** An *affine space* is a pair (A, U) where A is a non-empty set and U is a vector space acting on A by a fixed-point free and transitive action. Write the action as $+ : A \times U \to A$, $(a, u) \mapsto a + u$. We call U the *direction space* of A. When U is clear from context, we also say "A is an affine space".

Lemma 3.1. Let (A, U) be an affine space. For any $a, b \in A$, there is a unique $u \in U$ such that $a + u = b$. We denote this vector by $b - a$.

Proof. Existence follows from transitivity. If $a + u = b = a + v$, then $a + (u - v) = a$. Fixed-point freeness gives $u - v = 0$, so $u = v$. □

Let (A, U) be a d-dimensional affine space. An *affine frame* is a tuple $(a_0, \ldots, a_d)$ in A such that $\{a_1 - a_0, \ldots, a_d - a_0\}$ is a basis of U. Every $a \in A$ has unique coordinates $\lambda = (\lambda_1, \ldots, \lambda_d) \in \mathbb{F}^d$ with $a = a_0 + \sum_{i=1}^{d} \lambda_i (a_i - a_0)$.

Convention 3.2. For computation, we encode an affine space by an affine frame (one basepoint plus a basis of the direction space).

Let (A, U) and (B, V) be affine spaces. A map $f : A \to B$ is an *affine map* if there exists a linear map $L : U \to V$ such that $f(a + u) = f(a) + L(u)$ for all $a \in A$ and $u \in U$. We call L the underlying linear map of f. Below we list some properties of affine space and maps that we will need in following sections.

(1) Every vector space V is also an affine space with direction space V.
(2) If A, B are vector spaces and $f : A \to B$ is a linear map, then f is also an affine map.
(3) If (A, U) and (B, V) are affine spaces, then $(A \times B, U \times V)$ is an affine space with action $(a, b) + (u, v) := (a + u, b + v)$.
(4) An affine map $f : A \times B \to C$ has an underlying linear map $L : U \times V \to W$ with $f\big((a, b) + (u, v)\big) = f(a, b) + L(u, v)$. This is in contrast to when f is a bilinear map in two variables.
(5) Let U and V be vector spaces, and let $L : U \to V$ be linear. If (A, W) is an affine subspace of U, then $L|_A : A \to V$ is affine, with underlying linear map $L|_W$.

Example 3.3. Let $A = \mathbb{R}^3$, $a_0 = (1, 2, 3)$, and $W = \{(x, y, 0) : x, y \in \mathbb{R}\} \leq \mathbb{R}^3$. Then $a_0 + W$ is the plane $z = 3$, an affine subspace of $(\mathbb{R}^3, \mathbb{R}^3)$.

Finally, we describe how we analyze algorithms and our notation for tensors.

Convention 3.4. For functions $f, g : \mathbb{N} \to \mathbb{N}$, we write $f \in O(g)$ if there exist constants $C, n_0 \in \mathbb{N}$ such that $f(n) \leq C\, g(n)$ for all $n \geq n_0$.

Convention 3.5. We always use the standard basis for coordinate vector spaces $\mathbb{F}^n$. For $u = (u_1, \ldots, u_n)$ and $v = (v_1, \ldots, v_n)$, we use the bilinear pairing $\langle u, v \rangle := \sum_{i=1}^{n} u_i v_i$.

Convention 3.6. We analyze algorithms by their arithmetic complexity, meaning the number of field operations over $\mathbb{F}$ (addition, subtraction, multiplication, and division).

Definition 3.7. Let $T \in \mathbb{F}^{a \times b \times c}$ be a tensor, let $I \subseteq [a]$, $J \subseteq [b]$, and let $k \in [c]$. We use the slice notation

$$T[I, :, k] \in \mathbb{F}^{|I| \times b}, \qquad T[:, J, k] \in \mathbb{F}^{a \times |J|}, \qquad T[I, J, k] \in \mathbb{F}^{|I| \times |J|}.$$

For $M \in \mathbb{F}^{a \times b}$ a matrix, we write

$$M[I, :] \in \mathbb{F}^{|I| \times b}, \qquad M[:, J] \in \mathbb{F}^{a \times |J|}.$$

The indexing of sliced tensors and matrices is by the fixed ordering of the slicing sets. For instance, for $I = \{i_1 < \cdots < i_t\} \subset [a]$, the restricted matrix $X[I, :] \in \mathbb{F}^{t \times r}$ has $X[I, :][j, :] = X[i_j, :]$. We use this convention for all restricted tensor and matrix indexing.

Example 3.8. Let

$$T_1 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad T_2 = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix},$$

then define $T \in \mathbb{F}^{2 \times 2 \times 2}$ by $T[:, :, 1] = T_1$, $T[:, :, 2] = T_2$. For $I = \{1\}$, $J = \{2\}$, we have

$$T[I, :, 1] = \begin{bmatrix} 1 & 2 \end{bmatrix}, \qquad T[:, J, 1] = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

For $J = \{2, 3\}$, we have

$$M = \begin{bmatrix} 9 & 10 & 11 \\ 12 & 13 & 14 \end{bmatrix} \in \mathbb{F}^{2\times 3}, \qquad M[:, J] = \begin{bmatrix} 10 & 11 \\ 13 & 14 \end{bmatrix} \in \mathbb{F}^{2\times 2}.$$

Definition 3.9. We define vectorization for matrices and tensors. For $M = (m_{ij}) \in \mathbb{F}^{a\times b}$, $\mathrm{vec}(M) \in \mathbb{F}^{ab}$ for $(i, j) \in [a] \times [b]$ by

$$\mathrm{vec}(M)_{\,i+(j-1)a} = m_{ij}.$$

For $T = (t_{ijk}) \in \mathbb{F}^{a\times b\times c}$, define $\mathrm{vec}(T) \in \mathbb{F}^{abc}$ for $(i, j, k) \in [a] \times [b] \times [c]$ by

$$\mathrm{vec}(T)_{\,i+(j-1)a+(k-1)ab} = t_{ijk}.$$

Example 3.10. For

$$M = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \in \mathbb{F}^{2\times 3}, \qquad T \in \mathbb{F}^{2\times 2\times 2}, \quad T[:, :, 1] = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad T[:, :, 2] = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix},$$

we have $\mathrm{vec}(M) = (1, 4, 2, 5, 3, 6)$, and $\mathrm{vec}(T) = (1, 3, 2, 4, 5, 7, 6, 8)$.

3.2. Linear Algebra subroutines.

Lemma 3.11. Let (A, U) and (B, V) be affine spaces, let $f : A \to B$ be affine with underlying linear map $L : U \to V$, and let (S, W) be an affine subspace of B. Then $f^{-1}(S) \subseteq A$ is either empty or an affine subspace of A.

Proof. If $f^{-1}(S) = \varnothing$, there is nothing to prove. Assume $f^{-1}(S) \neq \varnothing$, choose $a_0 \in f^{-1}(S)$, and set $U_0 := L^{-1}(W) \leq U$. For $u \in U$,

$$f(a_0 + u) = f(a_0) + L(u).$$

Since $f(a_0) \in S$ and S has direction space W,

$$f(a_0 + u) \in S \iff L(u) \in W \iff u \in U_0.$$

Hence $f^{-1}(S) = a_0 + U_0$, so $f^{-1}(S)$ is an affine subspace. □

Problem LinSolve($\mathbb{F}$).

Given: $\mathbb{F}$-vector spaces U, V, a linear map $L : U \to V$, and $v \in V$.
Return: A computational encoding of the solution set $\{u \in U : L(u) = v\}$.

Proposition 3.12. LinSolve($\mathbb{F}$) returns either an empty set or an affine subspace of U.

Proof. The map L is affine and $\{v\}$ is an affine subspace of V, so the solution set is $L^{-1}(\{v\})$. Applying Lemma 3.11 concludes the proposition. □

Proposition 3.13. A coordinatized instance of LinSolve($\mathbb{F}$), with

$$U = \mathbb{F}^{n\times \ell}, \qquad V = \mathbb{F}^{m\times \ell}, \qquad L(X) = MX, \qquad M \in \mathbb{F}^{m\times n},$$

is solvable in cost $\mathtt{T}_{p1}(m, n, \ell) \in O(\min\{m, n\}^2 \max\{m, n\} + mn\ell)$.

This is the standard dense Gaussian-elimination bound for one shared coefficient matrix and ℓ right-hand sides. Next is the workhorse for our solution strategy, allowing us to lift subproblem solutions to solutions of the whole problem.

Problem LinearEqualsAffine($\mathbb{F}$).

Given: $\mathbb{F}$-vector spaces U, W, a linear map $L : U \to W$, an affine space (Θ, V), and an affine map $N : \Theta \to W$.
Return: A computational encoding of the solution set $\{(u, \theta) \in U \times \Theta : L(u) = N(\theta)\}$.

Proposition 3.14. The solution set of LinearEqualsAffine($\mathbb{F}$) is either empty or an affine subspace of $U \times \Theta$.

Proof. Define $F : U \times \Theta \to W$ by $F(u, \theta) := L(u) - N(\theta)$. Then F is affine and the solution set is $F^{-1}(\{0\})$. Apply Lemma 3.11. □

Proposition 3.15. Let $\rho : W \to W/\operatorname{im}(L)$ be the quotient map and define

$$\Theta_{\text{feas}} := \{\theta \in \Theta : \rho(N(\theta)) = 0\}.$$

Then (u, θ) solves $L(u) = N(\theta)$ if and only if $\theta \in \Theta_{\text{feas}}$ and $u \in u_0(\theta) + \operatorname{Ker}(L)$ for some fixed solution $u_0(\theta)$ of $L(u) = N(\theta)$.

Proof. For fixed θ, solvability is $N(\theta) \in \operatorname{im}(L)$, which is exactly $\rho(N(\theta)) = 0$. If $u_0(\theta)$ is one solution, then $L(u) = N(\theta)$ if and only if $L(u - u_0(\theta)) = 0$ if and only if $u - u_0(\theta) \in \operatorname{Ker}(L)$. □

Proposition 3.16. Given a coordinatized instance of LinearEqualsAffine($\mathbb{F}$), with

$$U = \mathbb{F}^{n\times\ell}, \qquad W = \mathbb{F}^{m\times\ell}, \qquad L(X) = MX, \qquad M \in \mathbb{F}^{m\times n},$$

and with affine-frame coordinates $\theta \in \mathbb{F}^p$ for Θ and $N(\theta) = N_0 + \sum_{i=1}^p \theta_i N_i$, where

$$\kappa := \dim \operatorname{Ker}(M^T), \qquad q := \dim \Theta_{\text{feas}} \text{ if } \Theta_{\text{feas}} \neq \varnothing, \text{ and } 0 \text{ otherwise.}$$

The problem is solvable in cost

$$\mathtt{T}_{p2}(m, n, \ell, p, q, \kappa)$$

with

$$\begin{aligned}\mathtt{T}_{p2}(m, n, \ell, p, q, \kappa) \in O\Big(&mn^2 + p\kappa m\ell + (\kappa\ell)p^2 \\ &+ mn^2 + n^2\ell \\ &+ (q+1)p\,m\ell + mn^2 + (q+1)mn\ell\Big).\end{aligned}$$

Proof. The first line is the cost to solve for feasible parameters. Compute a basis $w_1, \ldots, w_\kappa$ of $\operatorname{Ker}(M^T)$ in $O(mn^2)$. By the standard pairing,

$$x \in \operatorname{im}(M) \iff \langle w_i, x\rangle = 0 \text{ for all } i \in [\kappa].$$

Let C be the matrix with rows $w_1, \ldots, w_\kappa$. Applying columnwise to $N(\theta)$ means $\rho(N(\theta)) = 0$ exactly when $CN(\theta) = 0$. Recall $N(\theta)$ is coordinatized as $N(\theta) = N_0 + \sum_{i=1}^p \theta_i N_i$. Thus we solve the inhomogeneous system $A_0 + \sum_{i=1}^p \theta_i A_i = 0$ with $A_i := CN_i$. Hence the three costs in line 1 are $O(mn^2)$ to compute w_i, $O(p\kappa m\ell)$ to form A_i, and $O((\kappa\ell)p^2)$ to solve the $(\kappa\ell) \times p$ system.

The second line is to compute a basis for the kernel of L. Compute a basis $u_1, \ldots, u_\nu$ of $\operatorname{Ker}(M)$ in $O(mn^2)$, where $\nu := \dim \operatorname{Ker}(M)$. Since $L(X) = MX$ acts columnwise, a basis for $\operatorname{Ker}(L)$ is given by the matrices $u_i e_j^T$ for $i \in [\nu]$ and $j \in [\ell]$. This costs $O(\nu n\ell) \subseteq O(n^2\ell)$.

The third line is to solve on an affine frame of the feasible parameters. We evaluate N on this affine frame of $(q+1)$ points in $O((q+1)p\,m\ell)$, row-reduce M once in $O(mn^2)$, and solve the $(q+1)$ right-hand sides with that reduction in $O((q+1)mn\ell)$. □

Example 3.17. In the following example, we have $m = 3, n = 2, p = 2$, and $\ell = 2$. And set

$$M = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \in \mathbb{F}^{3\times 2}, \qquad N : \mathbb{F}^2 \to \mathbb{F}^{3\times 2}, \qquad N(\theta_1, \theta_2) = N_0 + \theta_1 N_1 + \theta_2 N_2,$$

$$N_0 = \begin{bmatrix} 2 & 3 \\ 1 & 4 \\ -1 & -2 \end{bmatrix}, \qquad N_1 = \begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 1 & 2 \end{bmatrix}, \qquad N_2 = \begin{bmatrix} 0 & -1 \\ 0 & 2 \\ 1 & 2 \end{bmatrix}.$$

In the three-part process

(1) **Feasible parameters.** Here $\operatorname{im}(M) = \{(u, v, 0)^T : u, v \in \mathbb{F}\}$, so feasibility is exactly vanishing of the third row of $N(\theta_1, \theta_2)$. This is accomplished by the condition

$$\theta_1 + \theta_2 = 1.$$

Hence $\Theta_{\text{feas}} = \{(\theta_1, \theta_2) \in \mathbb{F}^2 : \theta_1 + \theta_2 = 1\}$.

(2) **Kernel.** $\operatorname{Ker}(M) = \{0\}$, so $\operatorname{Ker}(L) = \{0\}$.

(3) **Solve on an affine frame of Θ_{feas}.** Take frame points $\theta^{(0)} = (1, 0)$ and $\theta^{(1)} = (0, 1)$. Solving $MX = N(\theta^{(0)})$ and $MX = N(\theta^{(1)})$ gives

$$X^{(0)} = \begin{bmatrix} 3 & -5 \\ 0 & 4 \end{bmatrix}, \qquad X^{(1)} = \begin{bmatrix} 0 & -10 \\ 1 & 6 \end{bmatrix}.$$

Therefore parametrizing $\theta \in \Theta_{\text{feas}}$ as $\theta = (1 - \lambda, \lambda)$,

$$X(\theta) = X^{(0)} + \lambda\big(X^{(1)} - X^{(0)}\big).$$

Remark 3.18. If L has a left inverse G, then for each $\theta \in \Theta_{\text{feas}}$, $L(u) = N(\theta)$ has exactly one solution, $u = G(N(\theta))$.

The compatibility criteria that arise when lifting subproblem solutions are instances of the following problem.

Problem AffinePullback($\mathbb{F}$).

Given: $\mathbb{F}$-vector spaces U, V, W and affine maps $f : U \to W$, $g : V \to W$.
Return: A computational encoding of $P := \{(u, v) \in U \times V : f(u) = g(v)\}$.

The set P is either empty or an affine subspace of $U \times V$, as $P = h^{-1}(\{0\})$ for $h(u, v) := f(u) - g(v)$.

Proposition 3.19. Given a coordinate instance of AffinePullback($\mathbb{F}$) with

$$U = \mathbb{F}^d, \quad V = \mathbb{F}^e, \quad W = \mathbb{F}^m, \quad f(u) = Au + \alpha, \quad g(v) = Bv + \beta,$$

and $A \in \mathbb{F}^{m \times d}$ and $B \in \mathbb{F}^{m \times e}$, it is solvable in cost $\mathtt{T}_{p1}(m, d + e, 1)$.

Proof. The condition $f(u) = g(v)$ is the block linear system

$$\begin{bmatrix} A & -B \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \beta - \alpha,$$

which is one instance of LinSolve($\mathbb{F}$). □

3.3. **Simultaneous Sylvester Equations for $\mathrm{Adj}(T)$.** We work towards Theorem 3.21 in this section. Throughout, a, b, c, r, s will be natural numbers, and we use the notation

$$\mathcal{X} := \mathbb{F}^{a \times r}, \qquad \mathcal{Y} := \mathbb{F}^{s \times b},$$

to denote vector spaces. This notation may have subscripts to indicate restricted solution spaces. As an example, for $I \subseteq [a]$, $\mathcal{X}_I$ is the space $\mathbb{F}^{|I| \times r}$.

Problem SimultaneousSylvesterSystem.

Given: Tensors $R \in \mathbb{F}^{r \times b \times c}$, $S \in \mathbb{F}^{a \times s \times c}$, and $T \in \mathbb{F}^{a \times b \times c}$.
Return: Either return $\varnothing$, or the affine subspace

$$\{(X, Y) \in \mathcal{X} \times \mathcal{Y} : (\forall k \in [c])\ X\, R[:, :, k] + S[:, :, k]\, Y = T[:, :, k]\}.$$

This problem has a naive solution using LinSolve($\mathbb{F}$) with $m = abc$, $n = ar + bs$, and $\ell = 1$. When $R = -S$ and $T = 0$, solutions give the adjoint for the tensor R.

Example 3.20. Below is an example of laying out the equations for SimultaneousSylvesterSystem as a naive matrix vector system. Take $a = 2$, $b = 2$, $c = 2$, $r = 3$, and $s = 4$, with slices

$$R[:, :, 1] = \begin{bmatrix} 1 & 5 \\ 0 & -1 \\ 2 & 11 \end{bmatrix}, \qquad R[:, :, 2] = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ -1 & 1 \end{bmatrix},$$

$$S[:, :, 1] = \begin{bmatrix} 1 & 0 & 3 & 0 \\ 1 & 1 & 10 & -1 \end{bmatrix}, \qquad S[:, :, 2] = \begin{bmatrix} 0 & 1 & 7 & -1 \\ 1 & -1 & -4 & 1 \end{bmatrix},$$

$$T[:, :, 1] = \begin{bmatrix} 1 & 19 \\ -11 & 4 \end{bmatrix}, \qquad T[:, :, 2] = \begin{bmatrix} -10 & 4 \\ 8 & 2 \end{bmatrix}.$$

The corresponding augmented linear system is shown in Figure 1.

ijk	x_{00}	x_{01}	x_{02}	x_{10}	x_{11}	x_{12}	y_{00}	y_{10}	y_{20}	y_{30}	y_{01}	y_{11}	y_{21}	y_{31}	t_{ijk}
000	1	0	2				1	0	3	0					1
001	0	1	-1				0	1	7	-1					-10
010	5	-1	11								1	0	3	0	19
011	1	1	1								0	1	7	-1	4
100				1	0	2	1	1	10	-1					-11
101				0	1	-1	1	-1	-4	1					8
110				5	-1	11					1	1	10	-1	4
111				1	1	1					1	-1	-4	1	2

FIGURE 1. An augmented matrix for the Simultaneous Sylvester System in Example 3.20.

We describe an alternative approach, culminating in the following.

Theorem 3.21. For cubic ($n \times n \times n$) tensors, under the regularity conditions (established in Definition 3.32), SimultaneousSylvesterSystem is solved in the same cost as the full deterministic verification. This is $O(n^4)$ cost, as compared to the naive $O(n^7)$ cost by solving as a linear system of equations.

Example 3.22. We use the following running example throughout Section 3.3. Work over $\mathbb{F} = \mathbb{Q}$, with $a = b = r = s = 2$ and $c = 2$. Define

$$R[:,:,1] = S[:,:,1] = T[:,:,1] = \mathrm{diag}(1,0),$$
$$R[:,:,2] = S[:,:,2] = \mathrm{diag}(0,1), \quad T[:,:,2] = \mathrm{diag}(0,2).$$

The full solution family is

$$X = \mathrm{diag}(\alpha,\beta), \qquad Y = \mathrm{diag}(1-\alpha, 2-\beta), \qquad \alpha,\beta \in \mathbb{F}.$$

Our strategy for SimultaneousSylvesterSystem is to solve smaller systems with a subset of the constraints, and then lift them. We start with the following.

Problem DoubleRestricted.

Given: Tensors $R \in \mathbb{F}^{r\times b\times c}$, $S \in \mathbb{F}^{a\times s\times c}$, $T \in \mathbb{F}^{a\times b\times c}$, and index sets $I \subseteq [a]$, $J \subseteq [b]$ with $|I| = a'$ and $|J| = b'$.

Return: Return $\varnothing$, or the affine subspace

$$\mathcal{S}_{\mathrm{DR}} := \{(X_I, Y_J) \in \mathcal{X}_I \times \mathcal{Y}_J : (\forall k \in [c])\ X_I R[:,J,k] + S[I,:,k]Y_J = T[I,J,k]\}.$$

To solve, vectorizing yields an instance of LinSolve($\mathbb{F}$) which costs

$$\mathtt{T}_{p1}(m = a'b'c,\ n = a'r + b's,\ \ell = 1) \in O\big((a'b'c)(a'r+b's)^2\big).$$

We refer to this cost as $\mathtt{T}_{\mathrm{DR}}(a,b,c,r,s,a',b')$. When context is clear, we drop the parameters and the cost is denoted $\mathtt{T}_{\mathrm{DR}}$. By Proposition 3.12, the returned solution set is affine.

Example 3.23. In the running example, take $I = J = \{1\}$. Then $a' = b' = 1$, and

$$X_I = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \in \mathbb{F}^{1\times 2}, \qquad Y_J = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \in \mathbb{F}^{2\times 1}.$$

Here DoubleRestricted $x_1 + y_1 = 1$, with no other constraints on (x_2, y_2). Hence

$$\mathcal{S}_{\mathrm{DR}} = \left\{ \left(\begin{bmatrix} \alpha & \beta \end{bmatrix}, \begin{bmatrix} 1-\alpha \\ \gamma \end{bmatrix} \right) : \alpha,\beta,\gamma \in \mathbb{F} \right\}.$$

Problem RowRestricted.

Given: Tensors $R \in \mathbb{F}^{r\times b\times c}$, $S \in \mathbb{F}^{a\times s\times c}$, $T \in \mathbb{F}^{a\times b\times c}$, and index sets $I \subseteq [a]$, $J \subseteq [b]$ with $|I| = a'$ and $|J| = b'$.

Return: Return $\varnothing$, or the affine subspace

$$\mathcal{S}_{\mathrm{R}} := \{(X_I, Y) \in \mathcal{X}_I \times \mathcal{Y} : (\forall k \in [c])\ X_I R[:,:,k] + S[I,:,k]Y = T[I,:,k]\}.$$

Proposition 3.24. For fixed $I \subseteq [a]$ and $J \subseteq [b]$, RowRestricted is solved by solving DoubleRestricted and then LinearEqualsAffine($\mathbb{F}$), with cost

$$\begin{aligned} O\big(&\mathtt{T}_{p1}(m = a'b'c,\ n = a'r + b's,\ \ell = 1) \\ &+ d\,c\,a'r(b - b') \\ &+ \mathtt{T}_{p2}(m = ca',\ n = s,\ \ell = b - b',\ p = d,\ q = q_R,\ \kappa = \kappa_R)\big). \end{aligned}$$

where $a' = |I|$, $b' = |J|$, d is the dimension of the solution set of DoubleRestricted, q_R is the feasible parameter dimension, and κ_R is the left nullspace dimension parameter in the LinearEqualsAffine($\mathbb{F}$) instance (these three parameters are all 0 if $\varnothing$ is returned)

Proof. **Algorithm. Step 1.** Solve DoubleRestricted. If it returns $\varnothing$, return $\varnothing$ and stop.

Step 2. Set $\widehat{J} := [b] \setminus J$ and define an affine map $N_R : \mathcal{S}_{\mathrm{DR}} \to \mathbb{F}^{(ca')\times(b-b')}$ as well as an element $M_R \in \mathbb{F}^{(ca')\times s}$ by

$$N_R(X_I, Y_J) := \begin{bmatrix} T[I, \widehat{J}, 1] - X_I R[:, \widehat{J}, 1] \\ \vdots \\ T[I, \widehat{J}, c] - X_I R[:, \widehat{J}, c] \end{bmatrix} \in \mathbb{F}^{(ca')\times(b-b')}, \qquad M_R := \begin{bmatrix} S[I, :, 1] \\ \vdots \\ S[I, :, c] \end{bmatrix} \in \mathbb{F}^{(ca')\times s} \tag{3.1}$$

Apply LinearEqualsAffine($\mathbb{F}$) to

$$\begin{aligned} &U = \mathcal{Y}_{\widehat{J}}, \quad W = \mathbb{F}^{(ca')\times(b-b')}, \\ &L_R : U \to W, \quad \widehat{Y} \mapsto M_R\widehat{Y}, \quad \Theta = \mathcal{S}_{\mathrm{DR}}, \quad N = N_R. \end{aligned}$$

This returns either $\varnothing$, in which case our algorithm stops, or an affine subspace

$$\mathcal{A}_R \subseteq \mathcal{S}_{\mathrm{DR}} \times \mathcal{Y}_{\widehat{J}}$$

of pairs $((X_I, Y_J), \widehat{Y})$ satisfying $M_R\widehat{Y} = N_R(X_I, Y_J)$.

Step 3. For each $((X_I, Y_J), \widehat{Y}) \in \mathcal{A}_R$, form the unique $Y \in \mathcal{Y}$ such that $Y[:, J] = Y_J$ and $Y[:, \widehat{J}] = \widehat{Y}$. Return the affine subspace of $\mathcal{X}_I \times \mathcal{Y}$ obtained by sending each $((X_I, Y_J), \widehat{Y}) \in \mathcal{A}_R$ to (X_I, Y).

Timing. Step 1 is one call to DoubleRestricted, costing $\mathtt{T}_{\mathrm{DR}}(a, b, c, r, s, a', b')$. Step 2 has two contributions. Recall $\dim(\mathcal{S}_{\mathrm{DR}}) = d$, and we are given an affine frame $(X_{I,i}, Y_{J,i})_{i=0}^{d}$. Building the resulting coordinatized affine map N_R from Equation (3.1) costs

$$O\big(d\,c\,a'r(b - b')\big),$$

because each of the d matrices has size $a' \times r$, each $R[:, \widehat{J}, k]$ has size $r \times (b - b')$, and we compute c multiplications between them. Invoking Proposition 3.16 on (M_R, N_R) adds $\mathtt{T}_{p2}(m = ca',\ n = s,\ \ell = b - b',\ p = d,\ q = q_R,\ \kappa = \kappa_R)$. So the total cost is

$$O\big(\mathtt{T}_{\mathrm{DR}} + d\,c\,a'r(b - b') + \mathtt{T}_{p2}\big).$$

Correctness. Step 1 gives exactly $\mathcal{S}_{\mathrm{DR}}$, i.e. all (X_I, Y_J) that satisfy the (I, J)-restricted equations. Step 2 gives exactly the elements $\widehat{Y}$ for which the combined Y and $\widehat{Y}$ satisfies the row restricted equation. Step 3 combines $(Y_J, \widehat{Y})$ into Y, so $\mathcal{S}_{\mathrm{R}}$ is exactly the set of row-restricted solutions. □

Example 3.25. Continue the running example with $I = J = \{1\}$, so $\widehat{J} = \{2\}$. From Example 3.23, write

$$X_I = \begin{bmatrix} \alpha & \beta \end{bmatrix}, \qquad Y_J = \begin{bmatrix} 1 - \alpha \\ \gamma \end{bmatrix}.$$

Step 2 of Proposition 3.24 uses

$$M_R = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad N_R(X_I, Y_J) = \begin{bmatrix} 0 \\ -\beta \end{bmatrix}.$$

Solving $M_R\widehat{Y} = N_R(X_I, Y_J)$ forces $\widehat{Y} = \begin{bmatrix} 0 \\ \delta \end{bmatrix}$ and $\beta = 0$. Therefore

$$\mathcal{S}_{\mathrm{R}} = \left\{ \left(\begin{bmatrix} \alpha & 0 \end{bmatrix}, \begin{bmatrix} 1-\alpha & 0 \\ \gamma & \delta \end{bmatrix} \right) : \alpha, \gamma, \delta \in \mathbb{F} \right\}.$$

Problem ColumnRestricted.

Given: Tensors $R \in \mathbb{F}^{r\times b\times c}$, $S \in \mathbb{F}^{a\times s\times c}$, $T \in \mathbb{F}^{a\times b\times c}$, and index sets $I \subseteq [a]$, $J \subseteq [b]$ with $|I| = a'$ and $|J| = b'$.

Return: Return $\varnothing$, or the affine subspace

$$\mathcal{S}_{\mathrm{C}} := \{(X, Y_J) \in \mathcal{X} \times \mathcal{Y}_J : (\forall k \in [c])\ XR[:, J, k] + S[:, :, k]Y_J = T[:, J, k]\}.$$

Proposition 3.26. For fixed $I \subseteq [a]$ and $J \subseteq [b]$, ColumnRestricted is solved by solving DoubleRestricted and then LinearEqualsAffine($\mathbb{F}$), in cost

$$\begin{aligned} O\big(&\mathtt{T}_{p1}(m = a'b'c,\ n = a'r + b's,\ \ell = 1) \\ &+ d\,c\,(a - a')sb' \\ &+ \mathtt{T}_{p2}(m = cb',\ n = r,\ \ell = a - a',\ p = d,\ q = q_C,\ \kappa = \kappa_C)\big). \end{aligned}$$

where $a' = |I|$, $b' = |J|$, d is the dimension of the solution set of DoubleRestricted, q_C is the feasible parameter dimension, and κ_C is the left nullspace dimension parameter in the LinearEqualsAffine($\mathbb{F}$) instance (these three parameters are all 0 if $\varnothing$ is returned)

Proof. **Algorithm. Step 1.** Solve DoubleRestricted. If it returns $\varnothing$, return $\varnothing$ and stop.

Step 2. Set $\widehat{I} := [a] \setminus I$ and define an affine map $N_C : \mathcal{S}_{\mathrm{DR}} \to \mathbb{F}^{(cb')\times(a-a')}$ as well as an element $M_C \in \mathbb{F}^{(cb')\times r}$ by

$$N_C(X_I, Y_J) := \begin{bmatrix} (T[\widehat{I}, J, 1] - S[\widehat{I}, :, 1]Y_J)^T \\ \vdots \\ (T[\widehat{I}, J, c] - S[\widehat{I}, :, c]Y_J)^T \end{bmatrix} \in \mathbb{F}^{(cb')\times(a-a')}, \qquad M_C := \begin{bmatrix} (R[:, J, 1])^T \\ \vdots \\ (R[:, J, c])^T \end{bmatrix} \in \mathbb{F}^{(cb')\times r}.$$

Apply LinearEqualsAffine($\mathbb{F}$) to

$$\begin{aligned} &U = \mathbb{F}^{r\times(a-a')}, \quad W = \mathbb{F}^{(cb')\times(a-a')}, \\ &L_C : U \to W, \quad \widehat{X} \mapsto M_C\widehat{X}, \quad \Theta = \mathcal{S}_{\mathrm{DR}}, \quad N = N_C. \end{aligned}$$

This returns either $\varnothing$, or an affine subspace

$$\mathcal{A}_C \subseteq \mathcal{S}_{\mathrm{DR}} \times \mathbb{F}^{r\times(a-a')}$$

of pairs $((X_I, Y_J), \widehat{X})$ satisfying $M_C\widehat{X} = N_C(X_I, Y_J)$. If it returns $\varnothing$, return $\varnothing$ for ColumnRestricted and stop.

Step 3. For each $((X_I, Y_J), \widehat{X}) \in \mathcal{A}_C$, form the unique $X \in \mathcal{X}$ such that $X[I, :] = X_I$ and $X[\widehat{I}, :] = \widehat{X}^T$. Return $\mathcal{S}_{\mathrm{C}}$, the affine subspace of $\mathcal{X} \times \mathcal{Y}_J$ obtained by sending each $((X_I, Y_J), \widehat{X}) \in \mathcal{A}_C$ to (X, Y_J).

Timing. The same cost accounting as Proposition 3.24. Step 1 costs $\mathtt{T}_{\mathrm{DR}}(a, b, c, r, s, a', b')$, Step 2 has cost in $O\big(d\,c\,(a - a')sb' + \mathtt{T}_{p2}(m = cb',\ n = r,\ \ell = a - a',\ p = d,\ q = q_C,\ \kappa = \kappa_C)\big)$. So the total cost is

$$O\big(\mathtt{T}_{\mathrm{DR}} + d\,c\,(a - a')sb' + \mathtt{T}_{p2}\big).$$

Correctness. Same as Proposition 3.24. □

Example 3.27. Continue the running example with $I = J = \{1\}$, so $\widehat{I} = \{2\}$. From Example 3.23, write

$$X_I = \begin{bmatrix} \alpha & \beta \end{bmatrix}, \qquad Y_J = \begin{bmatrix} 1-\alpha \\ \gamma \end{bmatrix}.$$

Step 2 of Proposition 3.26 gives

$$M_C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad M_C \widehat{X} = N_C(X_I, Y_J) = \begin{bmatrix} 0 \\ -\gamma \end{bmatrix},$$

with $\widehat{X}$ as the unknown. Solving gives $\widehat{X} = \begin{bmatrix} 0 \\ \epsilon \end{bmatrix}$ and $\gamma = 0$. Therefore

$$\mathcal{S}_{\mathrm{C}} = \left\{ \left(\begin{bmatrix} \alpha & \beta \\ 0 & \epsilon \end{bmatrix}, \begin{bmatrix} 1-\alpha \\ 0 \end{bmatrix} \right) : \alpha, \beta, \epsilon \in \mathbb{F} \right\}.$$

Problem LiftRestricted.

Given: Solutions to RowRestricted and ColumnRestricted.
Return: Solution to SimultaneousSylvesterSystem.

Solving RowRestricted and ColumnRestricted gives $\mathcal{S}_{\mathrm{R}} \subseteq \mathcal{X}_I \times \mathcal{Y}$ and $\mathcal{S}_{\mathrm{C}} \subseteq \mathcal{X} \times \mathcal{Y}_J$. For $z_R = (X_I, Y) \in \mathcal{S}_{\mathrm{R}}$ and $z_C = (X, Y_J) \in \mathcal{S}_{\mathrm{C}}$, define

$$\pi_J : \mathcal{X}_I \times \mathcal{Y} \to \mathcal{X}_I \times \mathcal{Y}_J, \qquad \pi_J(z_R) := (X_I, Y[:, J]),$$
$$\pi_I : \mathcal{X} \times \mathcal{Y}_J \to \mathcal{X}_I \times \mathcal{Y}_J, \qquad \pi_I(z_C) := (X[I, :], Y_J).$$

Proposition 3.28. Let $\mathcal{S}_{\mathrm{R}} \subseteq \mathcal{X}_I \times \mathcal{Y}$, and $\mathcal{S}_{\mathrm{C}} \subseteq \mathcal{X} \times \mathcal{Y}_J$ be the solutions of RowRestricted and ColumnRestricted. Let the pullback of the projection maps be defined as

$$P := \{(z_R, z_C) \in \mathcal{S}_{\mathrm{R}} \times \mathcal{S}_{\mathrm{C}} : \pi_J(z_R) = \pi_I(z_C)\},$$

the output map be defined as

$$\Psi : P \to \mathcal{X} \times \mathcal{Y}, \qquad \Psi\big((X_I, Y), (X, Y_J)\big) := (X, Y),$$

and the error map be defined as

$$\mathrm{Err} : \mathcal{X} \times \mathcal{Y} \to \mathbb{F}^{a \times b \times c}, \qquad \mathrm{Err}(X, Y) := \big(XR[:, :, k] + S[:, :, k]Y - T[:, :, k]\big)_{k \in [c]}.$$

Then $\{\Psi(z) : z \in P \text{ and } \mathrm{Err}(\Psi(z)) = 0\}$ is the solution set to LiftRestricted.

Proof. The restricted pairs agreeing from $\mathcal{S}_{\mathrm{R}}$ and $\mathcal{S}_{\mathrm{C}}$ are exactly P. For each $z \in P$, $\Psi(z)$ is a candidate solution. This candidate is in the solution set of SimultaneousSylvesterSystem exactly when its residual is zero. □

Proposition 3.29. Let $\mathcal{S}_{\mathrm{R}}, \mathcal{S}_{\mathrm{C}}$ be solutions to RowRestricted and ColumnRestricted, and let

$$d := \max\{\dim \mathcal{S}_{\mathrm{R}}, \dim \mathcal{S}_{\mathrm{C}}\}, \qquad P := \{(z_R, z_C) \in \mathcal{S}_{\mathrm{R}} \times \mathcal{S}_{\mathrm{C}} : \pi_J(z_R) = \pi_I(z_C)\},$$
$$\tau := \dim P \text{ if } P \neq \varnothing, \text{ and } 0 \text{ otherwise.}$$

Then LiftRestricted is solvable in cost

$$O\big((a'r + b's)d^2 + (\tau + 1)\, c\, a\, b\, (r + s) + \mathtt{T}_{p1}(m = abc,\ n = \tau,\ \ell = 1)\big).$$

Proof. **Algorithm.**

If either RowRestricted or ColumnRestricted returns $\varnothing$, then we return $\varnothing$ and stop.

Step 1 (compute the candidates). From the outputs of RowRestricted and ColumnRestricted, we get the affine frames

$$z_{R,0}, \ldots, z_{R,d_R} \in \mathcal{S}_{\mathrm{R}}, \qquad z_{C,0}, \ldots, z_{C,d_C} \in \mathcal{S}_{\mathrm{C}},$$

and thereby a coordinate realization of the pullback P, which is solved for via an instance of AffinePullback($\mathbb{F}$) with

$$U = \mathbb{F}^{d_R}, \quad V = \mathbb{F}^{d_C}, \quad W = \mathcal{X}_I \times \mathcal{Y}_J,$$
$$f : U \to W,\ \lambda \mapsto \pi_J(\chi_R(\lambda)), \quad g : V \to W,\ \mu \mapsto \pi_I(\chi_C(\mu)),$$

where χ_R, χ_C are the coordinate maps to $\mathcal{S}_{\mathrm{R}}$ and $\mathcal{S}_{\mathrm{C}}$.

If $P = \varnothing$, return $\varnothing$. Otherwise output the frame $(p_0, \ldots, p_\tau) \subseteq \mathcal{S}_{\mathrm{R}} \times \mathcal{S}_{\mathrm{C}}$ from the coordinate pullback solutions.

Step 2 (compute the error map on a frame). For each $t \in \{0, \dots, \tau\}$, compute

$$(X_t, Y_t) := \Psi(p_t), \qquad E_t := \mathrm{Err}(X_t, Y_t) \in \mathbb{F}^{a \times b \times c},$$

and vectorize $e_t := \mathrm{vec}(E_t) \in \mathbb{F}^{abc}$.

If $\tau = 0$, this means at most one solution exists. We check if $e_0 = 0$. In the case $e_0 \neq 0$, return $\varnothing$. When $e_0 = 0$, the solution is the unique point $\Psi(p_0)$.

Else, $\tau > 0$, so we solve the inhomogeneous linear system

$$Mx = -e_0, \qquad M := \begin{bmatrix} e_1 - e_0 & \cdots & e_\tau - e_0 \end{bmatrix} \in \mathbb{F}^{abc \times \tau}$$

in the frame coordinates via LinSolve($\mathbb{F}$). If it returns $\varnothing$, return $\varnothing$. Otherwise output the frame of solutions in $\mathcal{X} \times \mathcal{Y}$ by applying Ψ to the coordinate solutions of the linear system.

Timing. Step 1 is one instance of AffinePullback($\mathbb{F}$), with $d = d_R$, $e = d_C$, and $m = a'r + sb'$. Under the assumptions that d_R, d_C are $O(d)$, Proposition 3.19 takes $O(d^2(a'r + sb'))$ operations. Evaluating Err at one point costs $O(c\,a\,b\,(r+s))$, so all $\tau+1$ evaluations cost $O((\tau+1)\,c\,a\,b\,(r+s))$. Solving $Mx = -e_0$ is one call to LinSolve($\mathbb{F}$) with $m = abc$, $n = \tau$, and $\ell = 1$, so it costs $\mathtt{T}_{p1}(m = abc,\ n = \tau,\ \ell = 1)$. □

Example 3.30. From Examples 3.25 and 3.27, a compatible pair in the pullback has the form

$$X = \begin{bmatrix} \alpha & 0 \\ 0 & \epsilon \end{bmatrix}, \qquad Y = \begin{bmatrix} 1-\alpha & 0 \\ 0 & \delta \end{bmatrix}.$$

for $\alpha, \epsilon, \delta \in \mathbb{F}$. Step 2 checks

$$\mathrm{Err}(X, Y) = \big(XR[:,:,k] + S[:,:,k]Y - T[:,:,k]\big)_{k \in [2]},$$

and the remaining condition is $\epsilon + \delta = 2$. Therefore LiftRestricted returns the same as the naive solution after renaming ϵ to β, namely $X = \mathrm{diag}(\alpha, \beta)$, and $Y = \mathrm{diag}(1-\alpha, 2-\beta)$ for $\alpha, \beta \in \mathbb{F}$.

Now we are ready to solve SimultaneousSylvesterSystem.

Theorem 3.31. Let $I \subseteq [a]$ and $J \subseteq [b]$. Then SimultaneousSylvesterSystem is solvable in cost

$$\mathtt{T}_{\mathrm{DR}} + \mathtt{T}_{\mathrm{RR}} + \mathtt{T}_{\mathrm{CR}} + \mathtt{T}_{\mathrm{lift}},$$

where $\mathtt{T}_{\mathrm{DR}}$ is the cost for DoubleRestricted, $\mathtt{T}_{\mathrm{RR}}, \mathtt{T}_{\mathrm{CR}}$ are the costs from RowRestricted and ColumnRestricted, not including the double restricted solve, and $\mathtt{T}_{\mathrm{lift}}$ the cost for LiftRestricted.

Proof. By correctness of DoubleRestricted, we obtain the doubly-restricted solution family for (I, J). By Propositions 3.24 and 3.26, the row- and column-restricted families computed from that data are exactly the solutions of RowRestricted and ColumnRestricted. Then LiftRestricted is applied to these two families. Its correctness means we solve SimultaneousSylvesterSystem. □

An end-to-end example can be found in Example B.1.

Finally we describe a class of problems on which this solve-and-lift algorithm performs best. **TODO: "regularity condition" for quicksylver needs more precision.**

Definition 3.32. The *regularity* condition that is the hypothesis of Theorem 3.21 is when

$$a, b, c, r, s \in O(n),$$

$I \subseteq [a], J \subseteq [b]$ satisfies

$$a' = |I| \in O(1), \qquad b' = |J| \in O(1),$$

and $\dim \mathcal{S}_{\mathrm{DR}},\ q_R,\ q_C,\ \tau \in O(1)$. Furthermore, the row/column restricted systems have the property that

$$N_R(\mathcal{S}_{\mathrm{DR}}) \subseteq \mathrm{im}(M_R), \qquad N_C(\mathcal{S}_{\mathrm{DR}}) \subseteq \mathrm{im}(M_C), \qquad \mathrm{Ker}(M_R) = \mathrm{Ker}(M_C) = \{0\}.$$

The above assumptions are plausible in the cubic case ($n \times n \times n$ tensors, with $r = s = n$ as well). When $a', b' \in O(1)$, the matrices M_R and M_C have dimensions

$$M_R \in \mathbb{F}^{(a'c)\times s}, \qquad M_C \in \mathbb{F}^{(b'c)\times r}$$

respectively. In the cubic regime, both the row counts and column counts of these systems are in $O(n)$, and for generic dense slice data one expects full column rank, so $\mathrm{Ker}(M_R) = \mathrm{Ker}(M_C) = \{0\}$ is likely.

The inclusions $N_R(\mathcal{S}_{\mathrm{DR}}) \subseteq \mathrm{im}(M_R)$ and $N_C(\mathcal{S}_{\mathrm{DR}}) \subseteq \mathrm{im}(M_C)$ assert that all double restricted solutions can be lifted. Whether this condition is satisfied is particular to the specific instance of the question.

Proof of Theorem 3.21. Assume the conditions in Definition 3.32. For the doubly-restricted step,

$$\mathtt{T}_{\mathrm{DR}} \in O\big((a'b'c)(a'r + b's)^2\big),$$

which is $O(n^3)$ since $a', b' \in O(1)$ and $c, r, s \in O(n)$.

For the row-restricted step, Proposition 3.24 gives

$$\mathtt{T}_{\mathrm{RR}} \in O\big(\mathtt{T}_{\mathrm{DR}} + d_{\mathrm{DR}}\, c\, a'r\, (b - b') + \mathtt{T}_{p2}(m = ca',\, n = s,\, \ell = (b - b'),\, p = d_{\mathrm{DR}},\, q = q_R,\, \kappa = \kappa_R)\big).$$

Here $ca', s, (b - b') \in O(n)$ and $d_{\mathrm{DR}}, q_R \in O(1)$. Also $\kappa_R \le ca'$, so in this regime $\kappa_R \in O(n)$, and the $\mathtt{T}_{p2}$ term is still $O(n^3)$. Therefore $\mathtt{T}_{\mathrm{RR}} \in O(n^3)$. The same argument gives $\mathtt{T}_{\mathrm{CR}} \in O(n^3)$.

For the lift step, Proposition 3.29 gives

$$\mathtt{T}_{\mathrm{lift}} \in O\big((a'r + b's)d^2 + (\tau + 1)\, c\, a\, b\, (r + s) + \mathtt{T}_{p1}(m = abc,\, n = \tau,\, \ell = 1)\big),$$

with $d = \max\{\dim \mathcal{S}_{\mathrm{R}}, \dim \mathcal{S}_{\mathrm{C}}\}$. Under Definition 3.32, we have $\dim \mathcal{S}_{\mathrm{R}} = \dim \mathcal{S}_{\mathrm{C}} = \dim \mathcal{S}_{\mathrm{DR}} \in O(1)$, so the first term is $O(n)$, the last is $O(n^3)$, and

$$(\tau + 1)\, c\, a\, b\, (r + s) \in O(n^4).$$

Hence $\mathtt{T}_{\mathrm{lift}} \in O(n^4)$, so the total solve-and-lift cost is $O(n^4)$.

Finally, full deterministic verification checks all abc constraints and has cost $O(cab(r + s)) = O(n^4)$. Therefore SimultaneousSylvesterSystem is solved in the same order of arithmetic complexity as full deterministic verification. □

3.4. **Derivation Equations for** $\mathrm{Der}(T)$**.** In this section, we work towards Theorem 3.33. Throughout, a, b, c, r, s, t will be natural numbers, and we denote the spaces

$$\mathcal{X} := \mathbb{F}^{a\times r}, \qquad \mathcal{Y} := \mathbb{F}^{s\times b}, \qquad \mathcal{Z} := \mathbb{F}^{t\times c}.$$

This notation may have subscripts to indicate restricted solution spaces. As an example, for $I \subseteq [a]$, $\mathcal{X}_I$ is the space $\mathbb{F}^{|I|\times r}$. For a matrix $Z \in \mathbb{F}^{t\times c}$, define T^Z, the outer action of Z on T, by specifying its slices for $k \in [c]$ as

$$(T^Z)[:, :, k] := \sum_{j\in[t]} T[:, :, j]\, Z[j, k]. \tag{3.2}$$

In addition, we shall need outer action on submatrices $Z_K \in \mathbb{F}^{ttimesc'}$. The above definition adapts to handle that case as well.

Problem DerivationSystem.

Given: Tensors $R \in \mathbb{F}^{r\times b\times c}$, $S \in \mathbb{F}^{a\times s\times c}$, $T \in \mathbb{F}^{a\times b\times t}$.
Return: Either return $\varnothing$, or the affine subspace

$$\big\{(X, Y, Z) \in \mathcal{X} \times \mathcal{Y} \times \mathcal{Z} : (\forall k \in [c])\ X\, R[:, :, k] + S[:, :, k]\, Y = (T^Z)[:, :, k]\big\}.$$

This problem has a naive solution using LinSolve($\mathbb{F}$) with $m = abc$, $n = ar + bs + ct$, and $\ell = 1$. When $R = S = T$, solutions to the equation correspond to the derivation algebra of a tensor T, as seen in Equation (1.1). We describe an alternative approach, culminating in the following.

Theorem 3.33. For cubic ($n \times n \times n$) tensors, under the regularity conditions (established in Definition 3.38), DerivationSystem is solved in cost $O(n^{4.5})$. This is an additional $n^{1/2}$ overhead compared to the deterministic verification cost of $O(n^4)$, as compared to the naive $O(n^7)$ cost by solving as a linear system of equations.

As in the above section, we solve the derivation system by solving on a subset of constraints, and then lifting those solutions to the full system.

Problem TripleRestrictedDer.

Given: Tensors $R \in \mathbb{F}^{r\times b\times c}$, $S \in \mathbb{F}^{a\times s\times c}$, $T \in \mathbb{F}^{a\times b\times t}$ and index sets $I \subseteq [a]$, $J \subseteq [b]$, and $K = \{k_1 < \cdots < k_{c'}\} \subseteq [c]$, with $|I| = a'$, $|J| = b'$, and $|K| = c'$.

Return: Either return $\varnothing$, or the affine subspace

$$\mathcal{D}_{I,J,K} := \left\{(X_I, Y_J, Z_K) \in \mathcal{X}_I \times \mathcal{Y}_J \times \mathcal{Z}_K : (\forall j \in [c'])\ X_I\, R[:, J, k_j] + S[I, :, k_j]\, Y_J = (T^{Z_K})[I, J, j]\right\}.$$

This is a dense linear system with $ra' + sb' + tc'$ unknowns and $a'b'c'$ equations. Hence it is solvable in cost

$$\mathtt{T}_{p1}(m = a'b'c',\ n = ra' + sb' + tc',\ \ell = 1). \tag{3.3}$$

We now solve three restricted systems which give solution families for X, Y, and Z, respectively.

Problem ColDepthRestrictedDer.

Given: Tensors $R \in \mathbb{F}^{r\times b\times c}$, $S \in \mathbb{F}^{a\times s\times c}$, $T \in \mathbb{F}^{a\times b\times t}$, and index sets $I \subseteq [a]$, $J \subseteq [b]$, and $K = \{k_1 < \cdots < k_{c'}\} \subseteq [c]$ with $|I| = a'$, $|J| = b'$, and $|K| = c'$.

Return: Either return $\varnothing$, or the affine subspace

$$\mathcal{D}_{[a],J,K} := \left\{(X, Y_J, Z_K) \in \mathcal{X} \times \mathcal{Y}_J \times \mathcal{Z}_K : (\forall j \in [c'])\ X\, R[:, J, k_j] + S[:, :, k_j]\, Y_J = (T^{Z_K})[:, J, j]\right\}.$$

Problem RowDepthRestrictedDer.

Given: Tensors $R \in \mathbb{F}^{r\times b\times c}$, $S \in \mathbb{F}^{a\times s\times c}$, $T \in \mathbb{F}^{a\times b\times t}$, and index sets $I \subseteq [a]$, $J \subseteq [b]$, and $K = \{k_1 < \cdots < k_{c'}\} \subseteq [c]$ with $|I| = a'$, $|J| = b'$, and $|K| = c'$.

Return: Either return $\varnothing$, or the affine subspace

$$\mathcal{D}_{I,[b],K} := \left\{(X_I, Y, Z_K) \in \mathcal{X}_I \times \mathcal{Y} \times \mathcal{Z}_K : (\forall j \in [c'])\ X_I\, R[:, :, k_j] + S[I, :, k_j]\, Y = (T^{Z_K})[I, :, j]\right\}.$$

Problem RowColRestrictedDer.

Given: Tensors $R \in \mathbb{F}^{r\times b\times c}$, $S \in \mathbb{F}^{a\times s\times c}$, $T \in \mathbb{F}^{a\times b\times t}$, and index sets $I \subseteq [a]$, $J \subseteq [b]$, $K \subseteq [c]$ with $|I| = a'$, $|J| = b'$, and $|K| = c'$.

Return: Either return $\varnothing$, or the affine subspace

$$\mathcal{D}_{I,J,[c]} := \left\{(X_I, Y_J, Z) \in \mathcal{X}_I \times \mathcal{Y}_J \times \mathcal{Z} : (\forall k \in [c])\ X_I\, R[:, J, k] + S[I, :, k]\, Y_J = (T^{Z})[I, J, k]\right\}.$$

Proposition 3.34. Each of ColDepthRestrictedDer, RowDepthRestrictedDer, and RowColRestrictedDer are solved via an instance of TripleRestrictedDer followed by an instance of LinearEqualsAffine($\mathbb{F}$) instance.

Proof. All three restricted problems are solved via computing $\mathcal{D}_{I,J,K}$ using TripleRestrictedDer (this is only done once and the solution is shared), and then LinearEqualsAffine($\mathbb{F}$) to find the missing block.

We detail RowDepthRestrictedDer. Let $\widehat{J} := [b] \setminus J$, and write $K = \{k_1 < \cdots < k_{c'}\}$. Define the affine map $N_R : \mathcal{D}_{I,J,K} \to \mathbb{F}^{(a'c')\times(b-b')}$ and $M_R \in \mathbb{F}^{a'c'\times s}$ by

$$N_R(X_I, Y_J, Z_K) := \begin{bmatrix} (T^{Z_K})[I, \widehat{J}, 1] - X_I\, R[:, \widehat{J}, k_1] \\ \vdots \\ (T^{Z_K})[I, \widehat{J}, c'] - X_I\, R[:, \widehat{J}, k_{c'}] \end{bmatrix} \in \mathbb{F}^{(a'c')\times(b-b')}, \qquad M_R := \begin{bmatrix} S[I, :, k_1] \\ \vdots \\ S[I, :, k_{c'}] \end{bmatrix} \in \mathbb{F}^{(a'c')\times s}. \tag{3.4}$$

Then, for each $(X_I, Y_J, Z_K) \in \mathcal{D}_{I,J,K}$, solve

$$M_R \widehat{Y} = N_R(X_I, Y_J, Z_K)$$

as an instance of LinearEqualsAffine($\mathbb{F}$).

ColDepthRestrictedDer is solved similarly, while RowColRestrictedDer requires some new notation to set up the system. Write $\widehat{K} = \{\ell_1 < \cdots < \ell_{c-c'}\} = [c] \setminus K$. Define a matrix $M_D \in \mathbb{F}^{a'b' \times t}$ by specifying its jth column $M_D[:, j] (j \in [t])$ as $\mathrm{vec}(T[I, J, j]) \in \mathbb{F}^{a'b'}$.

And define the affine map $N_D : \mathcal{D}_{I,J,K} \to \mathbb{F}^{a'b' \times (c-c')}$ by specifying the jth column of its output:

$$N_D(X_I, Y_J, Z_K)[:, j] := \mathrm{vec}\big(X_I\, R[:, J, \ell_j] + S[I, :, \ell_j]\, Y_J\big) \qquad (j \in [c - c'])$$

Now solving for $\widehat{Z} \in \mathbb{F}^{t \times (c-c')}$ in $M_D \widehat{Z} = N_D(X_I, Y_J, Z_K)$ as an instance of LinearEqualsAffine($\mathbb{F}$) concludes the solution. □

Proposition 3.35. Let $d := \dim \mathcal{D}_{I,J,K}$ and let (q_R, κ_R), (q_C, κ_C), (q_Z, κ_Z) be the feasible parameter and left nullspace dimensions respectively of the three LinearEqualsAffine($\mathbb{F}$) instances solving Proposition 3.34. Then

RowDepthRestrictedDer is solvable in cost

$$O\big(\mathtt{T}_{p1}(m = a'b'c',\ n = a'r + sb' + tc',\ \ell = 1) + d\, c'\, a'(b - b')\,(r + t) \\ + \mathtt{T}_{p2}(m = a'c',\ n = s,\ \ell = b - b',\ p = d,\ q = q_R,\ \kappa = \kappa_R)\big),$$

ColDepthRestrictedDer is solvable in cost

$$O\big(\mathtt{T}_{p1}(m = a'b'c',\ n = a'r + sb' + tc',\ \ell = 1) + d\, c'\,(a - a')\, b'\,(s + t) \\ + \mathtt{T}_{p2}(m = b'c',\ n = r,\ \ell = a - a',\ p = d,\ q = q_C,\ \kappa = \kappa_C)\big),$$

RowColRestrictedDer is solvable in cost

$$O\big(\mathtt{T}_{p1}(m = a'b'c',\ n = a'r + sb' + tc',\ \ell = 1) + d\,(c - c')\, a'b'\,(r + s) \\ + \mathtt{T}_{p2}(m = a'b',\ n = t,\ \ell = c - c',\ p = d,\ q = q_Z,\ \kappa = \kappa_Z)\big).$$

Proof. Each cost is read directly from the data needed to build the corresponding LinearEqualsAffine($\mathbb{F}$) instance. For RowDepthRestrictedDer, use Equation (3.4): for each $j \in [c']$, $X_I \in \mathbb{F}^{a' \times r}$ multiplies $R[:, \widehat{J}, k_j] \in \mathbb{F}^{r \times (b-b')}$, and $T^{Z_K}[I, \widehat{J}, j]$ is an $a' \times (b - b')$ block formed from $Z_K \in \mathbb{F}^{t \times c'}$ and the t slices of T. Hence each block costs $a'(b - b')(r + t)$, with c' blocks per frame point and d frame points, giving $d\, c'\, a'(b-b')(r+t)$. The ColDepthRestrictedDer and RowColRestrictedDer bounds are obtained in the same way from their stacked block dimensions. □

Problem LiftRestrictedDer.

Given: Solutions to ColDepthRestrictedDer, RowDepthRestrictedDer, and RowColRestrictedDer.

Return: The solution to DerivationSystem.

The overall strategy is the same as the solution of Problem 8. Each lifted solution space has a restriction map to $\mathcal{D}_{I,J,K}$.

$$\begin{aligned} &\rho_X : \mathcal{D}_{[a],J,K} \to \mathcal{D}_{I,J,K}, && (X, Y_J, Z_K) \mapsto (X[I, :], Y_J, Z_K) \\ &\rho_Y : \mathcal{D}_{I,[b],K} \to \mathcal{D}_{I,J,K}, && (X_I, Y, Z_K) \mapsto (X_I, Y[:, J], Z_K), \\ &\rho_Z : \mathcal{D}_{I,J,[c]} \to \mathcal{D}_{I,J,K}, && (X_I, Y_J, Z) \mapsto (X_I, Y_J, Z[:, K]). \end{aligned}$$

We first form the 2-way pullback as an instance of Problem 3. Solving it gives compatible partial solution triples (X, Y, Z_K).

$$Q_{XY} := \{(u, v) \in \mathcal{D}_{[a],J,K} \times \mathcal{D}_{I,[b],K} : \rho_X(u) = \rho_Y(v)\},$$

and then as a second pullback calculation, solve

$$Q := \{((u, v), w) \in Q_{XY} \times \mathcal{D}_{I,J,[c]} : \rho_X(u) = \rho_Y(v) = \rho_Z(w)\}.$$

Each point of Q determines a full candidate triple $(X, Y, Z) \in \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$.

What remains is to enforce zero residual on Q. Define

$$\mathrm{Err} : \mathcal{X} \times \mathcal{Y} \times \mathcal{Z} \to \mathbb{F}^{a \times b \times c}, \qquad \mathrm{Err}(X, Y, Z)[:, :, k] := X\, R[:, :, k] + S[:, :, k]\, Y - (T^Z)[:, :, k],$$

and solve for the candidate solutions that have zero residual error in a linear system.

An end-to-end example can be found in Example B.2.

Proposition 3.36. Let $\mathcal{D}_{I,J,K}$, $\mathcal{D}_{[a],J,K}$, $\mathcal{D}_{I,[b],K}$, and $\mathcal{D}_{I,J,[c]}$ be the affine spaces returned by TripleRestrictedDer, ColDepthRestrictedDer, and RowDepthRestrictedDer, and RowColRestrictedDer respectively. Let Q_{XY} and Q be the two pullbacks built in LiftRestrictedDer. Write

$$d := \dim \mathcal{D}_{I,J,K}, \quad d_X := \dim \mathcal{D}_{[a],J,K}, \quad d_Y := \dim \mathcal{D}_{I,[b],K}, \quad d_Z := \dim \mathcal{D}_{I,J,[c]},$$

$$d_{XY} := \dim Q_{XY}, \qquad \tau := \dim Q \text{ if } Q \neq \varnothing, \text{ and } 0 \text{ otherwise.}$$

Then LiftRestrictedDer is solvable in cost

$$\mathtt{T}_{p1}(m = d,\ n = d_X + d_Y,\ \ell = 1) + \mathtt{T}_{p1}(m = d,\ n = d_{XY} + d_Z,\ \ell = 1)$$
$$+ O\big((\tau + 1)\, abc(r + s + t)\big) + \mathtt{T}_{p1}(m = abc,\ n = \tau,\ \ell = 1).$$

Proof. The first two summands are the costs of the two applications of AffinePullback($\mathbb{F}$). In both pullbacks, we work in affine-frame coordinates on $\mathcal{D}_{I,J,K}$, so the common codomain has coordinate dimension d. After the second pullback, we evaluate the Err map on an affine frame

$$q_0, \ldots, q_\tau \in Q.$$

Each evaluation costs $O(abc(r + s + t))$, since for each $k \in [c]$ we must compute

$$X\, R[:, :, k], \qquad S[:, :, k]\, Y, \qquad (T^Z)[:, :, k].$$

Finally, solving for the coordinates of Q that satisfies $\mathrm{Err}(q) = 0$ requires an application of LinSolve($\mathbb{F}$) with

$$m = abc, \qquad n = \tau, \qquad \ell = 1.$$ □

Theorem 3.37. Fix $I \subseteq [a]$, $J \subseteq [b]$, and $K \subseteq [c]$. Then DerivationSystem is solvable in cost

$$\mathtt{T}_{\mathrm{core}} + \mathtt{T}_{\mathrm{col}} + \mathtt{T}_{\mathrm{row}} + \mathtt{T}_{\mathrm{slice}} + \mathtt{T}_{\mathrm{lift}},$$

where $\mathtt{T}_{\mathrm{core}}$ is the TripleRestrictedDer cost from Equation (3.3), $\mathtt{T}_{\mathrm{col}}$, $\mathtt{T}_{\mathrm{row}}$, and $\mathtt{T}_{\mathrm{slice}}$ are the three restricted lift costs from Proposition 3.35, not counting solving the triple restricted system, and $\mathtt{T}_{\mathrm{lift}}$ is the cost of computing compatible solutions from Proposition 3.36.

Proof. Same reasoning as Theorem 3.31. □

Finally we describe a class of problems on which this solve-and-lift algorithm performs best. **TODO: "regularity condition" for derivations needs more precision.**

Definition 3.38. The *regularity* condition that is the hypothesis of Theorem 3.33 is when

$$a, b, c, r, s, t \in O(n),$$

and for $I \subseteq [a], J \subseteq [b], K \subseteq [c]$,

$$a' = |I|,\ b' = |J|,\ c' = |K| \in O(n^{1/2}).$$

Furthermore,

$$\dim \mathcal{D}_{I,J,K},\ q_R,\ q_C,\ q_Z,\ \tau \in O(1),$$

and the three lift systems satisfy

$$N_R(\mathcal{D}_{I,J,K}) \subseteq \mathrm{im}(M_R), \qquad N_C(\mathcal{D}_{I,J,K}) \subseteq \mathrm{im}(M_C), \qquad N_D(\mathcal{D}_{I,J,K}) \subseteq \mathrm{im}(M_D),$$
$$\mathrm{Ker}(M_R) = \mathrm{Ker}(M_C) = \mathrm{Ker}(M_D) = \{0\}.$$

The above assumptions are plausible in the cubic case ($n \times n \times n$ tensors). The matrices used for lifting solutions have dimensions

$$M_R \in \mathbb{F}^{(a'c') \times s}, \qquad M_C \in \mathbb{F}^{(b'c') \times r}, \qquad M_D \in \mathbb{F}^{(a'b') \times t},$$

so in the cubic regime when $a', b', c' \in O(n^{1/2})$ they are all of order $n \times n$. For generic dense slice data this makes injectivity plausible, so $\mathrm{Ker}(M_R) = \mathrm{Ker}(M_C) = \mathrm{Ker}(M_D) = \{0\}$ is a reasonable condition.

The inclusions $N_R(\mathcal{D}_{I,J,K}) \subseteq \mathrm{im}(M_R)$, $N_C(\mathcal{D}_{I,J,K}) \subseteq \mathrm{im}(M_C)$, $N_D(\mathcal{D}_{I,J,K}) \subseteq \mathrm{im}(M_D)$ are lift-feasibility conditions. Whether they are satisfied is particular to the problem instance.

Proof of Theorem 3.33. Equation (3.3) gives

$$\mathtt{T}_{p1}(m = a'b'c',\ n = ra' + sb' + tc',\ \ell = 1).$$

Assuming Definition 3.38, we have $a', b', c' \in O(n^{1/2})$ and $r, s, t \in O(n)$, so the two matrix dimensions in Equation (3.3) are both in $O(n^{3/2})$. Therefore this term is $O(n^{4.5})$.

For the three restricted lift stages, Proposition 3.35 gives $O(n^4)$ under the same assumptions: here $d \in O(1)$, $q_R, q_C, q_Z \in O(1)$, and $\kappa_R \leq a'c'$, $\kappa_C \leq b'c'$, $\kappa_Z \leq a'b'$. For the final lifting stage, Proposition 3.36 gives

$$\mathtt{T}_{p1}(m = d,\ n = d_X + d_Y,\ \ell = 1) + \mathtt{T}_{p1}(m = d,\ n = d_{XY} + d_Z,\ \ell = 1)$$

$$+O\big((\tau + 1)abc(r + s + t)\big) + \mathtt{T}_{p1}(m = abc,\ n = \tau,\ \ell = 1).$$

Using $d = d_X = d_Y = d_Z = d_{XY} = \tau = \dim \mathcal{D}_{I,J,K} \in O(1)$, the two pullback solves are $O(1)$, the last term is $O(abc) = O(n^3)$, and the dominant term is $O(abc(r + s + t)) = O(n^4)$. Hence the total cost is $O(n^{4.5})$.

□

4. Nuclei of Product of Bimaps

In this section, we work towards the below theorem. As a reminder, for a bimap $b : U \times V \rightarrowtail W$, its left nucleus is $\mathcal{L}(b) = \{(\alpha, \gamma) \in \mathrm{End}(U) \times \mathrm{End}(W) : b \circ_2 \alpha = b \circ_0 \gamma\}$. The middle and right nuclei, $\mathcal{M}(b), \mathcal{R}(b)$, are defined in the same way by matching the corresponding axis actions.

Theorem 4.1. Let $s : U_2 \times U_1 \rightarrowtail U_0$ and $t : V_2 \times V_1 \rightarrowtail V_0$ be fully nondegenerate bimaps. Then

$$\mathcal{L}(s \otimes t) = \mathcal{L}(s) \boxtimes \mathcal{L}(t), \quad \mathcal{M}(s \otimes t) = \mathcal{M}(s) \boxtimes \mathcal{M}(t), \quad \mathcal{R}(s \otimes t) = \mathcal{R}(s) \boxtimes \mathcal{R}(t).$$

As a result of this theorem, we also have the following.

Corollary 4.2. Let $s : U_2 \times U_1 \rightarrowtail U_0$ and $t : V_2 \times V_1 \rightarrowtail V_0$ be fully nondegenerate bimaps. Then $\mathcal{C}(s \otimes t) = \mathcal{C}(s) \boxtimes \mathcal{C}(t)$.

4.1. Previous results.

Proposition 4.3 ([Wil09, Propositions 7.8 and 7.9])**.** Let $d : U \times U \rightarrowtail C$ be a nondegenerate Hermitian C-form and $b : V \times V \rightarrowtail W$ a k-bilinear map. Then $d \otimes b$ is Hermitian and

$$\mathrm{Adj}(d \otimes b) = \mathrm{Adj}(d) \otimes \mathrm{Adj}(b).$$

In our notation, $\mathrm{Adj}(b) = \mathcal{M}(b)$. In the Hermitian setting, the adjoint ring is faithfully represented on one axis, so the notation $\mathrm{Adj}(d) \otimes \mathrm{Adj}(b)$ is unambiguous. Theorem 4.1 recovers this case via $\mathcal{M}(d \otimes b) = \mathcal{M}(d) \boxtimes \mathcal{M}(b)$.

In the rest of this section, we'll use the key observation in Lemma 2.11 to convert a equality about the image of axis actions to a matrix factorization statement.

4.2. **Algebra preliminaries.**

Definition 4.4. Let U, V be finite-dimensional $\mathbb{F}$-vector spaces, and $t \in U \otimes V$. Define

$$\begin{aligned} \Theta_L &: U \otimes V \to \mathrm{Hom}(V^*, U), & \Theta_L(u \otimes v)(\varphi) &= \varphi(v)u, \\ \Theta_R &: U \otimes V \to \mathrm{Hom}(U^*, V), & \Theta_R(u \otimes v)(\psi) &= \psi(u)v \\ \mathrm{Im}_L(t) &:= \mathrm{Im}(\Theta_L(t)) \leq U, & \mathrm{Im}_R(t) &:= \mathrm{Im}(\Theta_R(t)) \leq V. \end{aligned}$$

Proposition 4.5. Let U, V be finite-dimensional $\mathbb{F}$-vector spaces and $t \in U \otimes V$. Then

$$t \in \mathrm{Im}_L(t) \otimes \mathrm{Im}_R(t) \leq U \otimes V.$$

Proof. Let $A := \mathrm{Im}_L(t) \leq U$, and $X := \mathrm{Im}_R(t) \leq V$. Choose a basis $v_1, \ldots, v_n$ of V with dual basis $\varphi_1, \ldots, \varphi_n$. Then

$$t = \sum_{j=1}^{n} \Theta_L(t)(\varphi_j) \otimes v_j \in A \otimes V,$$

since each $\Theta_L(t)(\varphi_j)$ lies in A. Choose a basis $u_1, \ldots, u_m$ of U with dual basis $\psi_1, \ldots, \psi_m$. Then

$$t = \sum_{i=1}^{m} u_i \otimes \Theta_R(t)(\psi_i) \in U \otimes X,$$

since each $\Theta_R(t)(\psi_i)$ lies in X. Therefore by Lemma 2.2, $t \in (A \otimes V) \cap (U \otimes X) = A \otimes X$. □

Remark 4.6. Let U, V be finite-dimensional $\mathbb{F}$-vector spaces. Let $\{u_i\}$ be an ordered bases of U and $\{v_j\}$ be an ordered basis V. Then t can be uniquely written as

$$t = \sum_{i,j} T_{ij}\, u_i \otimes v_j \in U \otimes V.$$

This identifies t with the matrix $T = (T_{ij})$.

Example 4.7. Let $U = \mathbb{F}^2$ with standard bases e_1, e_2 and $V = \mathbb{F}^3$ with standard basis f_1, f_2, f_3. Let $t = e_1 \otimes f_1 + e_2 \otimes f_1 + e_2 \otimes f_2$. Then t corresponds to the matrix

$$T = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Definition 4.8. Let U, V be finite-dimensional $\mathbb{F}$-vector spaces and $t \in U \otimes V$. The *rank* of t is the minimum number r such that t can be written as a sum $t = \sum_{i=1}^{r} u_i \otimes v_i$, for $u_i \in U$, and $v_i \in V$. The sum $\sum_{i=1}^{r} u_i \otimes v_i$ is called an *minimal rank decomposition* of t.

We'll use the observation that for $t \in U \otimes V$, the rank of t is less than or equal to the minimum of $\dim U$, and $\dim V$, as we can always expand t either in a basis of U or a basis of V.

Lemma 4.9. Let $t = \sum_{i=1}^{r} u_i \otimes v_i$ in $U \otimes V$, where $r = \mathrm{rank}(t)$. Then $\{u_1, \ldots, u_r\}$ and $\{v_1, \ldots, v_r\}$ are linearly independent, and

$$\mathrm{Im}_L(t) = \mathrm{span}\{u_1, \ldots, u_r\}, \qquad \mathrm{Im}_R(t) = \mathrm{span}\{v_1, \ldots, v_r\}.$$

Proof. Set $W := \mathrm{span}\{u_1, \ldots, u_r\}$ and $X := \mathrm{span}\{v_1, \ldots, v_r\}$. Since $t \in W \otimes V$, the above observation gives $\mathrm{rank}(t) \leq \dim W$. Since W is spanned by r vectors, $\dim W \leq r$. Together with $\mathrm{rank}(t) = r$, this gives $\dim W = r$, so $\{u_1, \ldots, u_r\}$ is linearly independent. An analogous argument gives $\dim X = r$, so $\{v_1, \ldots, v_r\}$ is linearly independent.

Choose $\varphi_i \in V^*$ with $\varphi_i(v_j) = \delta_{ij}$ for $1 \leq i, j \leq r$. Then

$$\Theta_L(t)(\varphi_i) = u_i,$$

so $W \leq \mathrm{Im}_L(t)$. For the reverse inclusion, we see for all $\varphi \in V^*$,

$$\Theta_L(t)(\varphi) = \sum_{i=1}^{r} \varphi(v_i)\, u_i \in W,$$

so $\mathrm{Im}_L(t) \leq W$. Hence $\mathrm{Im}_L(t) = W$. An analogous argument concludes $\mathrm{Im}_R(t) = X$. □

Remark 4.10. Suppose $\sum_{i=1}^r a_i \otimes x_i = \sum_{i=1}^r b_i \otimes y_i$ are two minimal rank decompositions of the same element in $U \otimes V$. Fix ordered bases of U and V, and let A, B be coordinates of a_i, b_i as columns respectively, and X, Y be coordinates of x_i, y_i as rows respectively. In these fixed coordinates, we get the equality $AX = BY$. Because A and B are both rank r, there is a unique $M \in \mathrm{GL}_r(\mathbb{F})$ such that

$$B = AM, \qquad Y = M^{-1}X.$$

Example 4.11. Take $\mathbb{F} = \mathbb{F}_5$, $U = V = \mathbb{F}^3$, and $r = 2$. Define in the standard basis

$$A = \left[\begin{array}{c|c} a_1 & a_2 \end{array}\right] = \left[\begin{array}{c|c} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{array}\right], \quad X = \left[\begin{array}{c} b_1 \\ \hline b_2 \end{array}\right] = \left[\begin{array}{ccc} 1 & 2 & 0 \\ \hline 0 & 1 & 1 \end{array}\right]$$

$$B = \left[\begin{array}{c|c} c_1 & c_2 \end{array}\right] = \left[\begin{array}{c|c} 1 & 2 \\ 0 & 3 \\ 3 & 4 \end{array}\right], \quad Y = \left[\begin{array}{c} d_1 \\ \hline d_2 \end{array}\right] = \left[\begin{array}{ccc} 3 & 2 & 1 \\ \hline 4 & 0 & 2 \end{array}\right].$$

A direct multiplication verifies

$$AX = BY = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix},$$

meaning $a_1 \otimes x_1 + a_2 \otimes x_2 = b_1 \otimes y_1 + b_2 \otimes y_2$. As per Remark 4.10, we find

$$M = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad M^{-1} = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix},$$

satisfying

$$AM = \begin{bmatrix} 1 & 2 \\ 0 & 3 \\ 3 & 4 \end{bmatrix} = B, \qquad M^{-1}X = \begin{bmatrix} 3 & 2 & 1 \\ 4 & 0 & 2 \end{bmatrix} = Y.$$

4.3. **Key reduction.** The next lemma powers the reduction used in the nuclei case.

Lemma 4.12. Let $s : U_2 \times U_1 \rightarrowtail U_0$ and $t : V_2 \times V_1 \rightarrowtail V_0$ be fully nondegenerate bimaps. Let $K \subseteq \{2, 1, 0\}$ be nonempty, and $w \in \mathrm{Bil}(U_2, U_1; U_0) \otimes \mathrm{Bil}(V_2, V_1; V_0)$ be an element such that for each $k \in K$, there is some $x_k \in \mathrm{End}(U_k \otimes V_k)$ satisfying

$$w = \Xi\big((s \otimes t) \circ_k x_k\big) \in \mathrm{Bil}(U_2, U_1; U_0) \otimes \mathrm{Bil}(V_2, V_1; V_0).$$

Fix a minimal rank decomposition $w = \sum_{i=1}^r u_i \otimes v_i$. Then for each $k \in K$, there exists $\{\alpha_{k,i} \in \mathrm{End}(U_k)\}_{i=1}^r$ and $\{f_{k,i} \in \mathrm{End}(V_k)\}_{i=1}^r$ such that $s \circ_k \alpha_{k,i} = u_i$, and $t \circ_k f_{k,i} = v_i$, and $x_k = \sum_{i=1}^r \kappa_k(\alpha_{k,i} \otimes f_{k,i})$.

Proof. Fix $k \in K$ and any decomposition $x_k = \sum_j \kappa_k(\alpha_j \otimes f_j)$. Equation (2.2), gives $w = \sum_j \varphi(s \circ_k \alpha_j, t \circ_k f_j)$, so

$$w \in \mathrm{Im}(\Lambda_k^s) \otimes \mathrm{Im}(\Lambda_k^t), \qquad \text{with } \mathrm{Im}_L(w) \leq \mathrm{Im}(\Lambda_k^s), \quad \mathrm{Im}_R(w) \leq \mathrm{Im}(\Lambda_k^t).$$

Since $w = \sum_i u_i \otimes v_i$ is a minimal rank decomposition, apply Lemma 4.9 to conclude for each i that the elements $u_i \in \mathrm{Im}_L(w) \leq \mathrm{Im}(\Lambda_k^s)$ and $v_i \in \mathrm{Im}_R(w) \leq \mathrm{Im}(\Lambda_k^t)$ are images of axis actions on s and t respectively. Specifically, there exist $\alpha_{k,i} \in \mathrm{End}(U_k)$ and $f_{k,i} \in \mathrm{End}(V_k)$ such that

$$u_i = s \circ_k \alpha_{k,i}, \qquad v_i = t \circ_k f_{k,i}.$$

Finally, for each k, set

$$x'_k := \sum_{i=1}^r \kappa_k(\alpha_{k,i} \otimes f_{k,i}).$$

Since $(s \otimes t) \circ_k x_k = (s \otimes t) \circ_k x'_k$ and $\Lambda_k^{s \otimes t}$ is injective, we conclude $x_k = x'_k$. $\square$

4.4. **Proof of Theorem 4.1.** We first prove the middle nucleus case.

Proof. ($\subseteq$) Let $(x, y) \in \mathcal{M}(s \otimes t)$, meaning $(s \otimes t) \circ_2 x = (s \otimes t) \circ_1 y$. Set $w := \Xi\big((s \otimes t) \circ_2 x\big) = \Xi\big((s \otimes t) \circ_1 y\big) \in \mathrm{Bil}(U_2, U_1; U_0) \otimes \mathrm{Bil}(V_2, V_1; V_0)$. Fix a minimal rank decomposition

$$w = \sum_{i=1}^{r} u_i \otimes v_i.$$

Apply Lemma 4.12 with $K = \{2, 1\}$. We obtain $\alpha_{2,i} =: \alpha_i \in \mathrm{End}(U_2),\ \alpha_{1,i} =: \beta_i \in \mathrm{End}(U_1),\ f_{2,i} =: f_i \in \mathrm{End}(V_2),\ f_{1,i} =: g_i \in \mathrm{End}(V_1)$ such that

$$s \circ_2 \alpha_i = u_i = s \circ_1 \beta_i, \qquad t \circ_2 f_i = v_i = t \circ_1 g_i,$$

and

$$x = \sum_{i=1}^{r} \kappa_2(\alpha_i \otimes f_i), \qquad y = \sum_{i=1}^{r} \kappa_1(\beta_i \otimes g_i).$$

Hence $(\alpha_i, \beta_i) \in \mathcal{M}(s)$ and $(f_i, g_i) \in \mathcal{M}(t)$ for each i, so

$$(x, y) = \sum_{i=1}^{r} \big((\alpha_i, \beta_i) \boxtimes (f_i, g_i)\big) \in \mathcal{M}(s) \boxtimes \mathcal{M}(t).$$

($\supseteq$) Let $(\alpha, \beta) \in \mathcal{M}(s)$ and $(f, g) \in \mathcal{M}(t)$. Then

$$(s \otimes t) \circ_2 \kappa_2(\alpha \otimes f) = (s \circ_2 \alpha) \otimes (t \circ_2 f) = (s \circ_1 \beta) \otimes (t \circ_1 g) = (s \otimes t) \circ_1 \kappa_1(\beta \otimes g).$$

So $(\alpha, \beta) \boxtimes (f, g) = (\kappa_2(\alpha \otimes f), \kappa_1(\beta \otimes g)) \in \mathcal{M}(s \otimes t)$. Since $\mathcal{M}(s) \boxtimes \mathcal{M}(t)$ is the span of such pure terms, we get $\mathcal{M}(s) \boxtimes \mathcal{M}(t) \subseteq \mathcal{M}(s \otimes t)$. The left and right nuclei case are analogous. □

A worked-out nucleus decomposition example is in the appendix (Example D.1).

Now we wrap up this chapter with the proof that $\mathcal{C}(s \otimes t) = \mathcal{C}(s) \boxtimes \mathcal{C}(t)$.

Proof of Corollary 4.2. We prove the direction $\mathcal{C}(s \otimes t) \subseteq \mathcal{C}(s) \boxtimes \mathcal{C}(t)$. Let $(x_2, x_1, x_0) \in \mathcal{C}(s \otimes t)$. Then

$$(s \otimes t) \circ_2 x_2 = (s \otimes t) \circ_1 x_1 = (s \otimes t) \circ_0 x_0.$$

Let $w := \Xi\big((s \otimes t) \circ_2 x_2\big) \in \mathrm{Bil}(U_2, U_1; U_0) \otimes \mathrm{Bil}(V_2, V_1; V_0)$ be the element in common. Fix a minimal rank decomposition $w = \sum_{i=1}^{r} u_i \otimes v_i$, and apply Lemma 4.12 with $K = \{2, 1, 0\}$, then follow the same argument as the proof of Theorem 4.1.

The direction $\mathcal{C}(s \otimes t) \supseteq \mathcal{C}(s) \boxtimes \mathcal{C}(t)$ is analogous to the proof of $\mathcal{M}(s \otimes t) \supseteq \mathcal{M}(s) \boxtimes \mathcal{M}(t)$ in Theorem 4.1. □

5. Derivation of Product of Bimaps

The main result of this section is the following.

Theorem 5.1. Let s and t be fully nondegenerate bimaps. Then

$$\begin{aligned} \mathrm{Der}(s \otimes t) = {} & \mathrm{Der}(s) \boxtimes \mathcal{C}(t) + \mathcal{C}(s) \boxtimes \mathrm{Der}(t) \\ & + \iota_{20}(\mathcal{L}(s) \boxtimes \mathcal{L}(t)) + \iota_{10}(\mathcal{R}(s) \boxtimes \mathcal{R}(t)) + \iota_{21}(\mathcal{M}(s) \boxtimes \mathcal{M}(t)), \end{aligned} \tag{5.1}$$

where $\iota_{20}(\alpha, \gamma) = (\alpha, 0, \gamma)$, $\iota_{10}(\beta, \gamma) = (0, \beta, \gamma)$, and $\iota_{21}(\alpha, \beta) = (-\alpha, \beta, 0)$.

The strategy is similar to the nuclei case, where we relate equations of axis actions on tensor product bimaps to equalities in tensor product vector spaces.

5.1. **Previous results.** In this subsection, algebras are finite-dimensional over $\mathbb{F}$, and not necessarily associative. For an algebra A, its derivation refers to

$$\mathcal{D}_{\text{alg}}(A) := \{d \in \operatorname{End}(A) : \forall x, y \in A,\ d(xy) = d(x)y + xd(y)\}.$$

Brešar proved the following.

Theorem 5.2 (Brešar, Theorem 3.1 [Bre17])**.** Let R and S be nonassociative unital algebras. Suppose that either at least one of R and S is finite-dimensional or they are both finitely generated. Then every derivation d of $R \otimes S$ can be written as

$$d = \operatorname{ad}(u) + \sum_{j=1}^{p} \lambda_{z_j} \otimes f_j + \sum_{i=1}^{q} g_i \otimes \lambda_{w_i}, \tag{5.2}$$

where $u \in \operatorname{Nuc}(R) \otimes \operatorname{Nuc}(S)$, $z_j \in Z(R)$, $w_i \in Z(S)$, $f_j \in \mathcal{D}_{\text{alg}}(S)$, and $g_i \in \mathcal{D}_{\text{alg}}(R)$.

Proposition 5.3. Let R and S be unital algebras over $\mathbb{F}$, with multiplication maps μ_R, μ_S. Then Theorem 5.1 implies Theorem 5.2 in the case both R and S are finite dimensional.

Proof. See Appendix C. □

Specialization to Matrix Algebras. Let A be a unital finite-dimensional $\mathbb{F}$-algebra. As a vector space, $\mathbb{M}_n(A) \cong \mathbb{M}_n(\mathbb{F}) \otimes A$, and the multiplication on $\mathbb{M}_n(A)$ is defined by $(X \otimes a)(Y \otimes b) = XY \otimes ab$.

Theorem 5.4 (Benkart–Osborn, Corollary 4.9 [BO81])**.** Assume $\operatorname{char} \mathbb{F} \neq 2, 3$ and $n > 3$. Then for any unital algebra A, $\mathcal{D}_{\text{alg}}(\mathbb{M}_n(A)) = \mathcal{D}_{\text{alg}}(L_n'(A)) = \mathcal{D}_{\text{alg}}(K_n(A)) = (\mathcal{D}_{\text{alg}}(A))^{\#} + \operatorname{ad}_{\mathbb{M}_n(N)}$.

Remark 5.5. In Theorem 5.4, N denotes the nucleus of A, $L_n(A)$ is $\mathbb{M}_n(A)$ under the Lie product, $L_n'(A) = [L_n(A), L_n(A)]$, and $K_n(A) = L_n'(A)/Z(L_n'(A))$. Also, $(\mathcal{D}_{\text{alg}}(A))^{\#}$ denotes entrywise application of derivations of A. Thus in the notation of this chapter,

$$N = \operatorname{Nuc}(A), \qquad (\mathcal{D}_{\text{alg}}(A))^{\#} = I_n \otimes \mathcal{D}_{\text{alg}}(A), \qquad \operatorname{ad}_{\mathbb{M}_n(N)} = \operatorname{ad}(\mathbb{M}_n(\operatorname{Nuc}(A))).$$

So Theorem 5.4 proves $\mathcal{D}_{\text{alg}}(\mathbb{M}_n(A)) = \big(I_n \otimes \mathcal{D}_{\text{alg}}(A)\big) + \operatorname{ad}(\mathbb{M}_n(\operatorname{Nuc}(A)))$.

Benkart–Osborn note that specifically for Corollary 4.9, no restrictions on $\operatorname{char} \mathbb{F}$ or n are needed. As a result, we have the following.

Proposition 5.6. Theorem 5.1 implies Theorem 5.4.

Proof. Theorem 5.1 implies Theorem 5.2 by Proposition 5.3, and Theorem 5.2 implies Theorem 5.4 by [Bre17, Cor. 4.2]. □

5.2. **Preliminaries.** The following decomposition definition features prominently. Throught this subsection, U and V are always finite dimensional $\mathbb{F}$-vector spaces.

Definition 5.7. Let $p, q, r \in U \otimes V$ be nonzero tensors satisfying $p + q = r$. Define

$$A := \operatorname{Im}_L(p), \quad B := \operatorname{Im}_L(q), \quad C := \operatorname{Im}_L(r),$$
$$X := \operatorname{Im}_R(p), \quad Y := \operatorname{Im}_R(q), \quad Z := \operatorname{Im}_R(r).$$

By Proposition 4.5, $p \in A \otimes X$, $q \in B \otimes Y$, $r \in C \otimes Z$.

A (p, q, r) *locally independent unified decomposition* (LIU decomposition) consists of a natural number n, and vectors

$$a_i \in A,\ b_i \in B,\ c_i \in C, \qquad x_i \in X,\ y_i \in Y,\ z_i \in Z,$$

for $i \in [n]$, satisfying the following:

- **Decomposition equality.**

$$p = \sum_{i=1}^{n} a_i \otimes x_i, \qquad q = \sum_{i=1}^{n} b_i \otimes y_i, \qquad r = \sum_{i=1}^{n} c_i \otimes z_i.$$

- **Local equality.** For all $i \in [n]$, $a_i \otimes x_i + b_i \otimes y_i = c_i \otimes z_i$.

Example 5.8. Let $U = V = \mathbb{F}^2$ with bases u_1, u_2 and v_1, v_2. Define

$$p := u_1 \otimes v_1 + u_1 \otimes v_2, \qquad q := u_2 \otimes v_1 - u_1 \otimes v_2, \qquad r := u_1 \otimes v_1 + u_2 \otimes v_1.$$

Then $p + q = r$, and

$$\begin{aligned} &A = \operatorname{span}\{u_1\}, && B = U, && C = U, \\ &X = V, && Y = V, && Z = \operatorname{span}\{v_1\}. \end{aligned}$$

Now let

$$\begin{aligned} &a_1 = u_1,\ b_1 = u_2,\ c_1 = u_1 + u_2, && x_1 = y_1 = z_1 = v_1, \\ &a_2 = u_1,\ b_2 = -u_1,\ c_2 = 0, && x_2 = y_2 = v_2,\ z_2 = v_1. \end{aligned}$$

Then for each $i \in [2]$, $a_i \otimes x_i + b_i \otimes y_i = c_i \otimes z_i$, and

$$\begin{aligned} p &= a_1 \otimes x_1 + a_2 \otimes x_2, \\ q &= b_1 \otimes y_1 + b_2 \otimes y_2, \\ r &= c_1 \otimes z_1 + c_2 \otimes z_2. \end{aligned}$$

Thus this is an LIU decomposition of $p + q = r$.

5.3. Existence of a LIU decomposition of the equation $p + q = r$.

Convention 5.9. For the remainder of this subsection, whenever given nonzero tensors $p, q, r \in U \otimes V$ satisfying $p + q = r$, write

$$\begin{aligned} &A := \operatorname{Im}_L(p), \quad B := \operatorname{Im}_L(q), \quad C := \operatorname{Im}_L(r), \\ &X := \operatorname{Im}_R(p), \quad Y := \operatorname{Im}_R(q), \quad Z := \operatorname{Im}_R(r). \end{aligned}$$

It is always the case that $A + B \leq U$ and $X + Y \leq V$. We also assume without loss of generality that

$$U := A + B, \qquad V := X + Y,$$

because it is always the case that

$$\begin{aligned} &p \in A \otimes X \leq (A + B) \otimes (X + Y) \leq U \otimes V \\ &q \in B \otimes Y \leq (A + B) \otimes (X + Y) \leq U \otimes V \\ &r \in \operatorname{Im}_L(r) \otimes \operatorname{Im}_R(r). \end{aligned}$$

Furthermore, $\operatorname{Im}_L(r), \operatorname{Im}_R(r)$ satisfies

$$\begin{aligned} \operatorname{Im}_L(r) &= \operatorname{Im}_L(p + q) \leq \operatorname{Im}_L(p) + \operatorname{Im}_L(q) = A + B, \\ \operatorname{Im}_R(r) &= \operatorname{Im}_R(p + q) \leq \operatorname{Im}_R(p) + \operatorname{Im}_R(q) = X + Y. \end{aligned}$$

So we may redefine U to the space $A + B$, and redefine V to the space $X + Y$.

Lemma 5.10. Let U be a $\mathbb{F}$-vector space, and let $A, B, C \leq U$ with $U = A + B$. Define $U^{(1)} \leq U$ by

$$U^{(1)} := A \cap B \cap C.$$

Define $U^{(2)} \leq A \cap B$, $U^{(3)} \leq A \cap C$, and $U^{(4)} \leq B \cap C$ by choosing complements

$$A \cap B = U^{(1)} \oplus U^{(2)}, \qquad A \cap C = U^{(1)} \oplus U^{(3)}, \qquad B \cap C = U^{(1)} \oplus U^{(4)}.$$

Define $U^{(5)} \leq A$ and $U^{(6)} \leq B$ by choosing complements

$$A = (U^{(1)} + U^{(2)} + U^{(3)}) \oplus U^{(5)}, \qquad B = (U^{(1)} + U^{(2)} + U^{(4)}) \oplus U^{(6)}.$$

Then

$$U = \bigoplus_{i=1}^{6} U^{(i)}, \qquad A = \bigoplus_{i \in \{1,2,3,5\}} U^{(i)}, \qquad B = \bigoplus_{i \in \{1,2,4,6\}} U^{(i)}.$$

Proof. Let $d_i := \dim U^{(i)}$ for $1 \le i \le 6$. By construction, $\dim(A \cap B) = d_1 + d_2$, $\dim(A \cap C) = d_1 + d_3$. Using the fact

$$\dim(X + Y) = \dim X + \dim Y - \dim(X \cap Y),$$

with $X = A \cap B$, $Y = A \cap C$, we get

$$\dim\big((A \cap B) + (A \cap C)\big) = (d_1 + d_2) + (d_1 + d_3) - d_1 = d_1 + d_2 + d_3.$$

Since $A = ((A \cap B) + (A \cap C)) \oplus U^{(5)}$, this gives

$$\dim A = d_1 + d_2 + d_3 + d_5.$$

Similarly, from $B = ((A \cap B) + (B \cap C)) \oplus U^{(6)}$,

$$\dim B = d_1 + d_2 + d_4 + d_6.$$

Finally, as $U = A + B$,

$$\dim U = \dim(A + B) = \dim A + \dim B - \dim(A \cap B) = \sum_{i=1}^{6} d_i.$$

Recall

$$U = A + B = \sum_{i=1}^{6} U^{(i)}, \qquad A = \sum_{i \in \{1,2,3,5\}} U^{(i)}, \qquad B = \sum_{i \in \{1,2,4,6\}} U^{(i)},$$

and the fact that for any finite family of subspaces $\{W_i\}$, $\dim(\sum_i W_i) \le \sum_i \dim W_i$, with equality if and only if the sum is direct. Applying this to the spaces U, A, B concludes

$$U = \bigoplus_{i=1}^{6} U^{(i)}, \qquad A = \bigoplus_{i \in \{1,2,3,5\}} U^{(i)}, \qquad B = \bigoplus_{i \in \{1,2,4,6\}} U^{(i)}.$$

□

The above lemma is to prepare for below, which is the key one in proving the existence of a LIU decomposition for the equation $p + q = r$.

Lemma 5.11. Let $p, q, r \in U \otimes V$ be nonzero elements satisfying $p + q = r$, and

$$A := \mathrm{Im}_L(p), \quad B := \mathrm{Im}_L(q), \quad C := \mathrm{Im}_L(r),$$
$$X := \mathrm{Im}_R(p), \quad Y := \mathrm{Im}_R(q), \quad Z := \mathrm{Im}_R(r).$$

Recall via Convention 5.9 that $U = A + B$, $V = X + Y$. With respect to the decomposition in Lemma 5.10, p, q, r satisfy:

$$p \in U^{(1)} \otimes V^{(1)} \oplus U^{(1)} \otimes V^{(\bullet)} \oplus U^{(\bullet)} \otimes V^{(1)} \oplus U^{(2)} \otimes V^{(2)} \oplus U^{(3)} \otimes V^{(3)},$$
$$q \in U^{(1)} \otimes V^{(1)} \oplus U^{(1)} \otimes V^{(\bullet)} \oplus U^{(\bullet)} \otimes V^{(1)} \oplus U^{(2)} \otimes V^{(2)} \oplus U^{(4)} \otimes V^{(4)},$$
$$r \in U^{(1)} \otimes V^{(1)} \oplus U^{(1)} \otimes V^{(\bullet)} \oplus U^{(\bullet)} \otimes V^{(1)} \oplus U^{(3)} \otimes V^{(3)} \oplus U^{(4)} \otimes V^{(4)},$$

where $U^{(\bullet)} := U^{(2)} \oplus \cdots \oplus U^{(6)}$ and $V^{(\bullet)} := V^{(2)} \oplus \cdots \oplus V^{(6)}$.

Proof. Apply Lemma 5.10 to the spaces $A, B, C \le U$ and to the spaces $X, Y, Z \le V$.

From the definitions, $A = U^{(1)} \oplus U^{(2)} \oplus U^{(3)} \oplus U^{(5)}$ and $B = U^{(1)} \oplus U^{(2)} \oplus U^{(4)} \oplus U^{(6)}$, and similarly $X = V^{(1)} \oplus V^{(2)} \oplus V^{(3)} \oplus V^{(5)}$ and $Y = V^{(1)} \oplus V^{(2)} \oplus V^{(4)} \oplus V^{(6)}$. Let $\pi_{ab} : U \otimes V \to U^{(a)} \otimes V^{(b)}$ be the block projections and write

$$\pi_{a\bullet} := \sum_{b=1}^{6} \pi_{ab}, \qquad \pi_{\bullet b} := \sum_{a=1}^{6} \pi_{ab}.$$

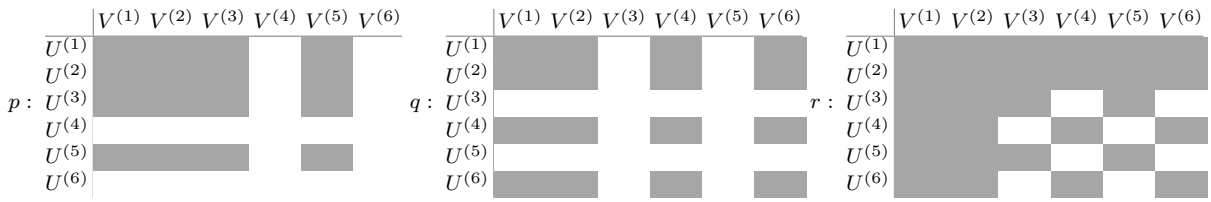
Hence $(\pi_{4\bullet} + \pi_{6\bullet})(p) = 0$ and $(\pi_{3\bullet} + \pi_{5\bullet})(q) = 0$, while $(\pi_{\bullet 4} + \pi_{\bullet 6})(p) = 0$ and $(\pi_{\bullet 3} + \pi_{\bullet 5})(q) = 0$. For $1 \le a, b \le 6$, set

$$p_{ab} := \pi_{ab}(p), \quad q_{ab} := \pi_{ab}(q), \quad r_{ab} := \pi_{ab}(r).$$

Applying π_{ab} to $p + q = r$ gives

$$p_{ab} + q_{ab} = r_{ab} \qquad (1 \le a, b \le 6).$$

This is a blockwise $p + q = r$ equality: if two entries in a block are zero, then the third is zero. Taking stock we visualize these relations below, where grey means *potentially nonzero*, white means *guaranteed to be zero*.



Row projections from known zero rows. Since $\pi_{3\bullet}(q) = 0$, define

$$s_3 \coloneqq \pi_{3\bullet}(p) = \pi_{3\bullet}(r).$$

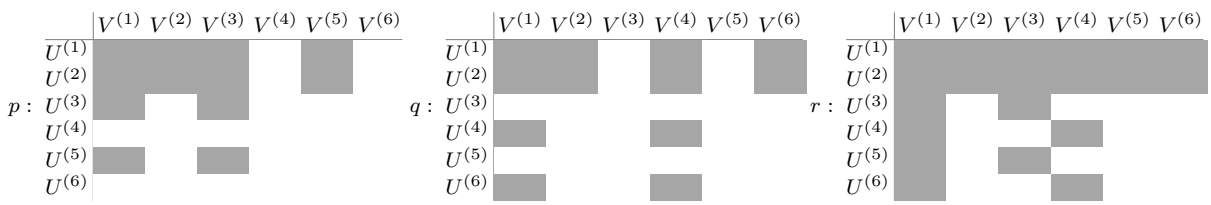
Then $s_3 \in U^{(3)} \otimes X$ and $s_3 \in U^{(3)} \otimes Z$, hence $s_3 \in U^{(3)} \otimes (X \cap Z) = U^{(3)} \otimes \big(V^{(1)} \oplus V^{(3)}\big)$. So row $U^{(3)}$ of p, r has support only in columns $V^{(1)}, V^{(3)}$. The same argument with $U^{(5)}$ gives row $U^{(5)}$ of p, r also in $V^{(1)} \oplus V^{(3)}$.

Since $\pi_{4\bullet}(p) = 0$, define

$$t_4 \coloneqq \pi_{4\bullet}(q) = \pi_{4\bullet}(r).$$

Then $t_4 \in U^{(4)} \otimes Y$ and $t_4 \in U^{(4)} \otimes Z$, hence $t_4 \in U^{(4)} \otimes (Y \cap Z) = U^{(4)} \otimes \big(V^{(1)} \oplus V^{(4)}\big)$. So row $U^{(4)}$ of q, r has support only in columns $V^{(1)}, V^{(4)}$. The same argument with $U^{(6)}$ gives row $U^{(6)}$ of q, r also in $V^{(1)} \oplus V^{(4)}$.

After these row inferences, the potentially nonzero blocks are:



Column projections from known zero columns. Since $\pi_{\bullet 3}(q) = 0$, define

$$u_3 \coloneqq \pi_{\bullet 3}(p) = \pi_{\bullet 3}(r).$$

Then $u_3 \in A \otimes V^{(3)}$ and $u_3 \in C \otimes V^{(3)}$, hence $u_3 \in (A \cap C) \otimes V^{(3)} = \big(U^{(1)} \oplus U^{(3)}\big) \otimes V^{(3)}$. So column $V^{(3)}$ of p, r has support only in rows $U^{(1)}, U^{(3)}$.

Since $\pi_{\bullet 4}(p) = 0$, define

$$u_4 \coloneqq \pi_{\bullet 4}(q) = \pi_{\bullet 4}(r).$$

Then $u_4 \in B \otimes V^{(4)}$ and $u_4 \in C \otimes V^{(4)}$, hence $u_4 \in (B \cap C) \otimes V^{(4)} = \big(U^{(1)} \oplus U^{(4)}\big) \otimes V^{(4)}$. So column $V^{(4)}$ of q, r has support only in rows $U^{(1)}, U^{(4)}$.

Since $\pi_{\bullet 5}(q) = 0$, define

$$u_5 \coloneqq \pi_{\bullet 5}(p) = \pi_{\bullet 5}(r).$$

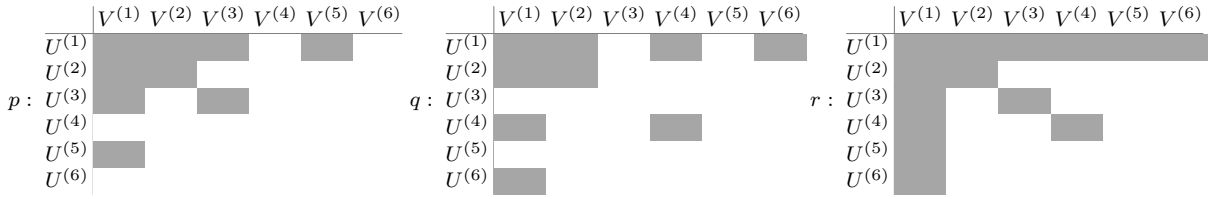
Then $u_5 \in (A \cap C) \otimes V^{(5)} = \big(U^{(1)} \oplus U^{(3)}\big) \otimes V^{(5)}$. Combining this with the row-$U^{(3)}$ and row-$U^{(5)}$ restrictions above forces support in column $V^{(5)}$ to lie only in row $U^{(1)}$.

Since $\pi_{\bullet 6}(p) = 0$, define

$$u_6 \coloneqq \pi_{\bullet 6}(q) = \pi_{\bullet 6}(r).$$

Then $u_6 \in (B \cap C) \otimes V^{(6)} = \big(U^{(1)} \oplus U^{(4)}\big) \otimes V^{(6)}$. Combining this with the row-$U^{(4)}$ and row-$U^{(6)}$ restrictions above forces support in column $V^{(6)}$ to lie only in row $U^{(1)}$.

After these column inferences, the potentially nonzero blocks are:



The $(2,2)$ block relation. Set $v_2 := \pi_{2\bullet}(r)$. By the previous row/column inferences, row $U^{(2)}$ of both p and q is supported in $V^{(1)} \oplus V^{(2)} = X \cap Y$, hence

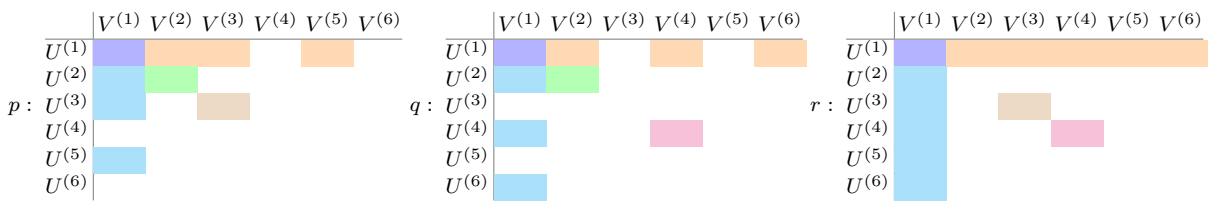
$$v_2 \in U^{(2)} \otimes (X \cap Y).$$

Also $r \in C \otimes Z$, so $v_2 \in U^{(2)} \otimes Z$. Therefore

$$v_2 \in U^{(2)} \otimes \big((X \cap Y) \cap Z\big) = U^{(2)} \otimes \big(X \cap Y \cap Z\big) = U^{(2)} \otimes V^{(1)}.$$

Thus $r_{22} = 0$, and the block equation gives $p_{22} + q_{22} = 0$.

What remains are the following support patterns, where blue is $U^{(1)} \otimes V^{(1)}$, orange is $U^{(1)} \otimes V^{(\bullet)}$, cyan is $U^{(\bullet)} \otimes V^{(1)}$, green is $U^{(2)} \otimes V^{(2)}$, brown is $U^{(3)} \otimes V^{(3)}$, and pink is $U^{(4)} \otimes V^{(4)}$, and blank entries are zero.



This is the claimed support pattern

$$p \in U^{(1)} \otimes V^{(1)} \oplus U^{(1)} \otimes V^{(\bullet)} \oplus U^{(\bullet)} \otimes V^{(1)} \oplus U^{(2)} \otimes V^{(2)} \oplus U^{(3)} \otimes V^{(3)},$$
$$q \in U^{(1)} \otimes V^{(1)} \oplus U^{(1)} \otimes V^{(\bullet)} \oplus U^{(\bullet)} \otimes V^{(1)} \oplus U^{(2)} \otimes V^{(2)} \oplus U^{(4)} \otimes V^{(4)},$$
$$r \in U^{(1)} \otimes V^{(1)} \oplus U^{(1)} \otimes V^{(\bullet)} \oplus U^{(\bullet)} \otimes V^{(1)} \oplus U^{(3)} \otimes V^{(3)} \oplus U^{(4)} \otimes V^{(4)}. \qquad \square$$

Example 5.12. Let $\mathbb{F} = \mathbb{F}_{997}$, let $U = V = \mathbb{F}^6$ with standard bases, and identify $U \otimes V$ with $\mathbb{M}_6(\mathbb{F})$ via $u \otimes v \mapsto [u][v]^T$ where $[u]$ is the column vector of coordinates.

Let $p, q, r \in U \otimes V$ correspond to the 6×6 matrices below.

$$p = \begin{bmatrix} 373 & 831 & 66 & 724 & 159 & 68 \\ 772 & 812 & 372 & 663 & 579 & 209 \\ 155 & 549 & 499 & 757 & 500 & 64 \\ 218 & 733 & 547 & 132 & 342 & 122 \\ 859 & 708 & 25 & 767 & 78 & 648 \\ 992 & 398 & 894 & 611 & 169 & 953 \end{bmatrix} \quad q = \begin{bmatrix} 176 & 355 & 631 & 671 & 759 & 955 \\ 252 & 645 & 740 & 978 & 500 & 638 \\ 308 & 485 & 963 & 867 & 212 & 39 \\ 146 & 830 & 155 & 854 & 264 & 14 \\ 30 & 375 & 193 & 887 & 38 & 692 \\ 107 & 281 & 218 & 640 & 676 & 450 \end{bmatrix} \quad r = \begin{bmatrix} 549 & 189 & 697 & 398 & 918 & 26 \\ 27 & 460 & 115 & 644 & 82 & 847 \\ 463 & 37 & 465 & 627 & 712 & 103 \\ 364 & 566 & 702 & 986 & 606 & 136 \\ 889 & 86 & 218 & 657 & 116 & 343 \\ 102 & 679 & 115 & 254 & 845 & 406 \end{bmatrix}$$

so that $p + q = r$ in $\mathbb{F}^{6\times 6}$. Choose change-of-basis matrices M, N so that the new coordinates are adapted to the decomposition $U = \bigoplus_{i=1}^{6} U^{(i)}$ and $V = \bigoplus_{i=1}^{6} V^{(i)}$:

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 310 & 34 & 931 & 898 & 971 & 773 \\ 104 & 130 & 918 & 780 & 84 & 647 \\ 795 & 834 & 145 & 316 & 939 & 574 \\ 386 & 64 & 11 & 618 & 494 & 754 \\ 732 & 237 & 949 & 667 & 810 & 127 \end{bmatrix}, \quad N = \begin{bmatrix} 1 & 655 & 49 & 786 & 153 & 994 \\ 0 & 757 & 318 & 920 & 875 & 310 \\ 0 & 304 & 386 & 307 & 423 & 363 \\ 0 & 988 & 354 & 652 & 88 & 712 \\ 0 & 780 & 227 & 987 & 917 & 291 \\ 0 & 59 & 338 & 600 & 869 & 429 \end{bmatrix},$$

the transformed matrices $p' = MpN$, $q' = MqN$, $r' = MrN$ have the support patterns as described by Lemma 5.11.

$p' =$

	$V^{(1)}$	$V^{(2)}$	$V^{(3)}$	$V^{(4)}$	$V^{(5)}$	$V^{(6)}$
$U^{(1)}$	373	16	259	0	968	0
$U^{(2)}$	119	905	0	0	0	0
$U^{(3)}$	966	0	123	0	0	0
$U^{(4)}$	0	0	0	0	0	0
$U^{(5)}$	647	0	0	0	0	0
$U^{(6)}$	0	0	0	0	0	0

$q' =$

	$V^{(1)}$	$V^{(2)}$	$V^{(3)}$	$V^{(4)}$	$V^{(5)}$	$V^{(6)}$
$U^{(1)}$	176	828	0	553	0	244
$U^{(2)}$	607	92	0	0	0	0
$U^{(3)}$	0	0	0	0	0	0
$U^{(4)}$	68	0	0	815	0	0
$U^{(5)}$	0	0	0	0	0	0
$U^{(6)}$	970	0	0	0	0	0

$r' =$

	$V^{(1)}$	$V^{(2)}$	$V^{(3)}$	$V^{(4)}$	$V^{(5)}$	$V^{(6)}$
$U^{(1)}$	549	844	259	553	968	244
$U^{(2)}$	726	0	0	0	0	0
$U^{(3)}$	966	0	123	0	0	0
$U^{(4)}$	68	0	0	815	0	0
$U^{(5)}$	647	0	0	0	0	0
$U^{(6)}$	970	0	0	0	0	0

Lemma 5.13. Let $p, q, r \in U \otimes V$ be nonzero tensors satisfying $p + q = r$. Then (p, q, r) admits an LIU decomposition.

Proof. Recall from Convention 5.9 the spaces $A, B, C \leq U$, and $X, Y, Z \leq V$ where $U = A + B$ and $V = X + Y$. Decompose U and V by Lemma 5.10

$$U \otimes V = \bigoplus_{a,b=1}^{6} U^{(a)} \otimes V^{(b)}.$$

Let $\pi_{ab} : U \otimes V \to U^{(a)} \otimes V^{(b)}$ be the corresponding projections. Define

$$p_{11} := \pi_{11}(p), \quad p_{1\bullet} := \sum_{b=2}^{6} \pi_{1b}(p), \quad p_{\bullet 1} := \sum_{a=2}^{6} \pi_{a1}(p), \quad p_{22} := \pi_{22}(p), \quad p_{33} := \pi_{33}(p),$$

and analogously define q_{11}, $q_{1\bullet}$, $q_{\bullet 1}$, q_{22}, q_{44}, and r_{11}, $r_{1\bullet}$, $r_{\bullet 1}$, r_{33}, r_{44}. By Lemma 5.11, all other blocks vanish, so

$$p = p_{11} + p_{1\bullet} + p_{\bullet 1} + p_{22} + p_{33},$$

$$q = q_{11} + q_{1\bullet} + q_{\bullet 1} + q_{22} + q_{44},$$

$$r = r_{11} + r_{1\bullet} + r_{\bullet 1} + r_{33} + r_{44}.$$

Applying the relevant projections to $p + q = r$ gives equations

$$p_{11} + q_{11} = r_{11}, \quad p_{1\bullet} + q_{1\bullet} = r_{1\bullet}, \quad p_{\bullet 1} + q_{\bullet 1} = r_{\bullet 1},$$

$$p_{22} + q_{22} = 0, \qquad p_{33} = r_{33}, \qquad q_{44} = r_{44}.$$

We build an LIU decomposition for each block. After, summing the decompositions gives the desired result.

In the blue block $U^{(1)} \otimes V^{(1)}$, choose bases of $U^{(1)}$ and $V^{(1)}$. Writing p_{11}, q_{11}, r_{11} in that basis, gives coefficientwise local equalities. The green ($U^{(2)} \otimes V^{(2)}$), brown ($U^{(3)} \otimes V^{(3)}$), and pink ($U^{(4)} \otimes V^{(4)}$) blocks are special cases of the blue-block argument.

In the orange block $U^{(1)} \otimes V^{(\bullet)}$, choose a basis $\{u_m\}_{m=1}^{d}$ of $U^{(1)}$ and write

$$p_{1\bullet} = \sum_{m=1}^{d} u_m \otimes f_m, \qquad q_{1\bullet} = \sum_{m=1}^{d} u_m \otimes g_m, \qquad r_{1\bullet} = \sum_{m=1}^{d} u_m \otimes h_m.$$

Since $p_{1\bullet} + q_{1\bullet} = r_{1\bullet}$, we get

$$\forall m \in [d], \qquad u_m \otimes f_m + u_m \otimes g_m = u_m \otimes h_m.$$

This gives local equalities with a common left factor in $A \cap B \cap C$. The cyan block ($U^{(\bullet)} \otimes V^{(1)}$) case is analogous. □

Example 5.14. We demonstrate this splitting for Example 5.12. Recall all coefficients are in $\mathbb{F}_{997}$.

Let u_i and v_j be basis vectors with $U^{(i)} = \langle u_i \rangle$ and $V^{(j)} = \langle v_j \rangle$.

For the blue block,

$$p_{11} = 373\, u_1 \otimes v_1, \qquad q_{11} = 176\, u_1 \otimes v_1, \qquad r_{11} = 549\, u_1 \otimes v_1,$$

so $p_{11} + q_{11} = (373 + 176)\, u_1 \otimes v_1 = 549\, u_1 \otimes v_1 = r_{11}$.

For the orange block,

$$p_{1\bullet} = u_1 \otimes (16v_2 + 259v_3 + 968v_5), \quad q_{1\bullet} = u_1 \otimes (828v_2 + 553v_4 + 244v_6),$$

so $p_{1\bullet} + q_{1\bullet} = u_1 \otimes (844v_2 + 259v_3 + 553v_4 + 968v_5 + 244v_6) = r_{1\bullet}$.

Likewise,

$$p_{\bullet 1} = (119u_2 + 966u_3 + 647u_5) \otimes v_1, \quad q_{\bullet 1} = (607u_2 + 68u_4 + 970u_6) \otimes v_1,$$

hence $p_{\bullet 1} + q_{\bullet 1} = (726u_2 + 966u_3 + 68u_4 + 647u_5 + 970u_6) \otimes v_1 = r_{\bullet 1}$.

5.4. **Proof of Theorem 5.1.** We apply the LIU decomposition to the images of axis actions on tensors and match each block to one summand in Equation (5.1). First, we need to describe the partitioning of the index sets of a LIU decomposition.

Remark 5.15. Given a LIU decomposition of $p + q = r$ produced by Lemma 5.13, with

$$p = \sum_{i=1}^{n} a_i \otimes x_i, \qquad q = \sum_{i=1}^{n} b_i \otimes y_i, \qquad r = \sum_{i=1}^{n} c_i \otimes z_i,$$

and local equalities

$$\forall i \in [n], \qquad a_i \otimes x_i + b_i \otimes y_i = c_i \otimes z_i.$$

Let $I_{11},\ I_{1\bullet},\ I_{\bullet 1},\ I_{22},\ I_{33},\ I_{44}$ be the index sets coming from the six blockwise decompositions in the proof of Lemma 5.13. Then these sets give a partition

$$[n] = I_{11} \sqcup I_{1\bullet} \sqcup I_{\bullet 1} \sqcup I_{22} \sqcup I_{33} \sqcup I_{44}.$$

Proof. ($\subseteq$) We prove the inclusion

$$\mathrm{Der}(s \otimes t) \subseteq \mathrm{Der}(s) \boxtimes \mathcal{C}(t) + \mathcal{C}(s) \boxtimes \mathrm{Der}(t) + \iota_{20}(\mathcal{L}(s) \boxtimes \mathcal{L}(t)) + \iota_{10}(\mathcal{R}(s) \boxtimes \mathcal{R}(t)) + \iota_{21}(\mathcal{M}(s) \boxtimes \mathcal{M}(t)).$$

Let $(\delta^{(2)}, \delta^{(1)}, \delta^{(0)}) \in \mathrm{Der}(s \otimes t)$, and define

$$p := \Xi((s \otimes t) \circ_2 \delta^{(2)}), \qquad q := \Xi((s \otimes t) \circ_1 \delta^{(1)}), \qquad r := \Xi((s \otimes t) \circ_0 \delta^{(0)}).$$

Then $p + q = r$ in $\mathrm{Bil}(U_2, U_1; U_0) \otimes \mathrm{Bil}(V_2, V_1; V_0)$.

If $r = 0$, then $(s \otimes t) \circ_2 \delta^{(2)} = -(s \otimes t) \circ_1 \delta^{(1)}$, so $(-\delta^{(2)}, \delta^{(1)}) \in \mathcal{M}(s \otimes t)$ and $(\delta^{(2)}, \delta^{(1)}, 0) \in \iota_{21}(\mathcal{M}(s \otimes t))$. The cases $p = 0$ and $q = 0$ are analogous, giving elements of $\iota_{10}(\mathcal{R}(s \otimes t))$ and $\iota_{20}(\mathcal{L}(s \otimes t))$ respectively. Applying Theorem 4.1 proves the desired inclusion in all three cases. Hence for the remainder of the argument, assume p, q, r are all nonzero.

The strategy is to apply Lemma 5.13 to $p + q = r$, obtain an LIU decomposition, then lift the local equalities to relations on the axis operators. With

$$A := \mathrm{Im}_L(p), \quad B := \mathrm{Im}_L(q), \quad C := \mathrm{Im}_L(r),$$
$$X := \mathrm{Im}_R(p), \quad Y := \mathrm{Im}_R(q), \quad Z := \mathrm{Im}_R(r).$$

By Proposition 4.5, $p \in A \otimes X$, $q \in B \otimes Y$, and $r \in C \otimes Z$. Let $(a_i, b_i, c_i, x_i, y_i, z_i)_{i=1}^n$ be the LIU decomposition by applying Lemma 5.13. By Remark 5.15, the LIU decomposition index set $[n]$ is partitioned into

$$I_{11} \sqcup I_{1\bullet} \sqcup I_{\bullet 1} \sqcup I_{22} \sqcup I_{33} \sqcup I_{44},$$

where each subset indexes local terms supported in the correspondingly colored block ($U^{(1)} \otimes V^{(1)}$, $U^{(1)} \otimes V^{(\bullet)}$, $U^{(\bullet)} \otimes V^{(1)}$, $U^{(2)} \otimes V^{(2)}$, $U^{(3)} \otimes V^{(3)}$, $U^{(4)} \otimes V^{(4)}$). We show each part gives an operator triple that's in one of the summands on the right-hand side of Equation (5.1).

- $i \in I_{1\bullet}$ ($U^{(1)} \otimes V^{(\bullet)}$): Each local equality satisfies

  $$a_i = b_i = c_i, \qquad x_i + y_i = z_i,$$

  with $a_i, b_i, c_i \in A \cap B \cap C$, $x_i \in X$, $y_i \in Y$, $z_i \in Z$. By Lemma 2.11 and Proposition 4.5, we have

  $$A \leq \mathrm{Im}(\Lambda_2^s), \quad B \leq \mathrm{Im}(\Lambda_1^s), \quad C \leq \mathrm{Im}(\Lambda_0^s),$$
  $$X \leq \mathrm{Im}(\Lambda_2^t), \quad Y \leq \mathrm{Im}(\Lambda_1^t), \quad Z \leq \mathrm{Im}(\Lambda_0^t).$$

  So each of $a_i, b_i, c_i, x_i, y_i, z_i$ has a preimage under the corresponding axis action maps. Choose

  $$\alpha_i \in \mathrm{End}(U_2),\ \beta_i \in \mathrm{End}(U_1),\ \gamma_i \in \mathrm{End}(U_0)$$
  $$f_i \in \mathrm{End}(V_2),\ g_i \in \mathrm{End}(V_1),\ h_i \in \mathrm{End}(V_0)$$

  such that $s \circ_2 \alpha_i = a_i,\ s \circ_1 \beta_i = b_i,\ s \circ_0 \gamma_i = c_i$, and $t \circ_2 f_i = x_i,\ t \circ_1 g_i = y_i,\ t \circ_0 h_i = z_i$.

  Then $(\alpha_i, \beta_i, \gamma_i) \in \mathcal{C}(s)$ and $(f_i, g_i, h_i) \in \mathrm{Der}(t)$, so

  $$(\kappa_2(\alpha_i \otimes f_i), \kappa_1(\beta_i \otimes g_i), \kappa_0(\gamma_i \otimes h_i)) \in \mathcal{C}(s) \boxtimes \mathrm{Der}(t).$$

- $i \in I_{\bullet 1}$ ($U^{(\bullet)} \otimes V^{(1)}$): Here each local equality satisfies $x_i = y_i = z_i$ and $a_i + b_i = c_i$. Choose
$$\alpha_i \in \mathrm{End}(U_2),\ \beta_i \in \mathrm{End}(U_1),\ \gamma_i \in \mathrm{End}(U_0)$$
$$f_i \in \mathrm{End}(V_2),\ g_i \in \mathrm{End}(V_1),\ h_i \in \mathrm{End}(V_0)$$
such that
$$s \circ_2 \alpha_i = a_i,\ s \circ_1 \beta_i = b_i,\ s \circ_0 \gamma_i = c_i, \qquad t \circ_2 f_i = x_i,\ t \circ_1 g_i = y_i,\ t \circ_0 h_i = z_i.$$
Then $(\alpha_i, \beta_i, \gamma_i) \in \mathrm{Der}(s)$ and $(f_i, g_i, h_i) \in \mathcal{C}(t)$, hence
$$(\kappa_2(\alpha_i \otimes f_i), \kappa_1(\beta_i \otimes g_i), \kappa_0(\gamma_i \otimes h_i)) \in \mathrm{Der}(s) \boxtimes \mathcal{C}(t).$$
- $i \in I_{22}, I_{33}, I_{44}$: these are the nucleus cases with one side equal to 0. We show the I_{22} case, which has $c_i \otimes z_i = 0$. The local equality is $a_i \otimes x_i + b_i \otimes y_i = 0$, with $a_i, b_i \in A \cap B$ and $x_i, y_i \in X \cap Y$. Choose operators
$$\alpha_i \in \mathrm{End}(U_2),\ \beta_i \in \mathrm{End}(U_1), \qquad f_i \in \mathrm{End}(V_2),\ g_i \in \mathrm{End}(V_1)$$
satisfying
$$s \circ_2 \alpha_i = a_i,\ s \circ_1 \beta_i = b_i, \qquad t \circ_2 f_i = x_i,\ t \circ_1 g_i = y_i.$$
Then for all i,
$$(s \otimes t) \circ_2 \kappa_2(\alpha_i \otimes f_i) = a_i \otimes x_i = -\, b_i \otimes y_i = -\,(s \otimes t) \circ_1 \kappa_1(\beta_i \otimes g_i).$$
Hence
$$\big(-\kappa_2(\alpha_i \otimes f_i),\ \kappa_1(\beta_i \otimes g_i)\big) \in \mathcal{M}(s \otimes t),$$
and therefore
$$(\kappa_2(\alpha_i \otimes f_i), \kappa_1(\beta_i \otimes g_i), 0) = \iota_{21}\big(-\kappa_2(\alpha_i \otimes f_i), \kappa_1(\beta_i \otimes g_i)\big) \in \iota_{21}(\mathcal{M}(s \otimes t)).$$
Applying Theorem 4.1, $\iota_{21}(\mathcal{M}(s \otimes t)) = \iota_{21}(\mathcal{M}(s) \boxtimes \mathcal{M}(t))$, so the operator triple lies in one of the summands on the right-hand side of Equation (5.1).

  The I_{33} and I_{44} cases are analogous.
- $i \in I_{11}$ ($U^{(1)} \otimes V^{(1)}$): This is the same as the $I_{1\bullet}$ case, now with $x_i, y_i, z_i \in X \cap Y \cap Z$ as well.

For each i, let $(\delta_i^{(2)}, \delta_i^{(1)}, \delta_i^{(0)}) \in \mathrm{End}(U_2 \otimes V_2) \times \mathrm{End}(U_1 \otimes V_1) \times \mathrm{End}(U_0 \otimes V_0)$ be the resulting operator triple constructed above. By construction, each triple lies in one of the summands on the right-hand side. Summing over i,
$$\Xi\left((s \otimes t) \circ_2 \sum_{i=1}^{n} \delta_i^{(2)}\right) = p = \Xi((s \otimes t) \circ_2 \delta^{(2)}),$$
$$\Xi\left((s \otimes t) \circ_1 \sum_{i=1}^{n} \delta_i^{(1)}\right) = q = \Xi((s \otimes t) \circ_1 \delta^{(1)}),$$
$$\Xi\left((s \otimes t) \circ_0 \sum_{i=1}^{n} \delta_i^{(0)}\right) = r = \Xi((s \otimes t) \circ_0 \delta^{(0)}).$$
Since Ξ is an isomorphism and $\Lambda_2^{s\otimes t}, \Lambda_1^{s\otimes t}, \Lambda_0^{s\otimes t}$ are injective, we get
$$\sum_{i=1}^{n} \delta_i^{(2)} = \delta^{(2)}, \qquad \sum_{i=1}^{n} \delta_i^{(1)} = \delta^{(1)}, \qquad \sum_{i=1}^{n} \delta_i^{(0)} = \delta^{(0)}.$$
Therefore $(\delta^{(2)}, \delta^{(1)}, \delta^{(0)}) = \sum_{i=1}^{n} (\delta_i^{(2)}, \delta_i^{(1)}, \delta_i^{(0)})$. Since each triple $(\delta_i^{(2)}, \delta_i^{(1)}, \delta_i^{(0)})$ lies in one summand on the right-hand side, their sum lies in that right-hand side subspace sum as well.

($\supseteq$) We prove $\mathrm{Der}(s \otimes t) \supseteq \mathrm{Der}(s) \boxtimes \mathcal{C}(t) + \mathcal{C}(s) \boxtimes \mathrm{Der}(t) + \iota_{20}(\mathcal{L}(s) \boxtimes \mathcal{L}(t)) + \iota_{10}(\mathcal{R}(s) \boxtimes \mathcal{R}(t)) + \iota_{21}(\mathcal{M}(s) \boxtimes \mathcal{M}(t))$ by proving each summand on the right belongs to $\mathrm{Der}(s \otimes t)$.

For $\mathrm{Der}(s) \boxtimes \mathcal{C}(t)$, take
$$(\alpha, \beta, \gamma) \in \mathrm{Der}(s), \qquad (f, g, h) \in \mathcal{C}(t).$$
Then $(\alpha, \beta, \gamma) \boxtimes (f, g, h) = \big(\kappa_2(\alpha \otimes f), \kappa_1(\beta \otimes g), \kappa_0(\gamma \otimes h)\big)$.

As $t \circ_2 f = t \circ_1 g = t \circ_0 h$ and $s \circ_2 \alpha + s \circ_1 \beta = s \circ_0 \gamma$,

$$\begin{aligned}(s \otimes t) \circ_2 \kappa_2(\alpha \otimes f) + (s \otimes t) \circ_1 \kappa_1(\beta \otimes g) &= \big((s \circ_2 \alpha) \otimes (t \circ_2 f)\big) + \big((s \circ_1 \beta) \otimes (t \circ_1 g)\big) \\ &= \big((s \circ_2 \alpha) + (s \circ_1 \beta)\big) \otimes t \circ_0 h \\ &= (s \circ_0 \gamma) \otimes (t \circ_0 h) \\ &= (s \otimes t) \circ_0 \kappa_0(\gamma \otimes h).\end{aligned}$$

Taking span, $\mathrm{Der}(s) \boxtimes \mathcal{C}(t)$ lies in $\mathrm{Der}(s \otimes t)$. Hence $\mathrm{Der}(s) \boxtimes \mathcal{C}(t) \subseteq \mathrm{Der}(s \otimes t)$ by linearity.

The argument for $\mathcal{C}(s) \boxtimes \mathrm{Der}(t) \subseteq \mathrm{Der}(s \otimes t)$ is identical, and the argument for nucleus terms is also analogous after setting one of the three terms to 0. □

A full derivation decomposition example is the Appendix (Example D.2).

5.5. **Non-existence in higher valence.** In this subsection, we demonstrate an obstruction to being able to decompose trilinear maps' derivations.

Let $\mathcal{U} := \mathrm{Tri}(U_3, U_2, U_1; U_0)$, and $\mathcal{V} := \mathrm{Tri}(V_3, V_2, V_1; V_0)$, where Tri indicates trilinear. Fix $t_1 \in \mathcal{U}$ and $t_2 \in \mathcal{V}$, and let $(\delta^{(3)}, \delta^{(2)}, \delta^{(1)}, \delta^{(0)}) \in \mathrm{Der}(t_1 \otimes t_2)$. This gives tensors $p, q, r, s \in \mathcal{U} \otimes \mathcal{V}$ with $p + q + r = s$. We begin with the four-term definition of the LIU decomposition.

Definition 5.16. Let U and V be finite dimensional $\mathbb{F}$-vector spaces. Let $p, q, r, s \in U \otimes V$ be nonzero elements satisfying $p + q + r = s$. Let

$$\begin{aligned} A &:= \mathrm{Im}_L(p), \quad B := \mathrm{Im}_L(q), \quad C := \mathrm{Im}_L(r), \quad D := \mathrm{Im}_L(s), \\ X &:= \mathrm{Im}_R(p), \quad Y := \mathrm{Im}_R(q), \quad Z := \mathrm{Im}_R(r), \quad W := \mathrm{Im}_R(s). \end{aligned}$$

By Proposition 4.5, $p \in A \otimes X$, $q \in B \otimes Y$, $r \in C \otimes Z$, and $s \in D \otimes W$.

A (p, q, r, s) *LIU decomposition* consists of a natural number n, and vectors

$$a_i \in A,\ b_i \in B,\ c_i \in C,\ d_i \in D, \quad x_i \in X,\ y_i \in Y,\ z_i \in Z,\ w_i \in W,$$

for $i \in [n]$, satisfying the following:

- **Decomposition equality.**

$$p = \sum_{i=1}^{n} a_i \otimes x_i, \quad q = \sum_{i=1}^{n} b_i \otimes y_i, \quad r = \sum_{i=1}^{n} c_i \otimes z_i, \quad s = \sum_{i=1}^{n} d_i \otimes w_i.$$

- **Local equality.**

$$\forall i \in [n], \quad a_i \otimes x_i + b_i \otimes y_i + c_i \otimes z_i = d_i \otimes w_i.$$

We shall show when $p + q + r = s$, there may not exist a LIU decomposition of the quadruple. As a LIU decomposition corresponds to decomposing a derivation of product of trilinear map, its non-existence means no analgous theorem to Theorem 5.1 exists.

A natural question is whether one can reduce the $p + q + r = s$ case to the three-term case by setting $\tilde{p} = p + q$, and applying Lemma 5.13 to both $\tilde{p} + r = s$ and $p + q = \tilde{p}$. However, separate LIU decompositions of $(\tilde{p}, r, s)$ and $(p, q, \tilde{p})$ does not guarantee a LIU decomposition of (p, q, r, s). Applying Lemma 5.13 to $(\tilde{p}, r, s)$ gives

$$\tilde{p} = \sum_i \tilde{a}_i \otimes \tilde{x}_i, \quad \tilde{a}_i \otimes \tilde{x}_i \in \mathrm{Im}_L(\tilde{p}) \otimes \mathrm{Im}_R(\tilde{p}).$$

The equation $p + q = \tilde{p}$ means $\mathrm{Im}_L(\tilde{p}) \leq \mathrm{Im}_L(p) + \mathrm{Im}_L(q)$ and $\mathrm{Im}_R(\tilde{p}) \leq \mathrm{Im}_R(p) + \mathrm{Im}_R(q)$. To recover a (p, q, r, s) LIU decomposition, we need the following equation to be satisfied for each i:

$$\tilde{a}_i \otimes \tilde{x}_i = a_i \otimes x_i + b_i \otimes y_i \quad (a_i \in \mathrm{Im}_L(p),\ x_i \in \mathrm{Im}_R(p),\ b_i \in \mathrm{Im}_L(q),\ y_i \in \mathrm{Im}_R(q)),$$

but this is not guaranteed. Rank-one terms in $(A + B) \otimes (X + Y)$ can contain cross blocks terms in $A \otimes Y$ and $B \otimes X$. For example, with $A = \langle e_1 \rangle$, $B = \langle e_2 \rangle$, $X = \langle f_1 \rangle$, $Y = \langle f_2 \rangle$,

$$(e_1 + e_2) \otimes (f_1 + f_2) = e_1 \otimes f_1 + e_1 \otimes f_2 + e_2 \otimes f_1 + e_2 \otimes f_2,$$

so this rank-one tensor has nonzero $A \otimes Y$ and $B \otimes X$ parts and cannot be of the form $a \otimes x + b \otimes y$ with $a \in A$, $x \in X$, $b \in B$, $y \in Y$.

Lemma 5.17. Assume $s \neq 0$. Let p, q, r, s be as the above definitions. If a LIU decomposition exists, then the intersection space

$$H := (A \otimes X + B \otimes Y + C \otimes Z) \cap (D \otimes W)$$

contains a nonzero rank-one tensor.

Proof. Each local equality has right-hand side $d_i \otimes w_i \in D \otimes W$ and left-hand side in $A \otimes X + B \otimes Y + C \otimes Z$, hence $d_i \otimes w_i \in H$ for every i. Since

$$s = \sum_{i=1}^{n} d_i \otimes w_i \neq 0,$$

at least one $d_i \otimes w_i$ is nonzero, giving a nonzero rank-one tensor in H. □

We now show that LIU decompositions may not exist.

Theorem 5.18. Let $\mathbb{F}$ be a field such that $\mathrm{char}(\mathbb{F}) \neq 2$, and let $J \in \mathbb{M}_2(\mathbb{F})$ be a matrix with an irreducible minimal polynomial. Let $U = V = \mathbb{F}^4$ be vector spaces. Then there exist nonzero elements $p, q, r, s \in U \otimes V$ satisfying $p + q + r = s$ such that the quadruple (p, q, r, s) admits no LIU decomposition.

Proof. Take $U = V = \mathbb{F}^4$ with standard bases $\{e_i\}$ and $\{f_i\}$, and set

$$D = \mathrm{span}\{e_1, e_2\}, \quad D' = \mathrm{span}\{e_3, e_4\}, \qquad W = \mathrm{span}\{f_1, f_2\}, \quad W' = \mathrm{span}\{f_3, f_4\}.$$

Define isomorphisms

$$\iota_D : D \to D',\ e_1 \mapsto e_3,\ e_2 \mapsto e_4, \qquad \iota_W : W \to W',\ f_1 \mapsto f_3,\ f_2 \mapsto f_4.$$

For $J \in \mathbb{M}_2(\mathbb{F})$, use the fixed bases (e_1, e_2) of D, (e_3, e_4) of D', (f_1, f_2) of W, and (f_3, f_4) of W' to define linear maps

$$J_D : D \to D', \qquad J_W : W \to W',$$

where J_D acts on columns vectors, and J_W on rows vectors using the entries of J. Write $Jd := J_D(d)$ and $wJ := J_W(w)$. With this notation, define the subspaces

$$A = \{(d, \iota_D d) : d \in D\}, \qquad B = \{(d, -\iota_D d) : d \in D\}, \qquad C = \{(d, Jd) : d \in D\},$$
$$X = \{(w, \iota_W w) : w \in W\}, \qquad Y = \{(w, -\iota_W w) : w \in W\}, \qquad Z = \{(w, wJ) : w \in W\},$$

and the intersection space $H := (A \otimes X + B \otimes Y + C \otimes Z) \cap (D \otimes W)$. We prove all nonzero elements in H are rank 2, and construct a nonzero element $s \in H$ equal to $p + q + r$ to show existence.

Let $P, Q, R, S \in \mathbb{M}_2(\mathbb{F})$. Relative to $U = D \oplus D'$ and $V = W \oplus W'$, for $p \in A \otimes X$, $q \in B \otimes Y$, $r \in C \otimes Z$, and $s \in D \otimes W$, we have

$$p = \begin{bmatrix} P & P \\ P & P \end{bmatrix}, \qquad q = \begin{bmatrix} Q & -Q \\ -Q & Q \end{bmatrix}, \qquad r = \begin{bmatrix} R & RJ \\ JR & JRJ \end{bmatrix}, \qquad s = \begin{bmatrix} S & 0 \\ 0 & 0 \end{bmatrix}. \tag{5.3}$$

Hence $p + q + r = s$ is equivalent to the equation

$$\begin{bmatrix} P + Q + R & P - Q + RJ \\ P - Q + JR & P + Q + JRJ \end{bmatrix} = \begin{bmatrix} S & 0 \\ 0 & 0 \end{bmatrix},$$

meaning

$$P + Q + R = S, \qquad P - Q = -RJ, \qquad P - Q = -JR, \qquad P + Q = -JRJ.$$

Using the second and third equations, $RJ - JR = 0$, so R and J commute. Because the minimal polynomial of J is irreducible over $\mathbb{F}_5$, the matrix $I_2 - J^2$ is invertible. Using this on the last equation, we get $P + Q = -RJ^2$. Substitution $-RJ^2$ for $P + Q$ in the first equation, we conclude $S = R - RJ^2$, or $R = S(I_2 - J^2)^{-1}$. This defines P, Q, and R in terms of the fixed J and S via

$$P = -\frac{1}{2}RJ(I_2 + J), \qquad Q = \frac{1}{2}RJ(I_2 - J), \qquad R = S(I_2 - J^2)^{-1}. \tag{5.4}$$

Since J and R commute, R is in the commutant of J. And because the minimal polynomial of J is irreducible, $R \in \mathbb{F}[J]$ so R is invertible whenever nonzero. From above, this means S is invertible too, concluding all nonzero elements in H are rank 2. For the existence portion, fix $S = I_2$. From (5.3), we recover a quadruple (p, q, r, s) whose intersection space has all nonzero elements of rank greater than 1.

□

Example 5.19. We apply Theorem 5.18 to get a specific example. Fix $\mathbb{F} = \mathbb{F}_5$, $J = \begin{bmatrix} 0 & 4 \\ 1 & 4 \end{bmatrix}$, and $S = I_2$. Compute P, Q, R from Equation (5.4):

$$P = \begin{bmatrix} 1 & 1 \\ 4 & 2 \end{bmatrix}, \qquad Q = \begin{bmatrix} 3 & 2 \\ 3 & 0 \end{bmatrix}, \qquad R = (I_2 - J^2)^{-1} = \begin{bmatrix} 2 & 2 \\ 3 & 4 \end{bmatrix}.$$

In the standard coordinates, we write down a quadruple (p, q, r, s) from Equation (5.3),

$$p = \left[\begin{array}{cc|cc} 1 & 1 & 1 & 1 \\ 4 & 2 & 4 & 2 \\ \hline 1 & 1 & 1 & 1 \\ 4 & 2 & 4 & 2 \end{array}\right], \quad q = \left[\begin{array}{cc|cc} 3 & 2 & 2 & 3 \\ 3 & 0 & 2 & 0 \\ \hline 2 & 3 & 3 & 2 \\ 2 & 0 & 3 & 0 \end{array}\right], \quad r = \left[\begin{array}{cc|cc} 2 & 2 & 2 & 1 \\ 3 & 4 & 4 & 3 \\ \hline 2 & 1 & 1 & 2 \\ 4 & 3 & 3 & 3 \end{array}\right], \quad s = \left[\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right],$$

where for instance, $p = \sum_{i,j=1}^{2} P_{ij} a_i \otimes x_j$ where $a_i = (e_i, \iota_D e_i)$, $x_i = (f_i, \iota_W f_i)$ and so on. One checks directly that $p + q + r = s$.

6. Conclusion

Write last: a summary of what was done.

6.1. Future Directions.

(1) Depending on what gets written for the quick der chapter, randomized algorithms analysis
(2) Quick der/quicksylver for higher valence tensors
(3) X-ray idea with decomposing tensors.
(4) Derivation closure and porting the result on the derivation algebra to one on tensor spaces

Appendix A. Quicksylver and Quickder implementation performance

TODO: Add some performance numbers/charts from Quicksylver/derivation calculations from the algorithm implementation in Julia

Appendix B. Quicksylver and Quickder examples

B.1. Simultaneous Sylvester Example. Back to main text.

Example B.1. Let $\mathbb{F} = \mathbb{Q}$, with $a = b = r = s = 2$ and $c = 4$. Define

$$R[:,:,1] = \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix}, \qquad R[:,:,2] = \begin{bmatrix} 3 & 1 \\ 4 & 3 \end{bmatrix}, \qquad R[:,:,3] = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}, \qquad R[:,:,4] = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix},$$

$$S[:,:,1] = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}, \qquad S[:,:,2] = \begin{bmatrix} 3 & 2 \\ 5 & 2 \end{bmatrix}, \qquad S[:,:,3] = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}, \qquad S[:,:,4] = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix},$$

$$T[:,:,1] = \begin{bmatrix} 11 & 7 \\ 24 & 25 \end{bmatrix}, \quad T[:,:,2] = \begin{bmatrix} 19 & 10 \\ 44 & 23 \end{bmatrix}, \quad T[:,:,3] = \begin{bmatrix} 3 & -1 \\ 20 & 15 \end{bmatrix}, \quad T[:,:,4] = \begin{bmatrix} 20 & 7 \\ 20 & 11 \end{bmatrix}.$$

The full system has the unique solution

$$X = \begin{bmatrix} -1 & 1 \\ 2 & 3 \end{bmatrix}, \qquad Y = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}.$$

Now run the solve-and-lift algorithm with $I = J = \{1\}$.

Double restricted. Set

$$X_I = \begin{bmatrix} x_1 & x_2 \end{bmatrix}, \qquad Y_J = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}.$$

The restricted system is

$$\begin{bmatrix} 1 & 1 & 2 & 1 \\ 3 & 4 & 3 & 2 \\ 3 & 2 & 1 & 0 \\ 0 & 1 & 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 11 \\ 19 \\ 3 \\ 20 \end{bmatrix}.$$

It has the unique solution

$$X_I = \begin{bmatrix} -1 & 1 \end{bmatrix}, \qquad Y_J = \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

Hence $\mathcal{S}_{\mathrm{DR}} = \{(X_I, Y_J)\}$.

Row restricted. With $\widehat{J} = \{2\}$,

$$M_R = \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 1 & 0 \\ 4 & 1 \end{bmatrix}, \qquad N_R(X_I, Y_J) = \begin{bmatrix} 5 \\ 8 \\ 2 \\ 9 \end{bmatrix}.$$

Solving $M_R \widehat{Y} = N_R(X_I, Y_J)$ gives

$$\widehat{Y} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad Y = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}.$$

So

$$\mathcal{S}_{\mathrm{R}} = \left\{ \left(\begin{bmatrix} -1 & 1 \end{bmatrix}, \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix} \right) \right\}.$$

Column restricted. With $\widehat{I} = \{2\}$,

$$M_C = \begin{bmatrix} 1 & 1 \\ 3 & 4 \\ 3 & 2 \\ 0 & 1 \end{bmatrix}, \qquad N_C(X_I, Y_J) = \begin{bmatrix} 5 \\ 18 \\ 12 \\ 3 \end{bmatrix}.$$

Solving $M_C \widehat{X} = N_C(X_I, Y_J)$ gives

$$\widehat{X} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \qquad X = \begin{bmatrix} -1 & 1 \\ 2 & 3 \end{bmatrix}.$$

So

$$\mathcal{S}_{\mathrm{C}} = \left\{ \left(\begin{bmatrix} -1 & 1 \\ 2 & 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 3 \end{bmatrix} \right) \right\}.$$

Lift to full. The pullback is

$$P = \{(z_R, z_C) \in \mathcal{S}_{\mathrm{R}} \times \mathcal{S}_{\mathrm{C}} : \pi_J(z_R) = \pi_I(z_C)\} = \{p_0\},$$

where

$$p_0 := \big((X_I, Y), (X, Y_J)\big), \qquad X_I = X[I, :], \;\; Y_J = Y[:, J].$$

Then

$$\mathrm{Err}(\Psi(p_0)) = \big(XR[:, :, k] + S[:, :, k]Y - T[:, :, k]\big)_{k \in [4]} = 0_{2 \times 2 \times 4}.$$

Therefore LiftRestricted returns the unique solution (X, Y) above.

B.2. **Derivation Example.** Back to main text.

Example B.2. We illustrate the solution strategy on a $2 \times 2 \times 2$ instance. Let $a = b = c = r = s = t = 2$, and $I = J = K = \{1\}$. Define the slices

$$R[:,:,1] = S[:,:,1] = T[:,:,1] = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad R[:,:,2] = S[:,:,2] = T[:,:,2] = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Write the unknowns as

$$X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}, \qquad Y = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix}, \qquad Z = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}.$$

We solve the derivation equations by brute force first. They are

$$X\, R[:,:,k] + S[:,:,k]\, Y = (T^Z)[:,:,k] \qquad (k \in [2])$$

For $k = 1$, we have

$$X \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} Y = z_{11} T[:,:,1] + z_{21} T[:,:,2].$$

This gives the relations

$$\begin{bmatrix} x_{11} + y_{11} & y_{12} \\ x_{21} & 0 \end{bmatrix} = \begin{bmatrix} z_{11} & 0 \\ 0 & z_{21} \end{bmatrix},$$

so $y_{12} = x_{21} = z_{21} = 0$, and $x_{11} + y_{11} = z_{11}$.

And for $k = 2$,

$$X \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} Y = z_{12} T[:,:,1] + z_{22} T[:,:,2],$$

gives the relations

$$\begin{bmatrix} 0 & x_{12} \\ y_{21} & x_{22} + y_{22} \end{bmatrix} = \begin{bmatrix} z_{12} & 0 \\ 0 & z_{22} \end{bmatrix},$$

so $z_{12} = x_{12} = y_{21} = 0$, and $x_{22} + y_{22} = z_{22}$. Combined, the solution space is

$$X = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix}, \qquad Y = \begin{bmatrix} \gamma & 0 \\ 0 & \delta \end{bmatrix}, \qquad Z = \begin{bmatrix} \alpha + \gamma & 0 \\ 0 & \beta + \delta \end{bmatrix}, \qquad \alpha, \beta, \gamma, \delta \in \mathbb{F},$$

We now construct the solution space using the steps described in this section.

Step 1: Triple-restricted solve. With $I = J = K = \{1\}$, the unknown blocks are

$$X_I = \begin{bmatrix} x_{11} & x_{12} \end{bmatrix}, \qquad Y_J = \begin{bmatrix} y_{11} \\ y_{21} \end{bmatrix}, \qquad Z_K = \begin{bmatrix} z_{11} \\ z_{21} \end{bmatrix},$$

and the triple-restricted equation has left-hand and right-hand sides

$$\begin{aligned} & X_I R[:,J,1] + S[I,:,1] Y_J \\ & = \begin{bmatrix} x_{11} & x_{12} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} y_{11} \\ y_{21} \end{bmatrix} \\ & = x_{11} + y_{11}, \\ & (T^{Z_K})[I,J,1] = \big(z_{11} T[:,:,1] + z_{21} T[:,:,2]\big)[1,1] = z_{11}. \end{aligned}$$

Hence the triple-restricted constraint is $x_{11} + y_{11} = z_{11}$.

Step 2: Column-depth lift. Recall $\widehat{I} = [a] \setminus I = \{2\}$ and $\widehat{X} = X[\widehat{I},:]^T = \begin{bmatrix} x_{21} \\ x_{22} \end{bmatrix}$. Using the notation from above:

$$M_C = \big[(R[:,J,1])^T\big] = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad N_C(X_I, Y_J, Z_K) = \big[(T^{Z_K})[\widehat{I},J,1] - S[\widehat{I},:,1] Y_J\big]^T = \begin{bmatrix} 0 \end{bmatrix}.$$

So the equation $M_C \widehat{X} = N_C(X_I, Y_J, Z_K)$ corresponds to the system

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_{21} \\ x_{22} \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix},$$

concluding $x_{21} = 0$.

Step 3: Row-depth lift. Recall $\widehat{J} = [b] \setminus J = \{2\}$ and $\widehat{Y} = Y[:, \widehat{J}] = \begin{bmatrix} y_{12} \\ y_{22} \end{bmatrix}$.

$$M_R = \begin{bmatrix} S[I, :, 1] \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad N_R(X_I, Y_J, Z_K) = \begin{bmatrix} (T^{Z_K})[I, \widehat{J}, 1] - X_I R[:, \widehat{J}, 1] \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}.$$

So the equation $M_R \widehat{Y} = N_R(X_I, Y_J, Z_K)$ corresponds to the system

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} y_{12} \\ y_{22} \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix},$$

concluding $y_{12} = 0$.

Step 4: Row-col lift. Recall $\widehat{K} = [c] \setminus K = \{2\}$ and $\widehat{Z} = Z[:, \widehat{K}] = \begin{bmatrix} z_{12} \\ z_{22} \end{bmatrix}$. The row-col lift uses

$$M_D = \begin{bmatrix} \mathrm{vec}(T[I, J, 1]) & \mathrm{vec}(T[I, J, 2]) \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix},$$

$$N_D(X_I, Y_J, Z_K) = \mathrm{vec}\big(X_I R[:, J, 2] + S[I, :, 2] Y_J\big) = \begin{bmatrix} 0 \end{bmatrix}.$$

So the equation $M_D \widehat{Z} = N_D(X_I, Y_J, Z_K)$ corresponds to the system

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} z_{12} \\ z_{22} \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix},$$

concluding $z_{12} = 0$.

Step 5: Pullback and final residual solve. Using the three lifts and the pullback conditions, the compatible candidates are

$$X = \begin{bmatrix} x_{11} & x_{12} \\ 0 & x_{22} \end{bmatrix}, \qquad Y = \begin{bmatrix} y_{11} & 0 \\ y_{21} & y_{22} \end{bmatrix}, \qquad Z = \begin{bmatrix} x_{11} + y_{11} & 0 \\ z_{21} & z_{22} \end{bmatrix}.$$

Now apply the final lift condition $\mathrm{Err}(X, Y, Z) = 0$, where $\mathrm{Err}(X, Y, Z)[:, :, k] = X\, R[:, :, k] + S[:, :, k] Y - (T^Z)[:, :, k]$. For the candidate family above,

$$(T^Z)[:, :, 1] = \begin{bmatrix} x_{11} + y_{11} & 0 \\ 0 & z_{21} \end{bmatrix}, \qquad \mathrm{Err}(X, Y, Z)[:, :, 1] = \begin{bmatrix} 0 & 0 \\ 0 & -z_{21} \end{bmatrix},$$

$$(T^Z)[:, :, 2] = \begin{bmatrix} 0 & 0 \\ 0 & z_{22} \end{bmatrix}, \qquad \mathrm{Err}(X, Y, Z)[:, :, 2] = \begin{bmatrix} 0 & x_{12} \\ y_{21} & x_{22} + y_{22} - z_{22} \end{bmatrix}.$$

So $\mathrm{Err}(X, Y, Z) = 0$ requires

$$z_{21} = 0, \qquad x_{12} = 0, \qquad y_{21} = 0, \qquad z_{22} = x_{22} + y_{22}.$$

Hence the full derivation space matches what we computed initially as

$$X = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix}, \qquad Y = \begin{bmatrix} \gamma & 0 \\ 0 & \delta \end{bmatrix}, \qquad Z = \begin{bmatrix} \alpha + \gamma & 0 \\ 0 & \beta + \delta \end{bmatrix}, \qquad \alpha, \beta, \gamma, \delta \in \mathbb{F},$$

Appendix C. Proof of Proposition 5.3

Throughout, let R and S be algebras with multiplication bimaps μ_R and μ_S, and set

$$X \coloneqq \mathrm{Der}(\mu_R) \boxtimes \mathcal{C}(\mu_S), \qquad Y \coloneqq \mathcal{C}(\mu_R) \boxtimes \mathrm{Der}(\mu_S),$$
$$N \coloneqq \iota_{20}(\mathcal{L}(\mu_R) \boxtimes \mathcal{L}(\mu_S)) + \iota_{10}(\mathcal{R}(\mu_R) \boxtimes \mathcal{R}(\mu_S)) + \iota_{21}(\mathcal{M}(\mu_R) \boxtimes \mathcal{M}(\mu_S)).$$

The proof analyzes the decomposition from Theorem 5.1

$$\mathrm{Der}(\mu_R \otimes \mu_S) = X + Y + N.$$

Since algebra derivations are diagonal triples, it is enough to identify the diagonal of $X + Y + N$. We first show that diagonalization splits across the sum, and then identify the diagonal subspace of X, Y, and N with the three summands in Brešar's decomposition. We start with the following.

Lemma C.1. If A is a unital algebra and $\mu : A \times A \rightarrowtail A$ is its multiplication tensor, then μ is fully nondegenerate.

Proof. $\mu(a,1) = a$ and $\mu(1,a) = a$ force both the left and right radical to be 0, and $\operatorname{span}(\mu(A,A)) = A$ since every $a \in A$ equals $\mu(a,1)$. $\square$

The map $d \mapsto (d,d,d)$ identifies the derivation algebra of A, $\mathcal{D}_{\text{alg}}(A)$, with the diagonal of $\operatorname{Der}(\mu_A)$. For $x \in A$, write $\lambda_x(a) := xa$ and $\rho_x(a) := ax$ for the left and right multiplication operators. Recall the definitions for a unital algebra A [Sch95, Ch. I, §2, p. 10]:

$$\begin{aligned}
\text{(left nucleus)} \quad & \operatorname{Nuc}_\ell(A) := \{x \in A : \forall a,b \in A,\ (xa)b = x(ab)\},\\
\text{(middle nucleus)} \quad & \operatorname{Nuc}_m(A) := \{y \in A : \forall a,b \in A,\ (ay)b = a(yb)\},\\
\text{(right nucleus)} \quad & \operatorname{Nuc}_r(A) := \{z \in A : \forall a,b \in A,\ a(bz) = (ab)z\},\\
\text{(nucleus)} \quad & \operatorname{Nuc}(A) := \operatorname{Nuc}_\ell(A) \cap \operatorname{Nuc}_m(A) \cap \operatorname{Nuc}_r(A),\\
\text{(center)} \quad & Z(A) := \{c \in \operatorname{Nuc}(A) : \forall a \in A,\ ca = ac\}.
\end{aligned}$$

We do not separately define the centroid of an algebra A, since in the unital case it identifies canonically with the center. The tensor versions are as follows.

Proposition C.2. Let A be a unital algebra with multiplication tensor μ. Then

$$\begin{aligned}
\mathcal{M}(\mu) &= \{(\rho_y, \lambda_y) : y \in \operatorname{Nuc}_m(A)\}, & \mathcal{L}(\mu) &= \{(\lambda_x, \lambda_x) : x \in \operatorname{Nuc}_\ell(A)\},\\
\mathcal{R}(\mu) &= \{(\rho_z, \rho_z) : z \in \operatorname{Nuc}_r(A)\}, & \mathcal{C}(\mu) &= \{(\lambda_c, \lambda_c, \lambda_c) : c \in Z(A)\}.
\end{aligned}$$

Proof. We prove $\mathcal{M}(\mu) = \{(\rho_y, \lambda_y) : y \in \operatorname{Nuc}_m(A)\}$.

For $\subseteq$, let $(\alpha, \beta) \in \mathcal{M}(\mu)$. Then

$$(\forall a, b \in A) \quad (\alpha(a))b = a(\beta(b)).$$

Setting $b = 1$ gives $\alpha(a) = a\,\beta(1)$, so $\alpha = \rho_y$ with $y = \beta(1)$. Setting $a = 1$ gives $\beta(b) = \alpha(1)\,b$, so $\beta = \lambda_y$ with $y = \alpha(1)$. Hence $y = \alpha(1) = \beta(1)$ and $(ay)b = a(yb)$, so $y \in \operatorname{Nuc}_m(A)$. Therefore $(\alpha, \beta) = (\rho_y, \lambda_y)$ is in the right-hand side.

For $\supseteq$, let $y \in \operatorname{Nuc}_m(A)$. Then for all $a, b \in A$,

$$(\rho_y(a))b = (ay)b = a(yb) = a(\lambda_y(b)),$$

so $(\rho_y, \lambda_y) \in \mathcal{M}(\mu)$.

The equations for $\mathcal{L}(\mu)$ and $\mathcal{R}(\mu)$ are proved identically. Finally,

$$\begin{aligned}
\mathcal{C}(\mu) &= \{(\alpha, \beta, \gamma) : \mu \circ_2 \alpha = \mu \circ_1 \beta = \mu \circ_0 \gamma\}\\
&= \{(\alpha, \beta, \gamma) : (\alpha, \beta) \in \mathcal{M}(\mu),\ (\alpha, \gamma) \in \mathcal{L}(\mu),\ (\beta, \gamma) \in \mathcal{R}(\mu)\}.
\end{aligned}$$

Using the previous three parts, let $(\alpha, \beta, \gamma) \in \mathcal{C}(\mu)$. Then

$$(\alpha, \gamma) = (\lambda_x, \lambda_x), \quad (\beta, \gamma) = (\rho_z, \rho_z), \quad (\alpha, \beta) = (\rho_y, \lambda_y)$$

for some $x \in \operatorname{Nuc}_\ell(A)$, $z \in \operatorname{Nuc}_r(A)$, $y \in \operatorname{Nuc}_m(A)$. Hence $\lambda_x = \rho_z$, $\lambda_x = \rho_y$, and $\rho_z = \lambda_y$, so $x = y = z =: c$ and $\lambda_c = \rho_c$. Thus $c \in \operatorname{Nuc}_\ell(A) \cap \operatorname{Nuc}_m(A) \cap \operatorname{Nuc}_r(A)$ and $ca = ac$ for all a. This means $c \in Z(A)$.

Conversely, if $c \in Z(A)$ then $(\lambda_c, \lambda_c, \lambda_c) \in \mathcal{C}(\mu)$. $\square$

For $u \in \operatorname{Nuc}(A)$, write $\operatorname{ad}(u) := \lambda_u - \rho_u$; then $\operatorname{ad}(u) \in \mathcal{D}_{\text{alg}}(A)$.

Definition C.3. Let V be a vector space. For a subspace $W \leq \operatorname{End}(V)^3$, define

$$W^{\text{diag}} := \{d \in \operatorname{End}(V) : (d,d,d) \in W\}.$$

The following proposition is key to identifying pieces of a unital algebra derivation with pieces of tensor derivations.

Proposition C.4. Let A be a unital algebra with multiplication tensor μ, and let $N_A := \iota_{20}(\mathcal{L}(\mu)) + \iota_{10}(\mathcal{R}(\mu)) + \iota_{21}(\mathcal{M}(\mu))$. Then

$$N_A^{\text{diag}} = \operatorname{ad}(\operatorname{Nuc}(A)).$$

Proof. For $\supseteq$, let $u \in \mathrm{Nuc}(A)$. By Proposition C.2,

$$(\lambda_u, \lambda_u) \in \mathcal{L}(\mu), \qquad (\rho_u, \rho_u) \in \mathcal{R}(\mu), \qquad (\rho_u, \lambda_u) \in \mathcal{M}(\mu).$$

Hence $\iota_{20}(\lambda_u, \lambda_u) - \iota_{10}(\rho_u, \rho_u) + \iota_{21}(\rho_u, \lambda_u) = (\mathrm{ad}(u), \mathrm{ad}(u), \mathrm{ad}(u)) \in N_A$, so $\mathrm{ad}(u) \in N_A^{\mathrm{diag}}$.

For $\subseteq$, let $d \in N_A^{\mathrm{diag}}$. Then $(d, d, d) \in N_A$, so

$$(d, d, d) = \iota_{20}(\alpha, \gamma) + \iota_{10}(\beta, \gamma') + \iota_{21}(\alpha', \beta').$$

By Proposition C.2, there exist $a \in \mathrm{Nuc}_\ell(A)$, $b \in \mathrm{Nuc}_r(A)$, and $c \in \mathrm{Nuc}_m(A)$ with

$$(\alpha, \gamma) = (\lambda_a, \lambda_a), \quad (\beta, \gamma') = (\rho_b, \rho_b), \quad (\alpha', \beta') = (\rho_c, \lambda_c).$$

Therefore $(d, d, d) = (\lambda_a - \rho_c,\ \rho_b + \lambda_c,\ \lambda_a + \rho_b)$. Comparing components gives $\rho_b = -\rho_c$, and $\lambda_a = \lambda_c$. Since A is unital, left and right multiplication are injective, so $a = c$, and $b = -c$. Hence $c \in \mathrm{Nuc}_\ell(A) \cap \mathrm{Nuc}_m(A) \cap \mathrm{Nuc}_r(A) = \mathrm{Nuc}(A)$ and

$$(d, d, d) = (\mathrm{ad}(c), \mathrm{ad}(c), \mathrm{ad}(c)),$$

so $d \in \mathrm{ad}(\mathrm{Nuc}(A))$. □

For the remainder of the proof, set

$$N_R := \iota_{20}(\mathcal{L}(\mu_R)) + \iota_{10}(\mathcal{R}(\mu_R)) + \iota_{21}(\mathcal{M}(\mu_R)), \qquad N_S := \iota_{20}(\mathcal{L}(\mu_S)) + \iota_{10}(\mathcal{R}(\mu_S)) + \iota_{21}(\mathcal{M}(\mu_S)).$$

For a unital algebra A, define the *defect map*

$$\epsilon_A : \mathrm{End}(A)^3 \to A^2, \qquad \epsilon_A(\alpha, \beta, \gamma) := (\alpha(1_A), \beta(1_A)).$$

The kernel of this defect map for tensor derivations is the diagonal of algebra derivations. Specifically, we have the following.

Lemma C.5. $\mathrm{Der}(\mu_A) \cap \mathrm{Ker}(\epsilon_A) = \{(d, d, d) : d \in \mathcal{D}_{\mathrm{alg}}(A)\}$.

Proof. Let $(\alpha, \beta, \gamma) \in \mathrm{Der}(\mu_A)$ with $\alpha(1_A) = \beta(1_A) = 0$. Setting $y = 1_A$ in

$$\alpha(x)y + x\beta(y) = \gamma(xy)$$

gives $\alpha(x) = \gamma(x)$, and setting $x = 1_A$ gives $\beta(y) = \gamma(y)$. Hence $\alpha = \beta = \gamma = d$. Computing, $d(xy) = d(x)y + xd(y)$, so $d \in \mathcal{D}_{\mathrm{alg}}(A)$. The reverse inclusion is immediate. □

For $U \leq \mathrm{End}(A)^3$ a vector subspace, we also define $\epsilon_A(U) := \{\epsilon_A(u) : u \in U\}$.

Remark C.6. Let $\Delta : \mathcal{D}_{\mathrm{alg}}(A) \to \mathrm{Der}(\mu_A) \cap \mathrm{Ker}(\epsilon_A)$ be the diagonal map $d \mapsto (d, d, d)$. By Lemma C.5, Δ is an isomorphism. For any subspace $W \leq \mathrm{Der}(\mu_A)$, we have $\Delta(W^{\mathrm{diag}}) = W \cap \mathrm{Ker}(\epsilon_A)$, thus $W^{\mathrm{diag}} \xrightarrow{\sim} W \cap \mathrm{Ker}(\epsilon_A)$.

For $A = R \otimes S$, if $(\alpha, \beta, \gamma) \in \mathrm{End}(R)^3$ and $(f, g, h) \in \mathrm{End}(S)^3$, then evaluating at $(1_R \otimes 1_S)$ gives

$$\epsilon_A\big((\alpha, \beta, \gamma) \boxtimes (f, g, h)\big) = (\alpha(1_R) \otimes f(1_S),\ \beta(1_R) \otimes g(1_S)). \tag{C.1}$$

The next lemma allows for understanding the diagonal subspace of $\mathrm{Der}(\mu_R \otimes \mu_S)$.

Lemma C.7. With the notation above,

$$(X + Y + N)^{\mathrm{diag}} = X^{\mathrm{diag}} + Y^{\mathrm{diag}} + N^{\mathrm{diag}}.$$

Proof. The inclusion $X^{\mathrm{diag}} + Y^{\mathrm{diag}} + N^{\mathrm{diag}} \subseteq (X + Y + N)^{\mathrm{diag}}$ is immediate.

In the reverse direction, let $d \in (X + Y + N)^{\mathrm{diag}}$. Then $(d, d, d) \in X + Y + N$, so write $(d, d, d) = x + y + n$ with $x \in X$, $y \in Y$, and $n \in N$. Our goal is to find some correction factors $x_N \in X \cap N$, $y_N \in Y \cap N$ where $x - x_N \in X \cap \mathrm{Ker}(\epsilon_A)$, $y - y_N \in Y \cap \mathrm{Ker}(\epsilon_A)$ and $n + x_N + y_N \in N \cap \mathrm{Ker}(\epsilon_A)$. By Remark C.6, these three terms are in X^{diag}, Y^{diag}, and Z^{diag} respectively.

Let

$$E_R := \epsilon_R(\mathrm{Der}(\mu_R)), \quad F_R := \epsilon_R(N_R), \quad E_S := \epsilon_S(\mathrm{Der}(\mu_S)), \quad F_S := \epsilon_S(N_S).$$

Thus E_R and E_S are the defects of the full derivation algebra, while F_R and F_S are the defects of embedded nuclei. This gives the relations $F_R \leq E_R$ and $F_S \leq E_S$. Now view $E_R \otimes Z(S)$ as a subspace of A^2 by $(a,b) \otimes z \mapsto (a \otimes z, b \otimes z)$, and similarly for $Z(R) \otimes E_S$.

Our first goal is to show $\epsilon_A(X) = E_R \otimes Z(S)$ and $\epsilon_A(Y) = Z(R) \otimes E_S$. Since $X = \mathrm{Der}(\mu_R) \boxtimes \mathcal{C}(\mu_S)$, every element of X is a sum of terms $\delta \boxtimes (\lambda_z, \lambda_z, \lambda_z)$ with $\delta \in \mathrm{Der}(\mu_R)$ and $z \in Z(S)$. Writing $\delta = (\alpha, \beta, \gamma)$, applying ϵ_A gives

$$\epsilon_A\big(u \boxtimes (\lambda_z, \lambda_z, \lambda_z)\big) = (\alpha(1_R) \otimes z, \beta(1_R) \otimes z).$$

This shows $\epsilon_A(X) \leq E_R \otimes Z(S)$. Conversely, for $\delta \in \mathrm{Der}(\mu_R)$, $\epsilon_R(\delta) = (a,b) \in E_R$ and for $z \in Z(S)$, we get

$$(a \otimes z, b \otimes z) = \epsilon_A\big(\delta \boxtimes (\lambda_z, \lambda_z, \lambda_z)\big),$$

so $E_R \otimes Z(S) \leq \epsilon_A(X)$. Thus we concldue

$$\epsilon_A(X) = E_R \otimes Z(S), \qquad \epsilon_A(Y) = Z(R) \otimes E_S. \tag{C.2}$$

The next step is to constrain the spaces the correction factors come from. The key observation is if $(d,d,d) = x + y + n$, then $\epsilon_A(x + y + n) = 0$, so $\epsilon_A(x) = -\epsilon_A(y) - \epsilon_A(n)$. As we want a correction factor x_N such that $\epsilon_A(x - x_N) = 0$ (so it is in X^{diag}) and x_N is in N (so it can be added to n), this necessitates $\epsilon_A(X) \cap \big(\epsilon_A(Y) + \epsilon_A(N)\big) = \epsilon_A(X \cap N)$. Similarly, we need $\epsilon_A(Y) \cap \big(\epsilon_A(X) + \epsilon_A(N)\big) = \epsilon_A(Y \cap N)$.

Because $Y = \mathcal{C}(\mu_R) \boxtimes \mathrm{Der}(\mu_S)$, the first two R-components of $\epsilon_A(Y)$ lie in $Z(R)^2 \leq F_R$, so $\epsilon_A(Y) \leq F_R \otimes S$. Also, as N is built from N_R on the R-side, we get $\epsilon_A(N) \leq F_R \otimes S$. Hence,

$$\epsilon_A(Y) + \epsilon_A(N) \leq F_R \otimes S, \qquad \epsilon_A(X) + \epsilon_A(N) \leq R \otimes F_S. \tag{C.3}$$

Since $N_R \boxtimes \mathcal{C}(\mu_S) \leq X \cap N$, applying ϵ_A gives

$$\epsilon_A(N_R \boxtimes \mathcal{C}(\mu_S)) = F_R \otimes Z(S) \subseteq \epsilon_A(X \cap N). \tag{C.4}$$

And analogously, $Z(R) \otimes F_S \subseteq \epsilon_A(Y \cap N)$. Chaining it all together

$$\begin{aligned}
\epsilon_A(X) \cap \big(\epsilon_A(Y) + \epsilon_A(N)\big) &\subseteq (E_R \otimes Z(S)) \cap (F_R \otimes S) && \text{by (C.2) and (C.3)} \\
&= F_R \otimes Z(S) && \text{by Lemma 2.2, } F_R \leq E_R \text{, and } Z(S) \leq S \\
&\subseteq \epsilon_A(X \cap N) && \text{by (C.4)}
\end{aligned}$$

Since $X \cap N \leq X$ and $X \cap N \leq Y + N$, we also have

$$\epsilon_A(X \cap N) \subseteq \epsilon_A(X) \cap \big(\epsilon_A(Y) + \epsilon_A(N)\big).$$

Hence this argument, and the same argument with R and S exchanged gives

$$\epsilon_A(X) \cap \big(\epsilon_A(Y) + \epsilon_A(N)\big) = \epsilon_A(X \cap N) \qquad \epsilon_A(Y) \cap \big(\epsilon_A(X) + \epsilon_A(N)\big) = \epsilon_A(Y \cap N). \tag{C.5}$$

Finally, put everything together. Recall from above $(d,d,d) = x + y + n$ for $x \in X$, $y \in Y$, and $n \in N$. Since $(d,d,d) \in \mathrm{Ker}(\epsilon_A)$,

$$\epsilon_A(x) = -\epsilon_A(y) - \epsilon_A(n) \in \epsilon_A(X) \cap (\epsilon_A(Y) + \epsilon_A(N)).$$

By Equation (C.5), $\epsilon_A(X) \cap (\epsilon_A(Y) + \epsilon_A(N)) = \epsilon_A(X \cap N)$, so $\epsilon_A(x) \in \epsilon_A(X \cap N)$. Choose $x_N \in X \cap N$ with $\epsilon_A(x_N) = \epsilon_A(x)$. Similarly choose $y_N \in Y \cap N$ with $\epsilon_A(y_N) = \epsilon_A(y)$. Set

$$x_0 \coloneqq x - x_N, \qquad y_0 \coloneqq y - y_N, \qquad n_0 \coloneqq n + x_N + y_N.$$

Then $x_0 \in X \cap \mathrm{Ker}(\epsilon_A)$, $y_0 \in Y \cap \mathrm{Ker}(\epsilon_A)$, and also $n_0 \in N$ with

$$\epsilon_A(n_0) = \epsilon_A(n) + \epsilon_A(x_N) + \epsilon_A(y_N) = \epsilon_A(n) + \epsilon_A(x) + \epsilon_A(y) = \epsilon_A(x + y + n) = 0.$$

Hence $n_0 \in N \cap \mathrm{Ker}(\epsilon_A)$.

To conclude, we have the equation $(d,d,d) = x_0 + y_0 + n_0$. By Remark C.6 applied to X,Y,and N,

$$X \cap \mathrm{Ker}(\epsilon_A) = X^{\mathrm{diag}}, \qquad Y \cap \mathrm{Ker}(\epsilon_A) = Y^{\mathrm{diag}}, \qquad N \cap \mathrm{Ker}(\epsilon_A) = N^{\mathrm{diag}}.$$

Therefore $d \in X^{\mathrm{diag}} + Y^{\mathrm{diag}} + N^{\mathrm{diag}}$ as needed. □

Proof of Proposition 5.3. By Lemma C.1, μ_R and μ_S are fully nondegenerate, so Theorem 5.1 gives $\mathrm{Der}(\mu_R \otimes \mu_S) = X + Y + N$. Therefore

$$\begin{aligned}\mathcal{D}_{\text{alg}}(R \otimes S) &= \mathrm{Der}(\mu_R \otimes \mu_S)^{\text{diag}} \\ &= (X + Y + N)^{\text{diag}} \\ &= X^{\text{diag}} + Y^{\text{diag}} + N^{\text{diag}} \qquad \text{by Lemma C.7.}\end{aligned}$$

It remains to identify these three diagonal summands.

For X^{diag}, we show $\mathcal{D}_{\text{alg}}(R) \otimes Z(S) = X^{\text{diag}}$. The inclusion $\mathcal{D}_{\text{alg}}(R) \otimes Z(S) \subseteq X^{\text{diag}}$ is immediate. If $d \in \mathcal{D}_{\text{alg}}(R)$ and $c \in Z(S)$, then $(d, d, d) \in \mathrm{Der}(\mu_R)$, and $(\lambda_c, \lambda_c, \lambda_c) \in \mathcal{C}(\mu_S)$.

For the reverse inclusion, let $u \in X^{\text{diag}}$, so $(u, u, u) \in X$. Choose a basis $z_1, \ldots, z_m$ of $Z(S)$. By Proposition C.2, we have a basis of $\mathcal{C}(\mu_S)$ as $\{(\lambda_{z_i}, \lambda_{z_i}, \lambda_{z_i}) : 1 \le i \le m\}$. Hence there exist $d_i = (d_{2,i}, d_{1,i}, d_{0,i}) \in \mathrm{Der}(\mu_R)$ such that

$$\begin{aligned}(u, u, u) &= \sum_{i=1}^{m} d_i \boxtimes (\lambda_{z_i}, \lambda_{z_i}, \lambda_{z_i}) \\ &= \Big(\sum_{i=1}^{m} d_{2,i} \otimes \lambda_{z_i}, \sum_{i=1}^{m} d_{1,i} \otimes \lambda_{z_i}, \sum_{i=1}^{m} d_{0,i} \otimes \lambda_{z_i}\Big).\end{aligned}$$

Evaluating the three components on $r \otimes 1_S$ gives

$$\sum_{i=1}^{m} d_{2,i}(r) \otimes z_i = \sum_{i=1}^{m} d_{1,i}(r) \otimes z_i = \sum_{i=1}^{m} d_{0,i}(r) \otimes z_i$$

for all $r \in R$. Since $z_1, \ldots, z_m$ is a basis of $Z(S)$, we get $d_{2,i} = d_{1,i} = d_{0,i}$ for each i. Thus each d_i is of the form $d_i = (\delta_i, \delta_i, \delta_i)$ with $\delta_i \in \mathcal{D}_{\text{alg}}(R)$, and

$$u = \sum_{i=1}^{m} \delta_i \otimes \lambda_{z_i} \in \mathcal{D}_{\text{alg}}(R) \otimes Z(S).$$

Hence $X^{\text{diag}} = \mathcal{D}_{\text{alg}}(R) \otimes Z(S)$, as needed. The same argument with R and S exchanged gives $Y^{\text{diag}} = Z(R) \otimes \mathcal{D}_{\text{alg}}(S)$.

For N^{diag}, recall

$$N = \iota_{20}(\mathcal{L}(\mu_R) \boxtimes \mathcal{L}(\mu_S)) + \iota_{10}(\mathcal{R}(\mu_R) \boxtimes \mathcal{R}(\mu_S)) + \iota_{21}(\mathcal{M}(\mu_R) \boxtimes \mathcal{M}(\mu_S)).$$

By Theorem 4.1, this is

$$N = \iota_{20}(\mathcal{L}(\mu_R \otimes \mu_S)) + \iota_{10}(\mathcal{R}(\mu_R \otimes \mu_S)) + \iota_{21}(\mathcal{M}(\mu_R \otimes \mu_S)),$$

for which we apply Proposition C.4 to. Thus $N^{\text{diag}} = \mathrm{ad}(\mathrm{Nuc}(R \otimes S))$. Finally, using Brešar's nucleus identity $\mathrm{Nuc}(R \otimes S) = \mathrm{Nuc}(R) \otimes \mathrm{Nuc}(S)$ [Bre17, p. 2] gives

$$N^{\text{diag}} = \mathrm{ad}(\mathrm{Nuc}(R \otimes S)) = \mathrm{ad}(\mathrm{Nuc}(R) \otimes \mathrm{Nuc}(S)).$$

Therefore $\mathcal{D}_{\text{alg}}(R \otimes S) = \mathcal{D}_{\text{alg}}(R) \otimes Z(S) + Z(R) \otimes \mathcal{D}_{\text{alg}}(S) + \mathrm{ad}(\mathrm{Nuc}(R) \otimes \mathrm{Nuc}(S))$, which is the subspace form of Equation (5.2). □

Appendix D. Tensor product bimap decomposition examples

D.1. Nucleus decomposition example. Back to main text.

Example D.1. We decompose a term in $\mathcal{M}(s \otimes t)$ as sum of terms from $\mathcal{M}(s) \boxtimes \mathcal{M}(t)$. Let $\mathbb{F} = \mathbb{F}_5$, and let $s, t : \mathbb{F}^2 \times \mathbb{F}^2 \rightarrowtail \mathbb{F}$ be bilinear maps with

$$S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad T = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Given matrices

$$X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad Y = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \text{we verify indeed } X(S \otimes T) = (S \otimes T)Y = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Let x, y correspond to the matrices X and Y. Then $(s \otimes t) \circ_2 x = (s \otimes t) \circ_1 y$, so $(x, y) \in \mathcal{M}(s \otimes t)$. Set $w := \Xi\big((s \otimes t) \circ_2 x\big) = \Xi\big((s \otimes t) \circ_1 y\big)$. A direct calculation gives

$$\begin{aligned} &w = u_1 \otimes v_1 + u_2 \otimes v_2, \quad \text{where} \\ &u_1 = E_{11} + E_{12}, \quad v_1 = E_{11}, \quad u_2 = E_{22}, \quad v_2 = E_{22}. \end{aligned}$$

Choose

$$\begin{aligned} (\alpha_1, f_1) &= (E_{11}, E_{11}), & (\beta_1, g_1) &= (E_{11} + E_{12}, E_{11}), \\ (\alpha_2, f_2) &= (E_{22}, E_{22}), & (\beta_2, g_2) &= (4E_{12} + E_{22}, E_{22}). \end{aligned}$$

Then $u_i = s \circ_2 \alpha_i = s \circ_1 \beta_i$ and $v_i = t \circ_2 f_i = t \circ_1 g_i$ for $i \in \{1, 2\}$. Therefore

$$x = \kappa_2(\alpha_1 \otimes f_1) + \kappa_2(\alpha_2 \otimes f_2), \qquad y = \kappa_1(\beta_1 \otimes g_1) + \kappa_1(\beta_2 \otimes g_2),$$

and we've decomposed (x, y) as the sum $(\alpha_1, \beta_1) \boxtimes (f_1, g_1) + (\alpha_2, \beta_2) \boxtimes (f_2, g_2)$ in $\mathcal{M}(s) \boxtimes \mathcal{M}(t)$.

D.2. **Derivation decomposition example.** **Back to main text.**

Example D.2. We decompose $(\delta^{(2)}, \delta^{(1)}, \delta^{(0)}) \in \mathrm{Der}(s \otimes t)$ as sum of terms from $\mathrm{Der}(s) \boxtimes \mathcal{C}(t)$ and $\mathcal{M}(s) \boxtimes \mathcal{M}(t)$. Let $\mathbb{F} = \mathbb{F}_5$. Let $s, t : \mathbb{F}^2 \times \mathbb{F}^2 \rightarrowtail \mathbb{F}$ be given by 2×2 matrices

$$S = E_{11} + E_{21} + E_{22}, \qquad T = I_2,$$

where E_{ij} denote matrix units Consider $\Delta = (\Delta^{(2)}, \Delta^{(1)}, \Delta^{(0)})$.

$$\Delta^{(2)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \qquad \Delta^{(1)} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \Delta^{(0)} = \begin{bmatrix} 1 \end{bmatrix},$$

for which we verify $\Delta^{(2)}(S \otimes T) + (S \otimes T)\Delta^{(1)} = (S \otimes T)^{\Delta^{(0)}}$. Let $\delta = (\delta^{(2)}, \delta^{(1)}, \delta^{(0)}) \in \mathrm{Der}(s \otimes t)$ be the operators corresponding to Δ. Set

$$p := (s \otimes t) \circ_2 \delta^{(2)}, \qquad q := (s \otimes t) \circ_1 \delta^{(1)}, \qquad r := (s \otimes t) \circ_0 \delta^{(0)},$$

where $p + q = r$. A direct calculation gives

$$\begin{aligned} p &= (E_{11} + E_{21}) \otimes I_2 + E_{22} \otimes E_{11}, \\ q &= E_{22} \otimes I_2 - E_{22} \otimes E_{11}, \\ r &= (E_{11} + E_{21} + E_{22}) \otimes I_2 + 0 \otimes E_{11}, \end{aligned}$$

so $p + q = r$. We read off two local equalities:

$$\begin{aligned} p &= a_1 \otimes x_1 + a_2 \otimes x_2, & a_1 &= E_{11} + E_{21},\ x_1 = I_2,\ a_2 = E_{22},\ x_2 = E_{11}, \\ q &= b_1 \otimes y_1 + b_2 \otimes y_2, & b_1 &= E_{22},\ y_1 = I_2,\ b_2 = -E_{22},\ y_2 = E_{11}, \\ r &= c_1 \otimes z_1 + c_2 \otimes z_2, & c_1 &= S,\ z_1 = I_2,\ c_2 = 0,\ z_2 = E_{11}. \end{aligned}$$

They are chosen so $a_i \otimes x_i + b_i \otimes y_i = c_i \otimes z_i$ for all $i \in \{1, 2\}$.

Let $\alpha_D = E_{11} + E_{21},\ \beta_D = E_{22},\ \gamma_D = 1$, and $f_D = g_D = I_2,\ h_D = 1$. Then

$$a_1 = s \circ_2 \alpha_D,\ b_1 = s \circ_1 \beta_D,\ c_1 = s \circ_0 \gamma_D, \qquad x_1 = t \circ_2 f_D,\ y_1 = t \circ_1 g_D,\ z_1 = t \circ_0 h_D,$$

with $(\alpha_D, \beta_D, \gamma_D) \in \mathrm{Der}(s)$ and $(f_D, g_D, h_D) \in \mathcal{C}(t)$.

Similarly, let $\alpha_N = 4E_{21} + E_{22},\ \beta_N = E_{22}$, and $f_N = g_N = E_{11}$. Then $a_2 = s \circ_2 \alpha_N,\ b_2 = -s \circ_1 \beta_N$, and $x_2 = t \circ_2 f_N,\ y_2 = t \circ_1 g_N$, meaning $(\alpha_N, \beta_N) \in \mathcal{M}(s)$, $(f_N, g_N) \in \mathcal{M}(t)$.

Therefore

$$\delta^{(2)} = \kappa_2(\alpha_D \otimes f_D) + \kappa_2(\alpha_N \otimes f_N) \quad \delta^{(1)} = \kappa_1(\beta_D \otimes g_D) - \kappa_1(\beta_N \otimes g_N), \quad \delta^{(0)} = \kappa_0(\gamma_D \otimes h_D).$$

This demonstrates $(\delta^{(2)}, \delta^{(1)}, \delta^{(0)})$ is a sum of a term from $\mathrm{Der}(s) \boxtimes \mathcal{C}(t)$ and a term from $\iota_{21}\big(\mathcal{M}(s) \boxtimes \mathcal{M}(t)\big)$.

Appendix E. Extras (will be deleted before dissertation)

Everything below the line are writing I had in the prelim or written up elsewhere that I think can potentially go somewhere in this doc, but I haven't had the chance to update their notation or figure out what background is needed for them to be coherent. Each block of writing is delineated by dashed lines.

Higher valence nucs/ders/centroids - possibly in future directions

Thus far we have only been concerned about bimaps. As a challenge upon resolving Problem 9, we propose extending the above ideas to higher valence tensors. We frame the question as solving for algebraic invariants of tensors.

We first give notation for an endomorphism acting on a specific axis of the tensor.

Definition E.1. Let $\langle t| : \prod_i U_i \rightarrowtail K$. Let $\sigma_a \in \mathrm{End}(U_a)$. Then define the tensor $\langle t|\sigma_a$ as

$$\langle t|\sigma_a|u\rangle = \langle t|\sigma_a u_a, u_{\overline{a}}\rangle$$

Where $u = (u_a, u_{\overline{a}})$ splits $u \in \prod_i U_i$ as an element of $U_a \times \prod_{i \neq a} U_i$.

Let $\langle t| : \prod_{i \in I} U_i \rightarrowtail K$. For a 2-element subset $\{a, b\} \subset I$, we define the ab-nucleus of t as

$$\mathrm{Nuc}_{ab}(t) := \{(\sigma_a, \sigma_b) \in \mathrm{End}(U_a)^{\mathrm{op}} \times \mathrm{End}(U_b) : \langle t|\sigma_a u_a, u_{\overline{a}}\rangle = \langle t|\sigma_b u_b, u_{\overline{b}}\rangle\}. \tag{E.1}$$

For $J \subset I$, define the J-centroid as

$$\mathrm{Cen}_J(t) := \left\{ (\sigma_j)_{j \in J} \in \prod_j \mathrm{End}(U_j) : \langle t|\sigma_j u_j, u_{\overline{j}}\rangle = \langle t|\sigma_k u_k, u_{\overline{k}}\rangle \ \forall j, k \in J \right\}. \tag{E.2}$$

Similary, define the J-derivation as

$$\mathrm{Der}_J(t) := \left\{ (\delta_j)_{j \in J} \in \prod_j \mathfrak{gl}(U_j) : \sum_j \langle t|\delta_j u_j, u_{\overline{j}}\rangle = 0 \right\}. \tag{E.3}$$

These spaces are computed by linear equations, and gives higher valence tensors the analogue of the centroid, adjoint, and derivation algebra of 3-tensors. Now we ask

> **Challenge A.** Can the higher valence nuclei, centroid, and derivation algebras be computed in operations fewer than cubic in the number of variables?
> Specifically for a m-valent tensor framed by n-dimensional vector spaces, we target $O(n^3)$ to find nuclei and less than $O((mn^3)^2)$ for the full derivation algebra.

A nice simple example of the product of tensors - it might be useful for the specific chapter?

Example E.2. The Kronecker product of matrices is a case of Definition 2.8, being the coordinate matrix of the tensor product of linear maps. Let $M \in \mathrm{Mult}(K^2; K^2) = \mathrm{End}(K^2)$ and $N \in \mathrm{End}(K^2)$ be given as 2×2 matrices.

$$M = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \qquad N = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$$

Then

$$M \otimes N = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \otimes \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} = \left(\begin{array}{cc|cc} 3 & 4 & \multicolumn{2}{c}{\multirow{2}{*}{$\mathbf{0}$}} \\ 5 & 6 & & \\ \hline \multicolumn{2}{c|}{\multirow{2}{*}{$\mathbf{0}$}} & 6 & 8 \\ & & 10 & 12 \end{array}\right)$$

The 4×4 matrix $M \otimes N$ is given as an element of $\mathbb{M}_2(\mathbb{M}_2(K)) \cong \mathbb{M}_4(K)$. It has interpretation $\langle M \otimes N| : (K^2 \otimes K^2) \rightarrowtail (K^2 \otimes K^2)$ given by mapping basis element $e_i \otimes e_j$ (dual basis $\epsilon_i \otimes \epsilon_j$) to the $2(i-1)+j$th column (row) of the matrix. Specifically, $(\epsilon_k \otimes \epsilon_l)\langle M \otimes N|e_i \otimes e_j\rangle$ is row $2k-1+l$, column $2i-1+j$ of $M \otimes N$, given by the product of the (k,i)th entry of M with the (l,j)th entry of N.

Small example, possibly worthwhile incorporating after notation fix or put into slides

Example E.3. Let $\langle s| : K^2 \times K \rightarrowtail K^2$ be the right scalar action map, and $\langle t| : K \times K^3 \rightarrowtail K^3$ be the left scalar action map. The map $\langle s \otimes t| : (K^2 \otimes K) \times (K \otimes K^3) \rightarrowtail (K^2 \otimes K^3)$ is defined as

$$\langle s \otimes t|u \otimes k_1, k_2 \otimes v\rangle = \langle s|u, k_2\rangle \otimes \langle t|k_1, v\rangle = k_2 u \otimes k_1 v = k_1 k_2 (u \otimes v) \tag{E.4}$$

Let f be an isomorphism of $K^2 \otimes K$ with K^2 and g an isomorphism of $K \otimes K^3$ with K^3. Then $\langle s \otimes t|$ is identified with the outer product tensor $\langle r| : K^2 \times K^3 \rightarrowtail K^2 \otimes K^3$ by mapping the first input via f and the second via g. This is called an *isotopism* of tensors. [Wil16]

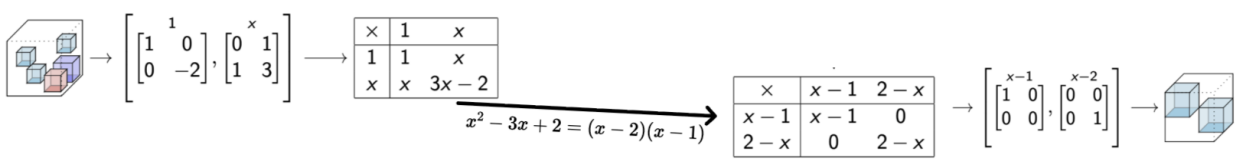
I don't think I'll have enough time to talk about the derivation closure results, but this could be a good example for the $\mathrm{Der}(s \otimes t)$ result too.

Example E.4. Let $\langle s| : \mathfrak{sl}_2 \times \mathfrak{sl}_2 \rightarrowtail \mathfrak{sl}_2$ be the multiplication tensor of $\mathfrak{sl}_2$. We compute $(\!|s|\!) = Ks$. Let $\langle t| : \mathbb{F}_3^2 \times \mathbb{F}_3^2 \rightarrowtail \mathbb{F}_3^2$ be the multiplication table of $\mathbb{F}_3[x]/(x^2+1)$. We compute $(\!|t|\!)$ as spanned by t and the tensor r given by the system of forms $\left[\begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\right]$.

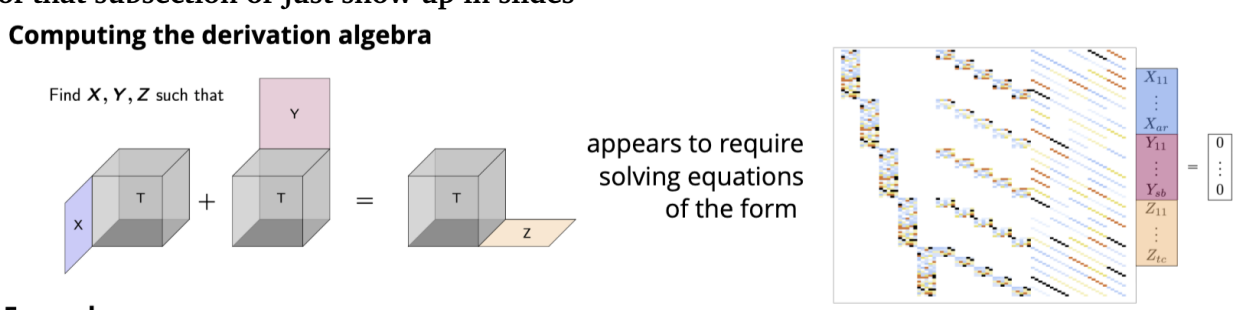
As expected, $(\!|s|\!) \otimes (\!|t|\!)$ and $(\!|s \otimes t|\!)$ are both 2 dimensional. By definition $(\!|s|\!) \otimes (\!|t|\!)$ is spanned by $s \otimes t$ and $s \otimes r$, while computation verifies $(\!|s \otimes t|\!)$ is spanned by these same tensors.

An example I created for quicksylver that could be useful to paste in here.

(In gradshow poster) Has nice visuals/can go into intro-ish chapter



(In gradshow poster) A picture for the derivation algebra that may go well in the intro part of that subsection or just show up in slides



(In gradshow poster) Small but concrete example that may be useful somewhere

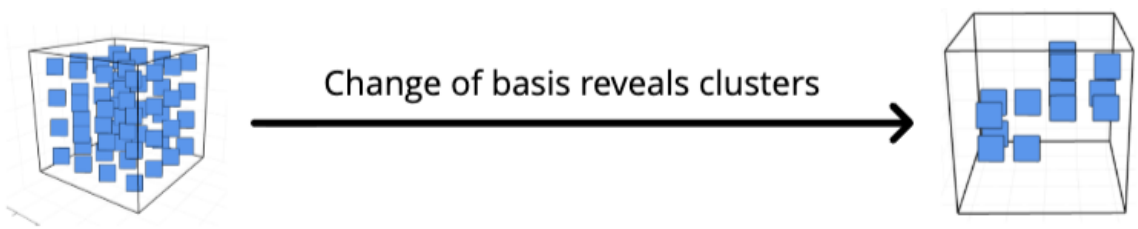
Example

$$t_1 = \begin{bmatrix} 10 & 5 \\ 5 & 6 \end{bmatrix} \quad t_2 = \begin{bmatrix} 9 & 3 \\ 3 & 7 \end{bmatrix}$$

$$T = [t_1, t_2]$$

$$\mathrm{Der}(T) = (X, Y, Z) = \mathrm{nullspace}\left(\begin{array}{cccc|cccc|cccc}
x_{11} & x_{12} & x_{21} & x_{22} & y_{11} & y_{21} & y_{12} & y_{22} & z_{11} & z_{12} & z_{21} & z_{22} \\
10 & 5 & & & 10 & 5 & & & -10 & & -9 & \\
9 & 3 & & & 9 & 3 & & & & -10 & & -9 \\
5 & 6 & & & & & 10 & 5 & -5 & & -3 & \\
3 & 7 & & & & & 9 & 3 & & -5 & & -3 \\
 & & 10 & 5 & 5 & 6 & & 0 & -5 & & -3 & \\
 & & 9 & 3 & 3 & 7 & & 0 & & -5 & & -3 \\
 & & 5 & 6 & & & 5 & 6 & -6 & & -7 & \\
 & & 3 & 7 & & & 3 & 7 & & -6 & & -7
\end{array}\right)$$

(In gradshow poster) Concrete change of basis clustering example, visual may help motivate in the intro.



Numerical example

$$T_1 = \begin{bmatrix} 5 & 6 & 6 & 8 \\ 5 & 6 & 6 & 8 \\ 0 & 0 & 0 & 0 \\ 2 & 4 & 4 & 4 \end{bmatrix} \quad T_2 = \begin{bmatrix} 9 & 12 & 14 & 14 \\ 5 & 10 & 10 & 14 \\ 4 & 10 & 12 & 8 \\ 5 & 14 & 14 & 16 \end{bmatrix}$$

$$T_3 = \begin{bmatrix} 10 & 4 & 4 & 12 \\ 10 & 4 & 4 & 12 \\ 0 & 0 & 0 & 0 \\ 10 & 12 & 12 & 16 \end{bmatrix} \quad T_4 = \begin{bmatrix} 16 & 17 & 20 & 20 \\ 14 & 14 & 16 & 18 \\ 6 & 9 & 12 & 6 \\ 11 & 16 & 18 & 16 \end{bmatrix}$$

$$R = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 0 \\ 2 & 2 & 2 & 0 \\ 2 & 0 & 2 & 2 \end{bmatrix}$$

$$\tilde{T}_1 = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \tilde{T}_2 = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\tilde{T}_3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 2 & 0 \end{bmatrix} \quad \tilde{T}_4 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 1 \end{bmatrix}$$